

A dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures

OpenAI

September 23, 2026

Abstract

We prove that every centered log-concave probability measure with a Lebesgue density on $\mathbb{R}^n$ and linear subgaussian parameter a satisfies $\text{Ent}_\mu(f^2) \leq Ca^2 \int |Df|^2 d\mu$ for compactly supported smooth f , with one universal constant C . This resolves positively the dimension-free logarithmic Sobolev conjecture for subgaussian log-concave measures.

1 Introduction

A logarithmic Sobolev inequality controls the entropy of a function by its Dirichlet energy. Besides implying a Poincaré inequality, it gives Gaussian concentration for Lipschitz functions. For a log-concave measure, a natural question is whether Gaussian tails of all *linear* functions already force this stronger conclusion. We prove that they do, with a constant independent of dimension.

For a probability measure ν and a nonnegative integrable function g , write

$$\text{Ent}_\nu(g) = \int g \log g d\nu - \left(\int g d\nu \right) \log \left(\int g d\nu \right),$$

with $0 \log 0 = 0$. A Lebesgue density ρ on $\mathbb{R}^n$ is log-concave if its positive set is convex and $\log \rho$ is concave there. We call a centered random vector X *linearly a -subgaussian* if

$$\sup_{|\theta|=1} \mathbb{E} \exp \left(\frac{\langle X, \theta \rangle^2}{a^2} \right) \leq 2. \quad (1.1)$$

This definition concerns only linear functions of X . We write D for the Euclidean gradient.

Theorem 1.1. *There is an absolute constant $C < \infty$ with the following property. For every $n \geq 1$, every centered log-concave probability measure μ with a Lebesgue density on $\mathbb{R}^n$, and every $a > 0$ satisfying (1.1) for $X \sim \mu$,*

$$\text{Ent}_\mu(f^2) \leq Ca^2 \int_{\mathbb{R}^n} |Df|^2 d\mu \quad \text{for every } f \in C_c^\infty(\mathbb{R}^n).$$

The same constant works in every dimension.

No smoothness of the density or positive lower curvature bound is required. In particular, the theorem includes uniform measures on convex bodies. The proof will use smooth positive densities only after a reduction that preserves any hypothetical sequence of counterexamples.

The Otto–Villani implication from logarithmic Sobolev to quadratic transport–entropy inequalities [17, Theorem 1] gives the following standard consequence, with the same constant. Write $\mathcal{P}_2(\mathbb{R}^n)$ for the Borel probabilities with finite second moment. For $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^n)$, set

$$W_2(\alpha, \beta)^2 = \inf_{\pi \in \Pi(\alpha, \beta)} \int |x - y|^2 d\pi(x, y),$$

where $\Pi(\alpha, \beta)$ is the set of their couplings. Relative entropy is $H(\alpha | \beta) = \int f \log f d\beta$ if $\alpha = f\beta$, with its possible value $+\infty$, and is $+\infty$ if $\alpha \not\ll \beta$.

Corollary 1.2 (Quadratic transport–entropy inequality). *Under the assumptions of Theorem 1.1, every $\nu \in \mathcal{P}_2(\mathbb{R}^n)$ satisfies*

$$W_2(\nu, \mu)^2 \leq Ca^2 H(\nu | \mu),$$

with the same absolute constant C as in that theorem.

The proof in Section 2.1 gives the approximation argument for nonsmooth log-concave densities.

1.1 History and significance

Gross’s work on logarithmic Sobolev inequalities connected the Gaussian inequality with hypercontractivity and established a framework in which the constants remain independent of dimension [8]. The diffusion approach of Bakry and Émery [2] gives the inequality for a density e^{-V} when $D^2V \geq \kappa \text{Id}$ with $\kappa > 0$: in the convention used below, its constant is at most κ^{-1} . Log-concavity alone asserts only $D^2V \geq 0$ when the potential is smooth. The question addressed here is whether a global assumption on the tails of linear functions can supply the logarithmic Sobolev bound without positive curvature.

A logarithmic Sobolev inequality implies Gaussian concentration for all Lipschitz functions through Herbst’s exponential-moment argument; see [14, Section 2.3]. Under log-concavity, Gaussian concentration for all Lipschitz functions also gives Gaussian isoperimetry by Milman’s theorem [16, Theorem 1.1], and hence a logarithmic Sobolev inequality [5, Lemma 3.3], with dimension-independent bounds. The question here asks whether control of linear functions suffices. Uniformly subgaussian linear marginals are necessary, up to universal constants, by the same Herbst argument. Bizeul formulated their sufficiency within the log-concave class in 2023 [4, Conjecture 2]. Theorem 1.1 resolves this conjecture for centered log-concave laws with a Lebesgue density, under the precise marginal condition (1.1).

Bobkov’s norm-based criterion [5, Theorem 1.3] bounds the entropy coefficient by a universal multiple of the squared subgaussian norm of $|X|$. Under (1.1), Hölder’s inequality gives $\mathbb{E} \exp(|X|^2/(na^2)) \leq 2$, so that criterion yields a coefficient of order na^2 . Bizeul’s improvement [4, Theorem 4] has dimension dependence $n^{1/4}$ in the parameter ρ_{LS} defined by $\text{Ent}(f^2) \leq 2\rho_{\text{LS}}^2 \int |Df|^2$. Thus for linear subgaussian scale a it gives an entropy coefficient of order $\sqrt{n}a^2$. Bizeul also obtained the dimension-free conclusion for rotationally invariant log-concave laws [4, Theorem 8]. Klartag and Lehec discuss the conjecture and these bounds in [12, Section 8.3, Conjecture 76 and Theorem 77]. The theorem’s marginal hypothesis concerns more than covariance: it controls the tails of every linear function, while its conclusion controls the entropy of nonlinear tests.

Our preliminary calculus uses the Bakry–Émery semigroup argument, including its entropy dissipation identity. We give the anisotropic and approximation forms needed for Gaussian posteriors in Section 2. The other classical input to that calculus comes from Gaussian channels. Guo, Shamaï, and Verdú relate the derivative of mutual information to minimum mean-square estimation error [9, Theorem 2]; their relative-entropy formulation [9, Theorem 5, Equation (66)]

gives the Gaussian prediction identity used here. We prove its conditional integrated version with our variance and diffusion-time conventions. The entropy-loss comparison in Lemma 4.2 requires an additional argument: repeated observations of independent copies make the cost of discarding conditional-mean directions negligible. That comparison links Gaussian prediction to the tensor near-equalities described next.

1.2 Proof strategy

Write $R(\nu)$ for the best constant in the convention

$$\text{Ent}_\nu(f^2) \leq 2R(\nu) \int |Df|^2 d\nu. \quad (1.2)$$

Failure of Theorem 1.1 would give, after truncation, a weak quadratic tilt, and dilation, a sequence of compactly supported log-concave laws μ such that $R(\mu) = 1$ while their linear subgaussian parameters tend to zero. The argument derives incompatible descriptions of fluctuations associated with these normalized laws.

From an extremal function to a predictable vector field. For a fixed $s > 0$, let G be a standard Gaussian independent of $X \sim \mu$. An extremizer for the entropy inequality of $X + \sqrt{s}G$ yields a second probability law η and a vector field

$$w = D \log \frac{d\eta}{d\mu_s}, \quad \mu_s = \text{Law}(X + \sqrt{s}G).$$

An exact change-of-reference identity shows that η itself satisfies a logarithmic Sobolev inequality. The Euler equation and translation tests make w an approximate first eigenfunction of the diffusion associated with η . If the entropy constant instead coincides with the Poincaré constant, take $\eta = \mu_s$ and $w = D\varphi$, where φ is a gap eigenfunction. Both cases are treated directly.

The crucial extra information concerns prediction of $w(Y)$ from a noisy observation of $Y \sim \eta$. With $I = \mathbb{E}_\eta |w|^2$ and a fresh independent standard Gaussian G , we prove the limiting identity

$$\frac{\mathbb{E}[\mathbb{E}[w(Y) \mid Y + \sqrt{r}G]^2]}{I} \longrightarrow e^{-r}$$

on a fixed positive interval of noise variances. It follows by matching entropy contraction with an entropy-loss comparison proved using repeated Gaussian observations. At each observation, we discard at most one direction: the difference between the conditional means under two competing priors. Independent copies make the accumulated cost of discarding these directions negligible.

From prediction to tensor near-equalities. Gaussian polynomial orthogonality turns the exact prediction curve into nearly sharp derivative estimates. Starting with $T_0 = w$, we construct symmetric tensor fields T_l whose derivatives approximate T_{l+1} , all with asymptotically the same squared norm and all concentrated near the first nonzero spectral value of the diffusion. The tensor estimates also control curvature and exclude concentration in the relevant small-dimensional subspaces.

For each tensor T_l , regard one index as a row and all remaining indices as a column, and form the positive matrix $T_l T_l^*$. Averaging $I^{-1} T_l T_l^*$ over a long band of levels gives a positive matrix field ρ of asymptotic mean trace one. Its square root is nearly an equality case in Hilbert–Schmidt Poincaré. Symmetry in the many tensor indices forces the infinitesimal matrix coefficients to

commute asymptotically in a weighted Hilbert–Schmidt norm. Gaussian linear combinations of these matrices, frozen at the initial diffusion point, consequently have scalar Gaussian moments under the weighted trace. A Hermite expansion along the stationary diffusion then gives covariance fluctuations with the lognormal factor

$$\exp(\sqrt{8t} Z - 4t), \quad Z \sim N(0, 1),$$

whose mean is one.

An information contradiction. Conditionally on the stationary point, observe a centered Gaussian vector whose covariance is a bounded increasing function of $T\rho$, where $T > 0$ is a scale and the covariance cap is adjustable. Its marginal law is a mixture of these Gaussian laws. Together with the square-root energy estimates, the logarithmic Sobolev inequality of η gives a universal asymptotic upper bound on the mutual information, integrated against dT/T^2 . The matrix fluctuation law gives a lower bound by a one-dimensional Gaussian mixture with the lognormal variance above. After taking the sequence limit at fixed diffusion time and covariance cap, that lower bound becomes arbitrarily large as these two parameters increase together.

The scale integral is essential. In a spectral block where ρ has size β^2 , the substitution $x = T\beta^2$ gives

$$\frac{dT}{T^2} = \beta^2 \frac{dx}{x^2}.$$

This is exactly the coefficient multiplying the unnormalized trace on that block. Consequently the information lower bound retains all of the matrix mass even when the spectrum is arbitrarily diffuse.

Figure 1 records the two constructions of the critical law and the main dependencies of the information comparison.

1.3 Components of the argument

Two ingredients have formulations beyond the final contradiction. The entropy-loss comparison of Lemma 4.2 relates relative entropy under common Gaussian convolution to conditional prediction of the original relative score. Its proof separates the convexity input from the finite-copy observation argument. The matrix reduction in Section 6 converts symmetric tensor near-equalities into a weighted scalar fluctuation law, without requiring simultaneous diagonalization in varying dimensions. Section 7 then tests that law through bounded covariance observations and integrates over scale.

For orientation, Section 2 records the analytic conventions and approximation facts. Section 3 constructs the critical laws and vector fields. Section 4 proves the prediction identity, and Section 5 constructs the tensor hierarchy. Sections 6 and 7 turn that hierarchy into the contradiction.

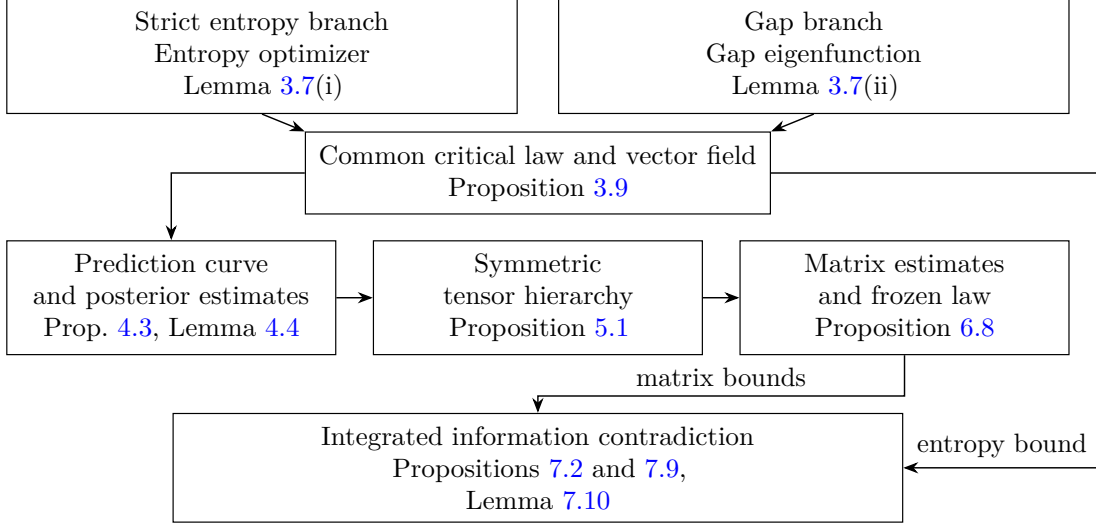


Figure 1: Main stages of the contradiction argument. At the regularized law, the strict entropy branch means that the logarithmic Sobolev constant exceeds the Poincaré constant; in the gap branch the two constants coincide. The logarithmic Sobolev inequality of the common critical law, together with the square-root energy estimates of Section 6, supplies the information upper bound. The frozen matrix law supplies the scalar lower bound. After the sequence limit at fixed parameters, its scale integral becomes arbitrarily large as the diffusion time and covariance cap grow together.

2 Curvature, diffusion, and entropy calculus

We first establish the analytic facts used to construct and study the probability measures in the proof. Two time conventions will be kept distinct: convolution with a Gaussian of variance r has generator $\Delta/2$, whereas a stationary diffusion has generator $-A$ and noise coefficient $\sqrt{2}$. These conventions determine every entropy dissipation factor below.

For probabilities p, π with $p \ll \pi$, define their relative entropy by

$$H(p \mid \pi) = \int \log \frac{dp}{d\pi} dp.$$

Set $H(p \mid \pi) = +\infty$ when absolute continuity fails. Write γ for standard Gaussian measure in the relevant dimension, and set

$$\mathcal{I}(p \mid \pi) = 4 \int \left| D \sqrt{\frac{dp}{d\pi}} \right|^2 d\pi.$$

The value is $+\infty$ unless the density exists and its square root belongs to the weighted Sobolev space. When the densities are positive and smooth this is $\int |D \log(dp/d\pi)|^2 dp$. Our convention is

$$\text{Ent}_\pi(f^2) \leq 2R(\pi) \int |Df|^2 d\pi, \quad 2H(p \mid \pi) \leq R(\pi) \mathcal{I}(p \mid \pi).$$

The Poincaré constant $R^P(\pi)$ uses the convention

$$\text{Var}_\pi(f) = \int f^2 d\pi - \left(\int f d\pi \right)^2 \leq R^P(\pi) \int |Df|^2 d\pi.$$

Linearization of the entropy inequality gives $R^P(\pi) \leq R(\pi)$.

For smooth positive densities the Sobolev space is the completion of C_c^∞ in the squared norm $\int (f^2 + |Df|^2) d\pi$. Cutoffs, truncation of the values, and smoothing on compact sets show that it also consists of the functions with a weak gradient in this norm. The same approximation and lower semicontinuity of entropy extend a compact-test logarithmic Sobolev inequality to this space. One convenient proof of the latter lower semicontinuity is the formula

$$H(p \mid \pi) = \sup_{g \text{ bounded}} \left\{ \int g dp - \log \int e^g d\pi \right\}. \quad (2.1)$$

Jensen's inequality proves the upper bound; bounded truncations of $\log(dp/d\pi)$ give equality. The same formula implies data processing under a measurable map or a Markov kernel, by conditional Jensen. Conditional entropy identities below are always averaged over the conditioning variable. In particular, disintegrating a joint density and splitting its logarithm gives the chain rule

$$H(P_{U,Z} \mid Q_U \otimes \nu) = H(P_U \mid Q_U) + \mathbb{E}_{P_U} H(P_{Z|U} \mid \nu).$$

It holds in the extended nonnegative sense by truncation. Mutual information is $I(U; Z) = H(P_{U,Z} \mid P_U \otimes P_Z)$. Taking $g(z) = v \cdot z$ in Equation (2.1) and optimizing over v also gives $H(p \mid \gamma) \geq |\mathbb{E}_p Z|^2/2$ when the mean exists.

The following is the anisotropic Bakry–Émery curvature criterion [2]. We include its semigroup proof and the convex approximation needed for Gaussian posteriors with nonsmooth support.

Lemma 2.1 (Positive curvature). *Let S be a positive definite matrix on $\mathbb{R}^n$, and let U be a proper lower semicontinuous convex function, possibly taking the value $+\infty$. Suppose*

$$d\pi(x) = Z^{-1} \exp\{-\tfrac{1}{2}\langle x, Sx \rangle - U(x)\} dx$$

is a probability with a full dimensional support. Then

$$\text{Ent}_\pi(f^2) \leq 2\mathbb{E}_\pi\langle Df, S^{-1}Df \rangle, \quad \text{Var}_\pi(f) \leq \mathbb{E}_\pi\langle Df, S^{-1}Df \rangle.$$

The inequalities hold first for compact smooth tests, and then for functions obtained by finite-energy approximation.

Proof. A linear change of variables reduces the assertion to $S = \text{Id}$. First suppose $V(x) = |x|^2/2 + U(x)$ is smooth with bounded Hessian. The diffusion with generator $L = \Delta - DV \cdot D$ has globally Lipschitz drift and invariant law π . If J_t is the derivative of its trajectory with respect to the starting point, then $\dot{J}_t = -D^2V(X_t)J_t$, whence $\|J_t\|_{\text{op}} \leq e^{-t}$. For smooth bounded positive q , bounded away from zero, weighted Cauchy–Schwarz gives

$$\frac{|DP_t q|^2}{P_t q} \leq e^{-2t} P_t \left(\frac{|Dq|^2}{q} \right).$$

Integration by parts yields

$$\frac{d}{dt} \int (P_t q) \log(P_t q) d\pi = - \int \frac{|DP_t q|^2}{P_t q} d\pi.$$

Couple a trajectory started at x to an independent stationary initial point with the same driving Brownian motion. Their distance is at most e^{-t} times their initial distance. Since π has Gaussian

tails, this proves convergence of $P_t q$ to $\mathbb{E}_\pi q$ for bounded Lipschitz q , and hence convergence of its entropy to zero. Integrating the preceding two displays gives $\text{Ent}_\pi(q) \leq \frac{1}{2} \int |Dq|^2 / q \, d\pi$. Apply this to $q = f^2 + \varepsilon$ and let $\varepsilon \downarrow 0$.

For general U , take its quadratic infimal convolutions

$$U_j(x) = \inf_z \{U(z) + j|x - z|^2\}.$$

The minimizer is unique, and monotonicity of the subgradient of U shows that DU_j is Lipschitz. The functions U_j are convex and increase to U . A fixed affine minorant $U(z) \geq \ell \cdot z + c$ gives $U_j(x) \geq \ell \cdot x + c - |\ell|^2/(4j)$. Mollify U_j with a smooth compactly supported kernel at radii tending to zero. The resulting functions are smooth, convex, and have bounded Hessian; the affine lower bound persists with a uniform additive error. Consequently their normalized densities converge almost everywhere and in L^1 , dominated by an integrable Gaussian times an exponential of a linear function. The compact-test inequality passes to the limit. Boundary values of U do not matter, since the boundary of a full dimensional convex support has Lebesgue measure zero. Finally, insert $f = 1 + \varepsilon g$, using finite-energy cutoffs of the constant, and compare the coefficients of ε^2 to obtain the variance inequality. $\square$

We shall also use two immediate consequences with their constants. If independent laws satisfy inequalities with quadratic energy forms Q_1, Q_2 , their product satisfies the inequality with energy $Q_1 + Q_2$. Indeed, decompose entropy into conditional entropy and the entropy of the fiber integral, and use

$$\left| D_x \left(\int f(x, y)^2 d\pi_2(y) \right)^{1/2} \right|^2 \leq \int |D_x f(x, y)|^2 d\pi_2(y).$$

Approximation removes zeros of the fiber norm. The variance version uses conditional variance and differentiation of the fiber mean. In particular, independent addition gives

$$R(\text{Law}(X + Z)) \leq R(\text{Law}(X)) + R(\text{Law}(Z)). \quad (2.2)$$

Second, Herbst's argument (see [14, Section 2.3]) shows that, if $R(\pi) \leq R$, every integrable 1-Lipschitz function u satisfies

$$\log \mathbb{E}_\pi e^{t(u - \mathbb{E}_\pi u)} \leq Rt^2/2. \quad (2.3)$$

For a bounded smooth u , applying the inequality to $e^{tu/2}$, and writing $\psi(t) = \log \mathbb{E} e^{tu}$, gives $t\psi'(t) - \psi(t) \leq Rt^2/2$. Integrate the derivative of $\psi(t)/t$ from zero. Bounded Lipschitz truncations and Fatou's lemma prove the assertion in general. In particular the conclusion applies to linear forms whenever their first moments are finite.

Lemma 2.2 (Dirichlet operator and stationary diffusion). *Let π be a smooth, strictly positive probability density on $\mathbb{R}^n$. On $C_c^\infty(\mathbb{R}^n)$ define*

$$A_\pi = -\Delta - D \log \pi \cdot D.$$

This nonnegative symmetric operator is essentially self-adjoint in $L^2(\pi)$, and C_c^∞ is a core for its closure. In particular, if $f \in L^2(\pi)$ and the distribution $A_\pi f$ belongs to $L^2(\pi)$, then $f \in \text{Dom}(A_\pi)$. Its semigroup $P_u = e^{-uA_\pi}$ preserves constants and is realized by the stationary diffusion

$$dY_u = \sqrt{2} dB_u + D \log \pi(Y_u) du, \quad Y_0 \sim \pi. \quad (2.4)$$

For $f \in \text{Dom}(A_\pi)$ the stationary Itô formula holds in L^2 , with drift $-A_\pi f$ and martingale integrand $\sqrt{2} Df$.

Proof. Integration by parts gives $\langle f, A_\pi g \rangle = \int Df \cdot Dg \, d\pi$. If $v \in L^2(\pi)$ is orthogonal to $(1 + A_\pi)C_c^\infty$, then $(1 + A_\pi)v = 0$ distributionally. Conjugating by $\sqrt{\pi}$ reduces this to a Poisson equation with a smooth local potential. The local Newtonian potential estimates and bootstrap make v smooth; see [20, Lecture 24, p. 2; Lecture 16, p. 2]. Take a compact cutoff χ_R , equal to one on the ball of radius R , with $|D\chi_R| \leq C/R$. Testing by $\chi_R^2 v$ and using $2ab \leq a^2/2 + 2b^2$ gives

$$\int \chi_R^2 v^2 \, d\pi + \frac{1}{2} \int \chi_R^2 |Dv|^2 \, d\pi \leq 2 \int |D\chi_R|^2 v^2 \, d\pi.$$

Thus $v = 0$. The range of $1 + A_\pi$ is dense. Nonnegativity gives $\|(1 + A_\pi)f\|_2 \geq \|f\|_2$, so the range of the closure is also closed and therefore equals L^2 . The closure is self-adjoint: for g in its adjoint domain, solve $(1 + \bar{A})h = (1 + A^*)g$; then $g - h$ is in the orthogonal complement of the range of $1 + A$, hence vanishes. This also proves the asserted maximal distributional domain characterization.

The cutoffs χ_R converge to 1 in the form norm with energy tending to zero. Hence 1 belongs to the nullspace of the closed form and $P_u 1 = 1$. To identify the diffusion, solve the stochastic equation locally and kill it on exiting each ball. Its kernels are symmetric and sub-Markov, as follows either from the Dirichlet heat equation or integration by parts. Increasing the balls gives the minimal sub-Markov semigroup. The stopped Itô formula on compact tests identifies its generator as an extension of $-A_\pi$. Essential self-adjointness identifies it with P_u , which loses no mass. Started in π , the resulting diffusion is stationary and nonexplosive. Finally approximate f in graph norm by compact smooth functions. Their gradients converge in $L^2(\pi)$ since $\|Dg\|_2^2 = \langle g, Ag \rangle$. Stationarity and the Itô isometry pass every term of their formulas to the limit. $\square$

Lemma 2.3 (Entropy dissipation). *For the semigroup of Lemma 2.2, if $p_0 = q_0\pi$ has finite relative entropy and $p_u = (P_u q_0)\pi$, then, for $0 \leq a \leq b$,*

$$H(p_a \mid \pi) - H(p_b \mid \pi) \geq \int_a^b \mathcal{I}(p_u \mid \pi) \, du. \quad (2.5)$$

*If instead both probability measures evolve by Gaussian convolution, $p_r = p_0 * \gamma_r$ and $q_r = q_0 * \gamma_r$, then*

$$H(p_a \mid q_a) - H(p_b \mid q_b) \geq \frac{1}{2} \int_a^b \mathcal{I}(p_r \mid q_r) \, dr \quad (2.6)$$

whenever the entropy on the left at time a is finite. Here γ_r is the centered Gaussian with covariance $r\text{Id}$, and time zero is allowed. For the standard Gaussian Ornstein–Uhlenbeck semigroup, Equation (2.5) is an equality, including time zero when the initial entropy is finite. Each assertion also holds after integration over conditioning labels whose noises are independent conditional on those labels.

Proof. For a bounded density q bounded away from zero, the form chain rule, first for smooth functions and then by form approximation, gives

$$\frac{d}{du} \int (P_u q) \log(P_u q) \, d\pi = - \int \frac{|DP_u q|^2}{P_u q} \, d\pi.$$

Spectral smoothing puts $P_u q$ in the operator domain for $u > 0$; the bounded range permits $1 + \log(P_u q)$ as a form test. Approximate an arbitrary finite-entropy density by truncating it above and below and normalizing. The initial entropies converge. The semigroup contracts L^1 , so densities converge at every later time. Lower semicontinuity of entropy and of $4 \int |D\sqrt{q}|^2 \, d\pi$,

followed by Fatou's lemma in time, gives Equation (2.5) from time zero. Restarting the argument gives arbitrary a .

For simultaneous heat evolution put $z = p_r/q_r$. Direct differentiation, using $\partial_r p_r = \Delta p_r/2$ and the same equation for q_r , gives

$$(\partial_r - \tfrac{1}{2}\Delta)\{q_r\Phi(z)\} = -\tfrac{1}{2}q_r\Phi''(z)|Dz|^2. \quad (2.7)$$

Here use smooth convex Φ_k with $\Phi_k(1) = \Phi'_k(1) = 0$, with second derivatives increasing to $1/z$, and with second derivatives supported in compact subsets of $(0, \infty)$. Such functions have at most linear growth and increase to $z \log z - z + 1$. Integrate Equation (2.7) against χ_R with $|\Delta\chi_R| \leq C/R^2$. At each fixed k the cutoff error vanishes because $q_r\Phi_k(p_r/q_r) \leq C_k(p_r + q_r)$. Let $R \rightarrow \infty$ and then $k \rightarrow \infty$. Monotone convergence gives the entropy and information terms, and in particular the stated inequality on positive time intervals. Data processing gives $H(p_\varepsilon | q_\varepsilon) \leq H(p_0 | q_0)$; let $\varepsilon \downarrow 0$ to include the initial time.

For Ornstein–Uhlenbeck evolution relative to γ , apply the same convex truncations to the ratio. Its generator is $\Delta - x \cdot D$. Expanding radial cutoffs have bounded generator, tending pointwise to zero, so the error vanishes by domination at fixed k . This gives equality on positive time intervals. At time zero, relative entropy is continuous: data processing gives its upper bound and lower semicontinuity under weak convergence gives its lower bound. Monotone convergence of the nonnegative time integral includes zero. Tonelli's theorem proves all conditional versions. $\square$

We use the relative-entropy form of the Gaussian-channel identity of Guo, Shamai, and Verdú [9, Theorem 5, Equation (66)]. The proof records explicitly its conditional integrated form.

Lemma 2.4 (Gaussian signals). *Let S be an $\mathbb{R}^d$ -valued random vector with finite second moment, let U be any side information, and let G be standard Gaussian independent of (S, U) . Write $Z_t = \sqrt{t}S + G$ and*

$$d(t) = \mathbb{E}_U H(\text{Law}(Z_t | U) | \gamma).$$

Then $d(0) = 0$, $d(t) \leq t\mathbb{E}|S|^2/2$, and

$$d(t) = \frac{1}{2} \int_0^t \mathbb{E} |\mathbb{E}[S | Z_v, U]|^2 dv. \quad (2.8)$$

The integrand is nondecreasing in v . The formula without side information is obtained by taking U constant.

Proof. Conditional convexity of relative entropy and $H(\gamma(\cdot - \sqrt{t}s) | \gamma) = t|s|^2/2$ give the upper bound. Gaussian kernel differentiation gives

$$D_z \log \frac{d \text{Law}(Z_t | U)}{d\gamma}(z) = \sqrt{t} \mathbb{E}[S | Z_t = z, U].$$

The differentiation is justified at each positive t by the Gaussian kernel and the finite second moment, or by truncating S and passing locally and in the conditional squared norm. Applying Ornstein–Uhlenbeck evolution for time u replaces t by te^{-2u} . Its entropy dissipation identity and $dt/du = -2t$ give $d'(t) = \frac{1}{2}\mathbb{E}|\mathbb{E}[S | Z_t, U]|^2$ almost everywhere on $(0, \infty)$. The upper bound implies $d(t) \rightarrow 0$ at zero, proving the integrated identity. For $0 < v < t$, the channel at v is obtained from the channel at t by multiplying its output by $\sqrt{v/t}$ and adding fresh Gaussian noise of covariance $(1 - v/t)\text{Id}$. Conditional Jensen then proves monotonicity of prediction energy. $\square$

2.1 The transport–entropy consequence

Proof of Corollary 1.2. Jensen’s inequality in each coordinate gives $\int |x|^2 d\mu \leq na^2 \log 2$, so $\mu \in \mathcal{P}_2(\mathbb{R}^n)$. Only the case $H(\nu \mid \mu) < \infty$ needs proof. For $s > 0$, put $\gamma_s = N(0, s\text{Id})$, $\mu_s = \mu * \gamma_s$, and $\nu_s = \nu * \gamma_s$. The density of μ_s is smooth and strictly positive. For $X \sim \mu$ and an independent standard Gaussian G , its potential V_s satisfies

$$D^2V_s(y) = s^{-1}\text{Id} - s^{-2}\text{Cov}(X \mid X + \sqrt{s}G = y) \geq 0.$$

Indeed, the posterior has the original convex potential plus $|x - y|^2/(2s)$, so Lemma 2.1 bounds its covariance by $s\text{Id}$. Gaussian differentiation is valid without compact support. Thus μ_s is centered and log-concave, with a smooth convex potential.

Set $\beta_s = \sqrt{8s/3}$ and $a_s = a + \beta_s$. For a scalar standard Gaussian Z , $\mathbb{E} \exp(sZ^2/\beta_s^2) = 2$. The inequality

$$\frac{(u+t)^2}{(a+\beta_s)^2} \leq \frac{a}{a+\beta_s} \frac{u^2}{a^2} + \frac{\beta_s}{a+\beta_s} \frac{t^2}{\beta_s^2}$$

and Hölder’s inequality show that μ_s satisfies (1.1) with parameter a_s . Theorem 1.1, with the Sobolev extension in Section 2, gives entropy coefficient Ca_s^2 . In the conventions of [17, Definitions 1–2], $\text{Ent}(f^2) \leq L \int |Df|^2$ is LSI(2/L), whose transport conclusion is $W_2^2 \leq LH$. Their Theorem 1 applies to the smooth convex potential of μ_s , and hence

$$W_2(\nu_s, \mu_s)^2 \leq Ca_s^2 H(\nu_s \mid \mu_s) \leq Ca_s^2 H(\nu \mid \mu).$$

The last inequality is data processing for the common Gaussian kernel, as in Section 2. Coupling a vector with itself plus independent Gaussian noise gives $W_2(\mu_s, \mu), W_2(\nu_s, \nu) \leq \sqrt{ns}$. Letting $s \downarrow 0$ at fixed n , with $a_s \rightarrow a$, proves the claim. $\square$

3 Critical laws after Gaussian regularization

Suppose that the conclusion of Theorem 1.1 is false. This section constructs, from a sequence of counterexamples, smooth probabilities η satisfying a logarithmic Sobolev inequality and vector fields w that nearly satisfy $A_\eta w = w$. There are two possible constructions. An attained nonlinear entropy quotient gives a score w ; equality of the logarithmic Sobolev and Poincaré constants gives the gradient of a gap eigenfunction. We shall retain both alternatives throughout.

3.1 Normalization and Gaussian regularization

Lemma 3.1 (Normalized counterexamples). *Failure of Theorem 1.1 produces a sequence of centered, compactly supported, full dimensional log-concave probabilities μ such that*

$$R(\mu) = 1, \quad \sup_{|\theta|=1} \mathbb{E}_\mu e^{\langle X, \theta \rangle^2 / a^2} \leq 2, \quad a \longrightarrow 0. \quad (3.1)$$

For an absolute constant C , they also satisfy

$$\log \mathbb{E}_\mu e^{v \cdot X} \leq Ca^2 |v|^2, \quad \text{Cov}_\mu(X) \leq Ca^2 \text{Id}. \quad (3.2)$$

Proof. Choose laws and compact smooth tests for which the entropy-to-energy ratio divided by the squared marginal parameter tends to infinity. The tests may be chosen with positive energy: a smooth test of zero energy is constant on the connected interior of the convex support, and its

entropy vanishes. Restriction of the law to a sufficiently large ball preserves each test quotient arbitrarily closely. Multiply the restricted density by $e^{-\varepsilon|x|^2/2}$ and normalize, with $\varepsilon > 0$ sufficiently small to preserve that quotient again. Both operations preserve log-concavity, their combined density relative to the original law can be bounded by 2, and Lemma 2.1 gives a finite logarithmic Sobolev constant for the resulting compact law.

A bounded density increase changes the exponential-square expectation by at most that factor. If $\mathbb{E}e^{Z^2/A^2} \leq 4$, Jensen gives $\mathbb{E}e^{Z^2/(2A^2)} \leq 2$. Centering costs only another absolute factor: $\mathbb{E}Z^2 \leq A^2 \log 4$, while $(Z - \mathbb{E}Z)^2 \leq 2Z^2 + 2(\mathbb{E}Z)^2$; increase the denominator first to make the resulting exponential expectation bounded by an absolute constant, and once more by Jensen to make it at most 2. These arguments apply separately to every unit linear form with the same constants. Translation preserves the test quotient. Thus the centered compact laws still have $R/A^2 \rightarrow \infty$ with $0 < R < \infty$. Dilation by $R^{-1/2}$ gives Equation (3.1).

For completeness, if a centered scalar Z obeys $\mathbb{E}e^{Z^2/a^2} \leq 2$, then $\mathbb{E}|Z|^k \leq 2a^k \Gamma(k/2 + 1)$, by integrating its tail. Expansion of e^{tZ} , with the linear term absent, therefore gives $\log \mathbb{E}e^{tZ} \leq Ca^2 t^2$ for $a|t| \leq c$. For $a|t| \geq c$, use $tZ \leq Z^2/a^2 + a^2 t^2/4$ and absorb $\log 2$ into $Ca^2 t^2$. Apply this to $Z = X \cdot \theta$. The second moment bound follows either from the exponential-square bound or by differentiating the centered moment generating function at zero. $\square$

We suppress the counterexample index, but all limits in this section refer to this sequence. Let G be a fresh standard Gaussian and set

$$\mu_s = \text{Law}(X + \sqrt{s}G), \quad R_s = R(\mu_s), \quad b_s(y) = \mathbb{E}[X \mid X + \sqrt{s}G = y], \quad V_s = -\log \mu_s.$$

Here and below μ_s also denotes its strictly positive density. Gaussian differentiation and posterior covariance give

$$DV_s(y) = \frac{y - b_s(y)}{s}, \quad M_s(y) = Db_s(y) = \frac{\text{Cov}(X \mid y)}{s}, \quad K_s(y) = D^2V_s(y) = \frac{\text{Id} - M_s(y)}{s}. \quad (3.3)$$

The posterior of X has a convex potential plus $|x - y|^2/(2s)$. Lemma 2.1, applied to its linear forms, therefore implies

$$0 \leq M_s \leq \text{Id}, \quad 0 \leq K_s \leq s^{-1}\text{Id}. \quad (3.4)$$

All these functions are smooth. Compact support of X also gives $|b_s| \leq \sup_{\text{supp } \mu} |x|$.

The monotonicity below follows from the heat-flow contraction of Klartag and Putterman [13, Theorem 1.2 and Section 3], which uses the transport construction of Kim and Milman [11]. We recall the argument for our regularized laws.

Lemma 3.2 (Bounds for the regularized constants). *The function $s \mapsto R_s$ is nondecreasing and 1-Lipschitz on $(0, \infty)$, and*

$$1 \leq R_s \leq 1 + s. \quad (3.5)$$

Proof. The heat equation for μ_s is the continuity equation with velocity $DV_s/2$. On a compact positive-time interval this velocity is globally Lipschitz and grows at most linearly, so its smooth flow is a diffeomorphism carrying μ_s to μ_t . Its Jacobian satisfies $\dot{J} = K_t J/2$. By positivity of K_t , the flow expands distances and its inverse is 1-Lipschitz when $t \geq s$. Transport a test by that inverse. Its entropy is unchanged and its Dirichlet energy cannot increase, proving $R_t \geq R_s$. Equation (2.2) gives $R_{s+h} \leq R_s + h$ and $R_s \leq 1 + s$. For every compact smooth test, its entropy and energy converge to those under μ as $s \downarrow 0$. Taking the supremum of its quotient gives $\liminf_{s \downarrow 0} R_s \geq R(\mu) = 1$. Monotonicity now proves the remaining bound. $\square$

3.2 Attainment above the Gaussian and Poincaré thresholds

On compact manifolds, Rothaus obtains nonconstant positive solutions of the logarithmic Euler equation when the sharp logarithmic Sobolev constant exceeds the Poincaré constant [18, p. 358]. The proof below must also control loss of entropy in the Gaussian tails. The next two lemmas establish this control and attainment. Their constants may depend on a single fixed law and its dimension. Uniform estimates along the counterexample sequence begin with the translation argument.

Lemma 3.3 (Compactness and the Gaussian entropy defect). *Fix $s > 0$ and a compactly supported probability μ on $\mathbb{R}^n$, and put $\nu = \mu * \gamma_s$. Its form domain embeds compactly into $L^2(\nu)$. For every $\varepsilon > 0$ there is a finite C_ε , depending on this fixed law, such that*

$$\text{Ent}_\nu(u^2) \leq 2(s + \varepsilon)\|Du\|_2^2 + C_\varepsilon\|u\|_2^2. \quad (3.6)$$

If $u_j \rightarrow u$ in $L^2(\nu)$, $\|u_j\|_2 = 1$, and $\|Du_j\|_2^2 \rightarrow E_* < \infty$, then

$$\limsup_j \text{Ent}_\nu(u_j^2) \leq \text{Ent}_\nu(u^2) + 2s(E_* - \|Du\|_2^2). \quad (3.7)$$

Proof. Write $r = d\nu/d\gamma_s$. The Gaussian kernel formula gives $D \log r = b_s/s$, which is bounded, and hence $|\log r(y)| \leq C(1 + |y|)$. For $v = u\sqrt{r}$,

$$\text{Ent}_\nu(u^2) = \text{Ent}_{\gamma_s}(v^2) - \int v^2 \log r \, d\gamma_s, \quad Dv = \sqrt{r}(Du + \tfrac{1}{2}uD \log r).$$

Gaussian integration by parts followed by Cauchy–Schwarz gives

$$\int |y|^2 v^2 d\gamma_s = sn\|v\|_2^2 + 2s \int v y \cdot Dv \, d\gamma_s \leq 2sn\|v\|_2^2 + 4s^2\|Dv\|_2^2.$$

Initially this calculation is for compact tests, and cutoffs extend it by lower semicontinuity. The linear-growth entropy correction is consequently bounded by $\delta\|Dv\|_2^2 + C_\delta\|v\|_2^2$ for every $\delta > 0$. Gaussian logarithmic Sobolev and the displayed derivative formula, with another arbitrarily small Cauchy–Schwarz parameter, prove Equation (3.6). The same moment estimate shows tightness of the $L^2(\nu)$ mass of any bounded form sequence. On a fixed ball, the smooth density is bounded above and away from zero, so the ordinary local Rellich theorem gives L^2 precompactness. A diagonal extraction and tightness prove compactness globally.

We give the tail argument for the sharp coefficient $2s$ in Equation (3.7). Put $v_{j,L} = (|u_j| - L)_+$ and $m_{j,L} = \|v_{j,L}\|_2^2$. Strong L^2 convergence implies $\lim_{L \rightarrow \infty} \sup_j m_{j,L} = 0$. The energies of $v_{j,L}$ and $|u_j| \wedge L$ sum to $\|Du_j\|_2^2$. Lower semicontinuity for the clipped functions therefore gives

$$\lim_{L \rightarrow \infty} \limsup_j \|Dv_{j,L}\|_2^2 \leq E_* - \|Du\|_2^2. \quad (3.8)$$

For a nonnegative v of squared norm $m \leq 1$, the negative part of $v^2 \log v^2$ has integral at most $C\sqrt{m}$ (use $-z \log z \leq C\sqrt{z}$ on $0 \leq z \leq 1$). Since $m \log m \leq 0$, Equation (3.6) thus gives

$$\int v^2 \log_+ v^2 \, d\nu \leq 2(s + \varepsilon)\|Dv\|_2^2 + C_\varepsilon m + C\sqrt{m}.$$

If $|u_j| > ML$, where $M > 1$ is fixed, then $|u_j| \leq c_M v_{j,L}$ with $c_M = M/(M - 1)$. Consequently

$$u_j^2 \log_+ u_j^2 \leq c_M^2 v_{j,L}^2 \log_+ v_{j,L}^2 + c_M^2 \log(c_M^2) v_{j,L}^2.$$

The integrals of bounded truncations of $z^2 \log z^2$ pass to the limit by strong L^2 convergence; its negative part does as well, being bounded and continuous. The last two bounds and Equation (3.8), with limits in the order j, L, M, ε , show that the lost positive entropy is at most $2s(E_* - \|Du\|_2^2)$. The normalization terms are zero because $\|u\|_2 = \|u_j\|_2 = 1$. $\square$

The threshold s in the entropy-defect estimate controls escape into the Gaussian tails. The Poincaré constant controls maximizing sequences that converge to constants. The next lemma proves attainment when the logarithmic Sobolev constant exceeds both thresholds.

Lemma 3.4 (Existence and positivity of an optimizer). *Let $\nu = \mu * \gamma_s$, where $s > 0$ and μ is compactly supported. Suppose its logarithmic Sobolev and Poincaré constants $R = R(\nu)$ and $P = R^P(\nu)$ satisfy $\infty > R > \max(P, s)$. Then there is a smooth strictly positive nonconstant f with*

$$\int f^2 d\nu = 1, \quad \text{Ent}_\nu(f^2) = 2R \int |Df|^2 d\nu, \quad A_\nu f = \lambda f \log f, \quad \lambda = R^{-1}. \quad (3.9)$$

The last equation is initially distributional and then classical.

Proof. Take nonnegative unit-norm maximizing tests f_j . Replacing a test by its absolute value preserves entropy and energy, so this choice loses nothing. Equation (3.6), with $s + \varepsilon < R$, bounds their energies. We first exclude energies tending to zero; this is the only place the strict inequality $R > P$ is required.

Suppose $\varepsilon_j^2 = \|Df_j\|_2^2 \rightarrow 0$, and write $c_j = \mathbb{E}_\nu f_j$ and $h_j = (f_j - c_j)/\varepsilon_j$. Poincaré gives $\|h_j\|_2^2 \leq P$, while $\|Dh_j\|_2^2 = 1$ and $1 - c_j^2 = \varepsilon_j^2 \|h_j\|_2^2$. Thus $c_j \rightarrow 1$. By compactness, after extraction, $h_j \rightarrow h$ in L^2 , where

$$\mathbb{E}h = 0, \quad \|Dh\|_2^2 \leq 1, \quad \|h\|_2^2 \leq P\|Dh\|_2^2.$$

For $c > 0$, $\varepsilon > 0$ and $z \geq -c/\varepsilon$ define

$$Q_{c,\varepsilon}(z) = \varepsilon^{-2} \left[(c + \varepsilon z)^2 \log \frac{(c + \varepsilon z)^2}{c^2} - (c + \varepsilon z)^2 + c^2 \right].$$

Normalization and $-\log c_j^2 - 1 + c_j^2 \geq 0$ give

$$\varepsilon_j^{-2} \text{Ent}(f_j^2) \leq \int Q_{c_j, \varepsilon_j}(h_j) d\nu. \quad (3.10)$$

Here is the scalar estimate needed to control unbounded h_j . For $1/2 \leq c \leq 1$ and $0 < \varepsilon \leq 1/2$,

$$0 \leq Q_{c,\varepsilon}(z) \leq (2 + 2 \log 2)z^2 + z^2 \log_+ z^2, \quad Q_{c,\varepsilon}(z) \rightarrow 2z^2 \quad (3.11)$$

uniformly on bounded z -intervals as $c \rightarrow 1$ and $\varepsilon \rightarrow 0$. To verify the bound, put $t = \varepsilon z/c$. Then $Q = z^2 B(t)$, where Taylor's integral formula gives

$$B(t) = 2 + 4 \int_0^1 (1-r) \log(1+rt) dr, \quad -1 \leq t < \infty.$$

For $t \leq 0$, $B(t) \leq 2$. For $t \geq 0$, use $t \leq z$ to obtain $B(t) \leq 2 + 2 \log(1+z) \leq 2 + 2 \log 2 + \log_+ z^2$. Continuity at $t = 0$ proves the compact convergence.

The tail proof in Lemma 3.3 did not use unit norm until its last sentence. In particular, for any bounded form sequence $u_j \rightarrow u$ in L^2 with energies tending to E_* , it proves

$$\lim_{K \rightarrow \infty} \limsup_j \int_{|u_j| > K} u_j^2 \log_+ u_j^2 d\nu \leq 2s(E_* - \|Du\|_2^2). \quad (3.12)$$

Apply the compact convergence in Equation (3.11) on $|h_j| \leq K$ and its upper bound on the complement. Strong L^2 convergence removes the quadratic tail as $K \rightarrow \infty$; Equation (3.12) controls the entropy tail. Equation (3.10) consequently gives

$$\limsup_j \varepsilon_j^{-2} \text{Ent}(f_j^2) \leq 2\|h\|_2^2 + 2s(1 - \|Dh\|_2^2) \leq 2\max(P, s) < 2R.$$

This contradicts the maximizing property.

We may therefore extract $f_j \rightarrow f$ strongly in L^2 with $\|Df_j\|_2^2 \rightarrow E_* > 0$. Put $E = \|Df\|_2^2$. The entropy defect and the sharp inequality give

$$2RE_* \leq \text{Ent}(f^2) + 2s(E_* - E) \leq 2RE + 2s(E_* - E).$$

Since $R > s$ and $E \leq E_*$, equality forces $E = E_* > 0$. Thus f is a nonnegative, nonconstant optimizer.

The homogeneous functional $\text{Ent}(g^2) - 2R\|Dg\|_2^2$ is nonpositive and is zero at f . Its derivative along every compact smooth variation ζ is

$$4 \int f \log f \zeta \, d\nu - 4R \int Df \cdot D\zeta \, d\nu = 0.$$

The entropy normalization term cancels the additional linear term. Differentiation is legitimate also at zeros because $z^2 \log z^2$ has derivative zero at zero. This proves the Euler equation in Equation (3.9).

We have obtained a nonnegative optimizer with positive energy. To use its logarithmic derivative, we still need local regularity and strict positivity. We spell out the weak-solution regularity step, using the interior Newtonian-potential L^p and Schauder estimates; see [20, Lecture 24, p. 2; Lecture 10, Proposition 1]. Write $\nu = e^{-V}$ and set $v = e^{-V/2}f \geq 0$. The function V is smooth, and direct differentiation of the weak equation gives

$$-\Delta v = \lambda v \log v + \left(\frac{\lambda V}{2} - \frac{|DV|^2}{4} + \frac{\Delta V}{2} \right) v.$$

The conjugating factor and its reciprocal are bounded on each fixed ball. Since $|z \log z| \leq C_\delta(1 + z^{1+\delta})$ for $z \geq 0$, local L^p integrability of v puts the right side in L_{loc}^q , where $q = p/(1+\delta) > 1$. Localize that right side on a larger ball and take its Newtonian potential with the sign solving $-\Delta u = F$. The cited L^q estimate gives $u \in W_{\text{loc}}^{2,q}$; the difference $v - u$ is distributionally harmonic and hence smooth by Weyl's lemma [20, Lecture 3, Theorem 2]. This proves the required $W^{2,q}$ gain directly for the weak solution.

For $n \geq 2$, start with $p = 2$ and choose $0 < \delta < \min(1, 2/n)$. Before boundedness is reached, Sobolev embedding improves reciprocal integrability from $1/p$ to $(1 + \delta)/p - 2/n$, a decrease greater than $1/n$ because $p \geq 2$. At a critical exponent use any sufficiently large finite integrability exponent and continue once more. After finitely many steps $q > n/2$, so v , and thus f , is locally bounded and Hölder continuous. In dimension one the same conclusion follows by twice integrating the local Poisson equation. The function $z \log z$ is locally Hölder of every exponent below one on $[0, \infty)$; reducing the exponent if necessary, the Schauder estimate therefore gives $f \in C_{\text{loc}}^{2,\alpha}$ for some $\alpha > 0$. All constants in this paragraph concern one fixed law and dimension; none enters a uniform estimate along the counterexample sequence.

If $f(x_0) = 0$, choose a ball B about x_0 with $0 \leq f \leq b < 1$ on its closure. Start the ν -diffusion at x_0 and stop at its exit time τ from B . For $Z_t = f(Y_{t \wedge \tau})$, $m(t) = \mathbb{E}Z_t$, and $g(z) = -\lambda z \log z$, the stopped Itô formula gives, for almost every t ,

$$m'(t) = \mathbb{E}[\mathbf{1}_{t < \tau} g(Z_t)] \leq \mathbb{E}g(Z_t) \leq g(m(t)), \quad m(0) = 0.$$

Indeed g is nonnegative and concave on $[0, b]$; the first inequality adds its nonnegative stopped boundary contribution. Also $m' \geq 0$. If $m(t_1) > 0$, let t_a be the first time at which $m = a$, for $0 < a < m(t_1)$. Integration gives

$$\int_a^{m(t_1)} \frac{dz}{-\lambda z \log z} \leq t_1 - t_a \leq t_1,$$

contradicting divergence as $a \downarrow 0$. Hence $m = 0$. The killed nondegenerate diffusion has positive probability of reaching each open subset of B while staying in B : one can follow a smooth interior path by requiring Brownian motion to remain in a small tube about its corresponding driving path, and use local Lipschitz continuity of the drift. Brownian tube probabilities are positive by its finite increment densities and bridge continuity. Consequently f vanishes throughout B . Its zero set is both open and closed, impossible in connected $\mathbb{R}^n$ for a unit-norm function. Thus $f > 0$, and smooth elliptic bootstrap makes f smooth everywhere. $\square$

An optimizer supplies a second probability with its own logarithmic Sobolev inequality. The next identity explains why no convexity assumption on this second probability is needed.

Lemma 3.5 (Change of reference at an optimizer). *Under Lemma 3.4, suppose in addition that $\nu = e^{-V}$ is log-concave with bounded $K = D^2V$. Define*

$$\eta = f^2 \nu, \quad h = \log f^2, \quad w = Dh, \quad I = \mathbb{E}_\eta |w|^2.$$

Then $I = 2\lambda H(\eta \mid \nu) > 0$, $R(\eta) \leq 1/\lambda$, each component of w belongs to $\text{Dom}(A_\eta)$, and

$$A_\nu h = \lambda h + \frac{1}{2}|w|^2, \quad A_\eta w = (\lambda \text{Id} - K)w. \quad (3.13)$$

Moreover,

$$\mathbb{E}_\eta \langle w, Kw \rangle \leq \lambda |\mathbb{E}_\eta w|^2. \quad (3.14)$$

Proof. Since $w = 2Df/f$, $I = 4\|Df\|_{L^2(\nu)}^2$; optimizer equality gives $I = 2\lambda H(\eta \mid \nu) > 0$. The chain rule applied to its Euler equation gives the first identity in Equation (3.13). Commuting one derivative with A_ν gives

$$A_\nu w = D(A_\nu h) - Kw = \lambda w + (Dw)w - Kw.$$

Since $A_\eta = A_\nu - w \cdot D$ and Dw is symmetric, the second identity follows distributionally. Its right side is in $L^2(\eta)$, since K is bounded and $w \in L^2(\eta)$. Lemma 2.2 therefore puts w in the operator domain; no unproved integrability of higher derivatives has been used.

Let $p = g^2 \eta$ have unit mass, where g is compactly supported and smooth. Direct expansion and compactly supported integration by parts give

$$\begin{aligned} \mathcal{I}(p \mid \nu) &= \mathcal{I}(p \mid \eta) + \int g^2 |w|^2 d\eta + 2 \int D(g^2) \cdot w d\eta \\ &= \mathcal{I}(p \mid \eta) + \int g^2 \{|w|^2 + 2A_\eta h\} d\eta \\ &= \mathcal{I}(p \mid \eta) + 2\lambda \int h dp. \end{aligned}$$

These formulas may be read in square-root form at zeros of g . On the other hand $H(p \mid \nu) = H(p \mid \eta) + \int h dp$. Thus

$$\mathcal{I}(p \mid \nu) - 2\lambda H(p \mid \nu) = \mathcal{I}(p \mid \eta) - 2\lambda H(p \mid \eta). \quad (3.15)$$

Apply the inequality for ν to the compact smooth function gf . Its deficit is nonnegative, so the right side is nonnegative. Homogeneity and the Sobolev approximation of Section 2 prove $R(\eta) \leq 1/\lambda$. Finally the domain identity gives

$$\mathbb{E}_\eta |Dw|^2 = \lambda I - \mathbb{E}_\eta \langle w, Kw \rangle,$$

while Poincaré for η gives the lower bound $\lambda(I - |\mathbb{E}_\eta w|^2)$. Subtract to obtain Equation (3.14). $\square$

3.3 Uniform estimates in the two alternatives

Lemma 3.6 (Translation gain). *Let μ satisfy Equation (3.1), with $a \leq 1$. Fix a compact interval $J \subset (0, 1/2)$. There is a constant C_J , independent of the dimension and of the support radius of μ , with the following property. For $s \in J$, put $\lambda = R_s^{-1}$. If σ is any probability with*

$$D_0 = H(\sigma \mid \mu_s) < \infty, \quad F_0 = \mathcal{I}(\sigma \mid \mu_s) < \infty, \quad p = \mathbb{E}_\sigma Y,$$

then

$$F_0 - 2\lambda D_0 \geq (s^{-2} - \lambda/s)|p|^2 - C_J \sqrt{a} (F_0 + D_0). \quad (3.16)$$

Proof. The centered linear moment generating functions under μ_s obey

$$\log \mathbb{E}_{\mu_s} e^{v \cdot Y} \leq C_J |v|^2, \quad \log \mathbb{E}_{\mu_s} e^{v \cdot b_s(Y)} \leq Ca^2 |v|^2.$$

The first follows by independent Gaussian addition and Equation (3.2); the second follows by conditional Jensen from that equation. Entropy duality, optimized over the scalar multiple of each unit vector, gives

$$|p| \leq C_J \sqrt{D_0}, \quad |\mathbb{E}_\sigma b_s(Y)| \leq Ca \sqrt{D_0}. \quad (3.17)$$

All expectations exist. More explicitly, compact support of X makes $\mathbb{E}_{\mu_s} e^{c|Y|^2} < \infty$ for some $c > 0$; entropy duality then implies $\mathbb{E}_\sigma |Y|^2 < \infty$. The latter fact is used only for integrability, not for a uniform bound in dimension.

Let σ_u be the translate of σ by $-up$, $0 \leq u \leq 1$, and let $D_u = H(\sigma_u \mid \mu_s)$. Change of variables in entropy and the Hessian bound give

$$\begin{aligned} D_u &= D_0 + \mathbb{E}_\sigma [V_s(Y - up) - V_s(Y)] \\ &\leq D_0 - u|p|^2/s + up \cdot \mathbb{E}_\sigma b_s(Y)/s + u^2|p|^2/(2s) \leq C_J D_0. \end{aligned}$$

Thus Equation (3.17), applied to σ_u , implies $|\mathbb{E}_{\sigma_u} b_s(Y - up)| \leq C_J a \sqrt{D_0}$ uniformly in u . Differentiating the exact entropy expression, and then integrating, gives

$$D_1 = D_0 - \frac{|p|^2}{2s} + O_J(aD_0). \quad (3.18)$$

Indeed its derivative is $-(1-u)|p|^2/s + p \cdot \mathbb{E}_{\sigma_u} b_s(Y - up)/s$.

Write $r(Y) = D \log(d\sigma/d\mu_s)(Y)$ in the weak square-root sense. Its mean is

$$\mathbb{E}_\sigma r = \mathbb{E}_\sigma DV_s = (p - \mathbb{E}_\sigma b_s)/s. \quad (3.19)$$

To justify this for finite Fisher information, the Lebesgue density q of σ has weak derivative $Dq = q(r - DV_s) \in L^1$: r is square integrable and DV_s has linear growth. Integrating against expanding cutoffs gives $\int Dq = 0$ and hence Equation (3.19).

In the moving coordinates Y , the score of σ_1 relative to μ_s is

$$r(Y) - p/s + d(Y)/s, \quad d(Y) = b_s(Y) - b_s(Y - p).$$

Since $0 \leq Db_s \leq \text{Id}$, integration along the segment gives

$$|d(Y)|^2 = \left| \int_0^1 M_s(Y - up)p \, du \right|^2 \leq \int_0^1 |M_s(Y - up)p|^2 \, du \leq p \cdot d(Y).$$

The mean estimates at the two endpoints therefore imply

$$\mathbb{E}_\sigma |d|^2 \leq C_J a D_0. \quad (3.20)$$

Expanding the score square and using Equation (3.19),

$$\mathbb{E}_\sigma |r - p/s|^2 = F_0 - |p|^2/s^2 + 2p \cdot \mathbb{E}_\sigma b_s/s^2.$$

The remaining cross term is bounded by $C_J \sqrt{aD_0} (\sqrt{F_0} + \sqrt{D_0})$, and the d -square term by $C_J a D_0$. Consequently the translated information is finite and satisfies

$$F_1 = F_0 - |p|^2/s^2 + O_J(\sqrt{a} (F_0 + D_0)). \quad (3.21)$$

These calculations require only the weak derivatives already justified, so they apply also when the density of σ vanishes. Apply logarithmic Sobolev to σ_1 , namely $F_1 \geq 2\lambda D_1$, and combine Equations (3.18) and (3.21). Since $2/3 \leq \lambda \leq 1$ on the time interval in question, this proves Equation (3.16). $\square$

We can now obtain estimates that are uniform along the counterexample sequence. Write R_s^P for the optimal Poincaré constant of μ_s . By linearization $R_s^P \leq R_s$. Since $R_s \geq 1 > s$ on $(0, 1/2)$, the strict alternative $R_s > R_s^P$ meets all the hypotheses of Lemma 3.4.

Lemma 3.7 (Critical fields in both alternatives). *Fix $J \subset (0, 1/2)$ compact and $s \in J$, and put $\lambda = 1/R_s$. There are a smooth positive probability η and a smooth vector field w , with $0 < I := \mathbb{E}_\eta |w|^2 < \infty$, such that*

$$R(\eta) \leq \lambda^{-1}, \quad w \in \text{Dom}(A_\eta)^n, \quad A_\eta w = (\lambda \text{Id} - K_s)w. \quad (3.22)$$

They can be chosen as follows.

(i) *If $R_s > R_s^P$, take the optimizer f of Lemma 3.4 and set*

$$\eta = f^2 \mu_s, \quad w = D \log f^2, \quad I = 2\lambda H(\eta \mid \mu_s).$$

(ii) *If $R_s = R_s^P$, there is a centered nonconstant gap eigenfunction φ satisfying $A_{\mu_s} \varphi = \lambda \varphi$. Take*

$$\eta = \mu_s, \quad w = D\varphi, \quad I = \lambda \mathbb{E}_{\mu_s} \varphi^2.$$

In either alternative,

$$|\mathbb{E}_\eta w|^2 + \mathbb{E}_\eta |K_s w|^2 \leq C_J \sqrt{a} I, \quad \mathbb{E}_\eta \langle w, K_s w \rangle \leq \lambda |\mathbb{E}_\eta w|^2. \quad (3.23)$$

In alternative (ii) the factor $\sqrt{a}$ can be replaced by a^2 . Coordinates belong to $\text{Dom}(A_\eta)$ in both alternatives.

Proof. In the first alternative the identities and the last inequality are Lemma 3.5. Apply Lemma 3.6 to $\sigma = \eta$, where $F_0 = I$ and $D_0 = I/(2\lambda)$, so the deficit vanishes. The coefficient $s^{-2} - \lambda/s$ is bounded below by a positive constant on J . Thus, with $p = \mathbb{E}_\eta Y$,

$$|p|^2 \leq C_J \sqrt{a} I, \quad |\mathbb{E}_\eta b_s| \leq C_J a \sqrt{I}.$$

Equation (3.19) applied to η now gives $|\mathbb{E}_\eta w|^2 \leq C_J \sqrt{a} I$. Since $K_s^2 \leq s^{-1} K_s$, the curvature-mean inequality gives the asserted estimate for $K_s w$.

In the second alternative, compact form embedding supplies a minimizer of the Rayleigh quotient among mean-zero functions of unit norm. Its positive eigenvalue is $1/R_s^P = \lambda$. The nullspace consists of constants, since zero weak gradient on $\mathbb{R}^n$ implies constancy. Compactness therefore makes this a positive gap, and interior elliptic regularity makes its eigenfunction φ smooth. Commuting a derivative gives $A_{\mu_s} D\varphi = (\lambda \text{Id} - K_s) D\varphi$. Its right side is in L^2 , so Lemma 2.2 gives the operator-domain assertion. Taking the energy pairing and then Poincaré gives, exactly as before,

$$\mathbb{E}\langle w, K_s w \rangle \leq \lambda |\mathbb{E} w|^2.$$

Coordinates belong to the operator domain because both Y and $A_{\mu_s} Y = (Y - b_s)/s$ belong to $L^2(\mu_s)$. Integration by parts, or self-adjointness with these coordinate tests, yields

$$\mathbb{E} w = \lambda \mathbb{E}[\varphi Y] = \frac{\mathbb{E}[\varphi Y] - \mathbb{E}[\varphi b_s]}{s}, \quad (1 - \lambda s) \mathbb{E}[\varphi Y] = \mathbb{E}[\varphi b_s].$$

Conditional Jensen gives

$$\mathbb{E}_{\mu_s} b_s b_s^* \leq \mathbb{E}_\mu X X^* \leq C a^2 \text{Id}.$$

For every unit vector v , Cauchy–Schwarz thus bounds $|\mathbb{E}[\varphi v \cdot b_s]|$ by $C a \|\varphi\|_2$. Taking the supremum over v introduces no dimension factor. Since $1 - \lambda s \geq 1/2$ and $I = \lambda \|\varphi\|_2^2$, it follows that $|\mathbb{E} w| \leq C_J a \sqrt{I}$. The bound for $K_s w$ follows from $K_s^2 \leq s^{-1} K_s$ once again.

It remains to justify the coordinate assertion for the first alternative. Finite $H(\eta \mid \mu_s)$ and the Gaussian tails of μ_s imply $\mathbb{E}_\eta |Y|^2 < \infty$, by the entropy argument in the proof of Lemma 3.6. The distributional identity

$$A_\eta Y = (Y - b_s)/s - w \tag{3.24}$$

has an $L^2(\eta)$ right side, because b_s is bounded for the fixed law and $w \in L^2(\eta)$. The maximal-domain assertion of Lemma 2.2 applies. $\square$

Proposition 3.8 (Limit of the regularized constants). *For the sequence in Lemma 3.1,*

$$R_s \longrightarrow 1 \quad \text{for every } 0 < s < 1/2. \tag{3.25}$$

At every differentiability point of R_s , the fields of Lemma 3.7 satisfy the contact identity

$$R'_s = R_s \frac{\mathbb{E}_\eta \langle w, K_s w \rangle}{I}. \tag{3.26}$$

In particular $0 \leq R'_s \leq C_J \sqrt{a}$ almost everywhere on each fixed compact $J \subset (0, 1/2)$.

Proof. Fix s , and let $\Phi_{s,t}$ be the heat continuity flow from Lemma 3.2, for t on either side of s . Transport a form-domain test g by $g_t = g \circ \Phi_{s,t}^{-1}$. Its distribution under μ_t is fixed, so both its entropy and its variance are constant. The Jacobian equation $\partial_t D\Phi_{s,t} = K_t D\Phi_{s,t}/2$ gives

$$\left. \frac{d}{dt} \right|_{t=s} \int |Dg_t|^2 d\mu_t = - \int \langle Dg, K_s Dg \rangle d\mu_s. \tag{3.27}$$

For compact smooth g this follows by differentiating the inverse-transpose Jacobian in the gradient. For a form-domain g , the change-of-variables formula still holds by Sobolev approximation. The flow derivatives and their time derivatives are uniformly bounded on a compact time interval, so the derivative of the integrand is dominated by $C|Dg|^2$. Dominated convergence proves Equation (3.27) without a smoothness or compact-support restriction on g .

In the strict alternative take $g = f$ and set $Q(t) = \text{Ent}_{\mu_t}(f_t^2)/(2 \int |Df_t|^2 d\mu_t)$. The denominator is positive near s , and $Q(t) \leq R_t$ for both signs of $t - s$, with $Q(s) = R_s$. At a differentiability point of R_s , this two-sided contact implies $Q'(s) = R'_s$. Equation (3.27) and $w = 2Df/f$ give Equation (3.26). In the gap alternative take $g = \varphi$ and instead use $Q(t) = \text{Var}_{\mu_t}(\varphi_t)/\int |D\varphi_t|^2 d\mu_t$. Now $Q(t) \leq R_t^P \leq R_t$ and $Q(s) = R_s^P = R_s$. The same contact argument gives the identical derivative formula, with $w = D\varphi$.

Equation (3.23) bounds this derivative by $C_J\sqrt{a}$ in either alternative, with constants independent of which alternative holds at each time. Lipschitz continuity makes R_s absolutely continuous. For fixed $0 < \delta < s < 1/2$, integration on $[\delta, s]$ gives

$$1 \leq R_s \leq R_\delta + C_{[\delta, s]}\sqrt{a}(s - \delta) \leq 1 + \delta + C_{[\delta, s]}\sqrt{a}(s - \delta).$$

First take the counterexample-sequence limit and then let $\delta \downarrow 0$. This proves Equation (3.25). $\square$

We finish by fixing the Gaussian time and collecting precisely the objects used in the following sections. The fixed time will not be sent to zero.

Proposition 3.9 (The critical sequence). *If Theorem 1.1 fails, there is a sequence of dimensions and centered compactly supported log-concave laws μ satisfying Equation (3.1). Fix $s = 1/8$, form μ_s, b_s, M_s, K_s as in Equation (3.3), and put $\lambda = R_s^{-1}$. After passage to a subsequence there are smooth positive probabilities η and smooth vector fields w such that, with $A = A_\eta$ and $I = \mathbb{E}_\eta|w|^2$,*

$$\lambda \longrightarrow 1, \quad R(\eta) \leq \lambda^{-1}, \quad 0 < I < \infty, \quad w \in \text{Dom}(A)^n, \quad (3.28)$$

$$Aw = (\lambda \text{Id} - K_s)w, \quad \|(A - 1)w\|_{L^2(\eta)} + |\mathbb{E}_\eta w| = o(\sqrt{I}), \quad (3.29)$$

$$\mathbb{E}_\eta|K_s w|^2 = o(I), \quad \mathbb{E}_\eta|Dw|^2 = (1 + o(1))I. \quad (3.30)$$

The spectral gap of A is at least λ . Each coordinate function belongs to $\text{Dom}(A)$. Exactly one of the following constructions can be used throughout the subsequence:

- (i) $R_s > R_s^P$, $\eta = f^2\mu_s$ for the positive unit-norm optimizer, $w = D \log f^2$, and $I = 2\lambda H(\eta \mid \mu_s)$. The exact deficit identity in Equation (3.15) holds for compact square-root tests relative to η , and $AY = (Y - b_s)/s - w$.
- (ii) $R_s = R_s^P$, $\eta = \mu_s$, $A\varphi = \lambda\varphi$, $\mathbb{E}_\eta\varphi = 0$, $w = D\varphi$, and $I = \lambda\mathbb{E}_\eta\varphi^2$. One may rescale φ so that $I \rightarrow 0$.

All o -terms refer to the counterexample sequence with fixed finite parameters. In particular no relation between dimension, I , and the rate of convergence is asserted or needed here.

Proof. Choose an infinite subsequence with the same alternative at $s = 1/8$. Lemma 3.7 gives all structural identities and the curvature and mean bounds. Proposition 3.8 gives $\lambda \rightarrow 1$. Consequently

$$(A - 1)w = (\lambda - 1)w - K_s w = o_{L^2}(\sqrt{I}).$$

The energy pairing yields $\mathbb{E}|Dw|^2 = \lambda I - \mathbb{E}\langle w, K_s w \rangle = (1 + o(1))I$. Poincaré with constant λ^{-1} is the asserted spectral gap. The coordinate and branch-specific identities have already been proved. In the gap alternative all estimates are homogeneous in φ , so multiplying it by positive scalars can enforce $I \rightarrow 0$ without changing any normalized assertion. $\square$

4 Efficient probes and the prediction curve

Fix $s = 1/8$ and take one of the two subsequences supplied by Proposition 3.9. We continue to suppress the counterexample index. Thus $\lambda \rightarrow 1$, η satisfies the logarithmic Sobolev inequality with constant at most $1/\lambda$, and, with $A = A_\eta$, w belongs coefficientwise to $\text{Dom}(A)$ and satisfies

$$I = \mathbb{E}_\eta |w|^2 > 0, \quad \|(A - 1)w\|_{L^2(\eta)} + |\mathbb{E}_\eta w| = o(\sqrt{I}). \quad (4.1)$$

In the strict case, $\eta = f^2 \mu_s$, $w = D \log f^2$, and $2H(\eta \mid \mu_s) = I/\lambda$. In the gap case, $\eta = \mu_s$, $w = D\varphi$, and $A_{\mu_s} \varphi = \lambda \varphi$, where $\mathbb{E}_{\mu_s} \varphi = 0$ and $I = \lambda \mathbb{E}_{\mu_s} \varphi^2$. All limits in this section are along the selected subsequence, with the displayed auxiliary parameters fixed unless a different order is specified.

Our objective is to determine how much of $w(Y_0)$ can be predicted from $Y_0 + \sqrt{r}G$. The strict case first requires an entropy estimate for an additional Gaussian observation of w . We then prove an entropy-loss comparison for arbitrary smooth alternatives to a log-concave law. These two estimates determine the prediction curve; linearization of the same comparison handles the gap case.

Lemma 4.1 (Efficient Gaussian probes). *In the strict case, let $Y_0 \sim \eta$ and let G' be an independent standard Gaussian vector in the same Euclidean space. For $T \geq 0$ define*

$$L_T = \sqrt{T}w(Y_0) + G', \quad d(T) = H(\text{Law}(L_T) \mid \gamma),$$

where γ is standard Gaussian measure. For every finite M ,

$$\sup_{0 \leq T \leq M} d(T) = o(I). \quad (4.2)$$

Proof. Run the stationary diffusion with law η , using Brownian noise independent of G' , and write $P_u = e^{-uA}$. Proposition 3.9 puts each coordinate in $\text{Dom}(A)$ and gives the identity

$$-Ay = -y/s + w + b_s/s. \quad (4.3)$$

In particular, $\mathbb{E}_\eta |Y_0|^2 < \infty$. The logarithmic Sobolev inequality for η then gives

$$\mathbb{E}_\eta e^{v \cdot (Y_0 - \mathbb{E} Y_0)} \leq e^{|v|^2/(2\lambda)}.$$

Integrating this inequality over a Gaussian v proves $\mathbb{E}_\eta e^{c'|Y_0|^2} < \infty$ for a sufficiently small $c' > 0$. Here the choice of c' may depend on the individual law and dimension. This exponential-square moment will also permit application of the entropy-loss comparison below.

For a fixed $T \leq M$, set

$$J_u = I(L_T; Y_u), \quad F_u = \mathbb{E} \mathcal{I}(\text{Law}(Y_u \mid L_T) \mid \eta), \quad Q_u = \mathbb{E} |\mathbb{E}[Y_u \mid L_T]|^2.$$

The Gaussian observation identity gives $J_0 = TI/2 - d(T)$. Entropy dissipation, including its conditional version, gives $-dJ_u \geq F_u du$, and in particular $\int_0^1 F_u du \leq J_0$. If $\sigma_{u,l} = \text{Law}(Y_u \mid L_T = l)$, then the entropy chain rule and the strict-case normalization give

$$\mathbb{E} H(\sigma_{u,L_T} \mid \mu_s) = J_u + \frac{I}{2\lambda}, \quad \mathbb{E} \mathcal{I}(\sigma_{u,L_T} \mid \mu_s) = F_u + I. \quad (4.4)$$

For the second equality, write the conditional relative score with respect to μ_s as the sum of w and the conditional relative score with respect to η . The latter scores have conditional average

zero at each Y_u , since the conditional likelihoods average to one. This cancels the cross term. At the almost every time where $F_u < \infty$, the assertion also follows weakly by differentiating that marginal identity: Cauchy–Schwarz gives locally integrable averaged derivatives.

Apply Lemma 3.6 to $\sigma_{u,l}$ and average over l . Since $s^{-2} - \lambda/s$ stays positive, Equations (4.4) give, on $[0, 1]$,

$$-dJ_u \geq (2\lambda J_u + c_s Q_u) du - d\mathcal{E}_u, \quad \mathcal{E}([0, 1]) = o_M(I), \quad (4.5)$$

where $c_s > 0$ is fixed and $\mathcal{E}$ is a nonnegative error measure. Indeed the error in that lemma is bounded by a number tending to zero times $F_u + I + J_u + I/(2\lambda)$, whose integral is $O_M(I)$. The measure formulation avoids imposing differentiability on J_u .

The spectral theorem and Equation (4.1) imply

$$\sup_{0 \leq u \leq 1} \|P_u w - e^{-u} w\|_2 \leq \|(A - 1)w\|_2 = o(\sqrt{I}). \quad (4.6)$$

Reversibility gives $\mathbb{E}[L_T | Y_u] = \sqrt{T} P_u w(Y_u)$. Relative entropy with respect to a standard Gaussian is at least half the square of the mean, by testing its variational formula with linear functions. Consequently

$$J_u + d(T) = \mathbb{E}H(\text{Law}(L_T | Y_u) | \gamma) \geq \frac{T}{2} \|P_u w\|_2^2 = \frac{TI}{2} e^{-2u} + o_M(I).$$

Integrating Equation (4.5) with factor $e^{2\lambda u}$ and comparing at $u = 1$ yields

$$\int_0^1 Q_u du \leq C_s d(T) + o_M(I). \quad (4.7)$$

Here the difference between $e^{-2\lambda}$ and e^{-2} contributes only $o_M(I)$.

It remains to turn control of the predicted position into control of the predicted score. The pushforward of μ_s by b_s has the centered linear moment bound

$$\log \mathbb{E}_{\mu_s} e^{v \cdot b_s} \leq C a^2 |v|^2,$$

by conditional Jensen and the corresponding bound for the original signal X . The entropy variational formula, optimized over v , thus gives

$$\mathbb{E}[\mathbb{E}[b_s(Y_v) | L_T]]^2 \leq C a^2 (J_v + I/(2\lambda)) = o_M(I).$$

Condition Equation (4.3) on L_T and use variation of constants. By Equation (4.6) and the contraction property of conditional expectation,

$$\mathbb{E}[Y_u | L_T] = e^{-u/s} \mathbb{E}[Y_0 | L_T] + \frac{e^{-u} - e^{-u/s}}{1/s - 1} \mathbb{E}[w(Y_0) | L_T] + o_{L^2, M}(\sqrt{I}), \quad (4.8)$$

uniformly for $0 \leq u \leq 1$. The two scalar functions multiplying the conditional means are linearly independent on $[0, 1]$: the first is nonzero at zero, whereas the second vanishes there and is nonzero for $u > 0$. Their 2×2 Gram matrix is therefore positive definite. Applying its smallest-eigenvalue bound to each coordinate and then averaging, Equations (4.7) and (4.8) show that

$$\mathbb{E}[\mathbb{E}[w(Y_0) | L_T]]^2 \leq C_s d(T) + o_M(I).$$

By Lemma 2.4, the left side is $2d'(T)$ for almost every T . The error is uniform for $T \in [0, M]$, and $d(0) = 0$. Integrating this differential inequality proves Equation (4.2). No lower bound on I is used. $\square$

The next comparison relates the entropy lost under heat flow to the conditional mean of the initial relative score. The conditional mean need not equal the relative score at positive heat time, so entropy dissipation alone does not give this comparison.

The choice of observation directions is related to function-preserving stochastic localization. Eldan–Koehler–Zeitouni [7, Section 2, Lemma 2] use controls approximating projection orthogonal to the covariance of the signal with a selected function. Chen and Eldan [6, Sections 2.4.2 and 3.2.3, Equation (27)] express the entropy drift through the difference of means under a measure and its normalized likelihood tilt, in a continuous-time setup with positive definite controls. Below we prove the cancellation and its error directly for finite Gaussian observations with singular projections, and use independent copies to control the discarded directions.

Lemma 4.2 (Entropy loss under a Gaussian observation). *Let p and q be smooth strictly positive probability densities on $\mathbb{R}^d$, with q log-concave. Suppose*

$$H_0 = H(p \mid q) < \infty, \quad w = D \log(p/q), \quad I = \mathbb{E}_p |w|^2 < \infty, \quad \mathbb{E}_p e^{c|Y|^2} < \infty$$

for some $c > 0$, where $Y \sim p$. Let G be a standard Gaussian vector independent of Y . For $r > 0$, let $U_r = Y + \sqrt{r}G$, let p_r and q_r be the Gaussian convolutions of p and q , and put $\bar{w}_r(U_r) = \mathbb{E}_p[w(Y) \mid U_r]$ and $H_r = H(p_r \mid q_r)$. Then, for every $z > 0$,

$$2(H_0 - H_z) \leq \int_0^z \mathbb{E}_p |\bar{w}_r(U_r)|^2 dr. \quad (4.9)$$

Proof. The proof adds weak observations that increase the curvature of the base posterior while losing asymptotically no relative entropy. At each step we remove the direction in which the two posterior means differ. The retained output means then agree, and we will show that its relative entropy, averaged over the retained history, is $O(\delta^2)$ for observation strength δ . Independent copies make the cost of removing these directions vanish after division by the number of copies. That last comparison concerns the score residuals after full observations, whose conditional independence will be essential.

We keep the one-copy problem fixed. The three successive limits will be refinement of a finite observation mesh, increase of the number of independent copies, and increase of the total additional precision. Set $b = 1/z$ and take N independent copies. Write $Y = (Y_1, \dots, Y_N)$ and $W = (w(Y_1), \dots, w(Y_N))$; the two possible priors are $p^{\otimes N}$ and $q^{\otimes N}$. Expectations in the proof use $p^{\otimes N}$.

Additional observations and their entropy cost. First observe $Z_0 = \sqrt{b}Y + G_0$. The chain rule says that the remaining posterior relative entropy, averaged over this observation, is $N(H_0 - H_z)$. Fix $u > 0$ and an integer m , and put $\delta = u/m$. For $i = 0, \dots, m-1$, let $\mathcal{H}_i$ be the retained history before step i . Compute the difference of the two posterior means given this history and let P_i be the orthogonal projection onto its span; set $P_i = 0$ when the difference vanishes. Retain only

$$(1 - P_i)Z_{i+1}, \quad Z_{i+1} = \sqrt{\delta}Y + G_{i+1},$$

where all Gaussian noises are independent. The same measurable rule is used under both priors. One may choose versions of the posterior means by their integral formulas; the positive densities make the rule well defined on all histories of positive probability.

For each realized retained history the likelihood, as a function of Y , is a Gaussian quadratic factor of precision

$$B = b\text{Id} + \delta \sum_{i=0}^{m-1} (1 - P_i) \succeq b\text{Id}. \quad (4.10)$$

There is no extra likelihood factor from the choices of P_i : each choice was already determined by the retained past. The factor is the same for both priors. Thus the posterior base has curvature at least B , and the posterior relative score is still W .

We next bound the relative entropy lost in the additional observations. Condition on a retained history at one step and put $Q = 1 - P_i$. The projected posterior means agree. After subtracting their common mean, let X_c denote either centered projected signal in $\text{ran } Q$, whose dimension is at most Nd . The base posterior has curvature at least $b\text{Id}$. Its centered linear moment bound and integration over a Gaussian vector give

$$\mathbb{E}_q[e^{\kappa|X_c|^2} \mid \mathcal{H}_i] \leq (1 - 2\kappa/b)^{-Nd/2}, \quad 0 < 2\kappa < b.$$

For the alternative, write $a_i = \mathbb{E}_p[Y \mid \mathcal{H}_i]$. For any sufficiently small $\kappa > 0$, conditional Jensen gives

$$\begin{aligned} \mathbb{E}_p[e^{\kappa|Q(Y-a_i)|^2} \mid \mathcal{H}_i] &\leq e^{2\kappa|a_i|^2} \mathbb{E}_p[e^{2\kappa|Y|^2} \mid \mathcal{H}_i] \\ &\leq (\mathbb{E}_p[e^{2\kappa|Y|^2} \mid \mathcal{H}_i])^2. \end{aligned}$$

Squaring and applying conditional Jensen once more shows that its square, averaged over $\mathcal{H}_i$, is bounded by $\mathbb{E}_p e^{8\kappa|Y|^2} < \infty$. These bounds hold for every history sigma-field, so their constants do not depend on the mesh or on the adaptive choices.

Let h_p, h_q be the output densities of $\sqrt{\delta}X_c + G$ in $\text{ran } Q$ and let ϕ be standard Gaussian density there. Centering and Jensen give the pointwise bound

$$\frac{h_q(x)}{\phi(x)} = \mathbb{E}_q e^{\sqrt{\delta}x \cdot X_c - \delta|X_c|^2/2} \geq e^{-\delta Nd/(2b)}.$$

For either conditional signal, using a conditionally independent copy X'_c and integrating the Gaussian kernel gives

$$\left\| \frac{h}{\phi} - 1 \right\|_{L^2(\phi)}^2 = \mathbb{E}[e^{\delta X_c \cdot X'_c} \mid \mathcal{H}_i] - 1.$$

The linear term vanishes because X_c is centered. The bound $|e^t - 1 - t| \leq t^2 e^{|t|}/2$, followed by $|X_c \cdot X'_c| \leq (|X_c|^2 + |X'_c|^2)/2$, bounds the last expression by $C\delta^2$ times the square of a small conditional exponential-square moment. The preceding moment bounds therefore give

$$\mathbb{E}H(h_p \mid h_q) \leq \mathbb{E}\chi^2(h_p \mid h_q) \leq 2e^{\delta Nd/(2b)} \mathbb{E} \left(\|h_p/\phi - 1\|_2^2 + \|h_q/\phi - 1\|_2^2 \right) \leq C_{N,b,p} \delta^2.$$

This statement includes the zero-dimensional observed subspace, for which the loss is zero. Summing the m steps loses $O_{N,b,p}(u\delta)$ entropy. Apply the anisotropic curvature inequality, Lemma 2.1, to the final posterior pair. For fixed N and u we obtain

$$2N(H_0 - H_z) - o_{\delta \downarrow 0}(1) \leq \mathbb{E}W^*B^{-1}W. \quad (4.11)$$

Comparison with full observations. We now estimate the effect of the rank-one exclusions. Work in the Hilbert space of square-integrable vector fields on the signal and all the full observations $Z_0, \dots, Z_m$. In particular, $\|W\|_2^2 = NI$. Completing the square pointwise gives

$$b\|W\|_2^2 - b^2\mathbb{E}W^*B^{-1}W = \min_v \left\{ b\|W - v\|_2^2 + \delta \sum_{i=0}^{m-1} \|(1 - P_i)v\|_2^2 \right\}. \quad (4.12)$$

Its minimizer is $v_* = bB^{-1}W$, so $\|v_*\|_2 \leq \|W\|_2$. Let $\mathcal{F}_i = \sigma(Z_0, \dots, Z_i)$ and let e_i be conditional expectation onto $\mathcal{F}_i$. The sigma-fields are nested, and P_i is $\mathcal{F}_i$ -measurable because the retained history is a function of this full history. We compare the original minimum with one that penalizes the part of v left unpredicted by the full observations: replace $1 - P_i$ by $1 - e_i$, and denote the resulting minimum by M_0 . The nested projections make this new minimum explicitly computable in terms of Gaussian prediction. We first find its minimizer, then use its residuals in a dual test to bound the original minimum below in terms of M_0 . Its minimizer is

$$v_0 = b \left(b + \delta \sum_{i=0}^{m-1} (1 - e_i) \right)^{-1} W.$$

Here is its explicit form. Set $\alpha_k = b/(b + k\delta)$ for $0 \leq k \leq m$. On $\text{ran } e_0$ the multiplier is one; on $\text{ran}(e_k - e_{k-1})$ it is α_k ; on $\text{ran}(1 - e_{m-1})$ it is α_m . Consequently

$$v_0 = \alpha_m W + \sum_{k=0}^{m-1} (\alpha_k - \alpha_{k+1}) e_k W. \quad (4.13)$$

All coefficients are nonnegative and sum to one. Put $g_i = (1 - e_i)v_0$. Equation (4.13) gives

$$g_i = \alpha_m (W - e_i W) + \sum_{k=i+1}^{m-1} (\alpha_k - \alpha_{k+1}) (e_k W - e_i W). \quad (4.14)$$

The weights in this formula sum to $\alpha_{i+1} \leq 1$.

We need a bound uniform over every subdivision. The retained histories need not factor across copies. We use independence only for the full observation arrays: those belonging to different copies are independent under $p^{\otimes N}$. Conditioning on $\mathcal{F}_i$ conditions each such array on its own past. Thus $e_k W_j$ depends only on copy j 's observations through step k . Conditional on $\mathcal{F}_i$, the blocks $g_{i,j}$ are independent and centered, even though they involve future full observations. To see this directly, factor the joint law into the independent copy arrays and condition each factor on its own past; the resulting conditional kernel remains a product. By conditional Jensen applied to Equation (4.14),

$$Z_{i,j} := \text{tr Cov}(g_{i,j} \mid \mathcal{F}_i) \leq \mathbb{E}[|W_j - e_i W_j|^2 \mid \mathcal{F}_i] \leq \mathbb{E}[|W_j|^2 \mid \mathcal{F}_i]. \quad (4.15)$$

For example, every term with $k > i$ is bounded in conditional squared mean by $\mathbb{E}[|W_j - e_i W_j|^2 \mid \mathcal{F}_i]$ by the tower property; the nonnegative weights of total at most one preserve this bound.

The right sides in Equation (4.15) are uniformly integrable over all these sigma-fields. Explicitly, if $Z \geq 0$ is integrable and $M = \mathbb{E}[Z \mid \mathcal{F}]$, then

$$\mathbb{E}[M \mathbf{1}_{\{M > K\}}] \leq 2\mathbb{E}[Z \mathbf{1}_{\{Z > K/2\}}]. \quad (4.16)$$

Indeed $M \leq K/2 + \mathbb{E}[Z \mathbf{1}_{\{Z > K/2\}} \mid \mathcal{F}]$, and on $\{M > K\}$ the last conditional expectation is at least $M/2$. The conditional covariance of g_i is block diagonal. Since P_i is measurable in $\mathcal{F}_i$ and has rank at most one,

$$\mathbb{E}|P_i g_i|^2 \leq \mathbb{E} \max_{1 \leq j \leq N} Z_{i,j}.$$

The maximum is at most $K + \sum_j Z_{i,j} \mathbf{1}_{\{Z_{i,j} > K\}}$. Equations (4.15) and (4.16) show that, uniformly in i, m , and δ ,

$$\mathbb{E}|P_i g_i|^2 \leq N q_N, \quad q_N := \inf_{K > 0} \left\{ \frac{K}{N} + 2\mathbb{E}_p[|w(Y)|^2 \mathbf{1}_{\{|w(Y)|^2 > K/2\}}] \right\} \longrightarrow 0. \quad (4.17)$$

No moment of the score beyond its second moment is required. In particular this bound permits P_i to depend adaptively on all copies.

For completeness, the comparison of the two minima follows from a dual test. For every v ,

$$\|(1 - P_i)v\|_2^2 \geq 2\langle v, (1 - P_i)g_i \rangle - \|g_i\|_2^2.$$

The normal equation for v_0 is $b(W - v_0) = \delta \sum_i g_i$. Therefore the minimum over v of

$$b\|W - v\|_2^2 + 2\delta \sum_i \langle v, g_i \rangle - \delta \sum_i \|g_i\|_2^2$$

is attained at v_0 and equals M_0 . Use the displayed linear bound at v_* in Equation (4.12). By Equation (4.17), the true minimum is at least

$$M_0 - 2\delta \sum_i \|v_*\|_2 \|P_i g_i\|_2 \geq M_0 - 2uN\sqrt{Iq_N}. \quad (4.18)$$

The prediction integral. It remains to compute M_0 and take the limits. The same orthogonal decomposition into nested projection increments gives

$$\frac{M_0}{N} = \sum_{i=0}^{m-1} \frac{b^2 \delta}{(b + i\delta)(b + (i+1)\delta)} \left(I - \|e_i W_1\|_2^2 \right). \quad (4.19)$$

Indeed a component first seen after j increments has penalty $bj\delta/(b + j\delta)$, and

$$\frac{bj\delta}{b + j\delta} = \sum_{i=0}^{j-1} \frac{b^2 \delta}{(b + i\delta)(b + (i+1)\delta)}.$$

Full observations through step i have the sufficient Gaussian statistic of precision $b + i\delta$. Completing the likelihood square proves sufficiency directly. Hence

$$\|e_i W_1\|_2^2 = \mathbb{E}_p |\bar{w}_{1/(b+i\delta)}(U_{1/(b+i\delta)})|^2.$$

This squared prediction norm is nondecreasing with precision, since a weaker observation can be obtained by adding independent Gaussian noise. It is bounded by I , so Equation (4.19) converges as $m \rightarrow \infty$ to

$$b^2 \int_0^u \frac{I - \mathbb{E}_p |\bar{w}_{1/(b+t)}(U_{1/(b+t)})|^2}{(b+t)^2} dt.$$

First take this mesh limit at fixed N, u . It removes the entropy error in Equation (4.11), while Equation (4.18) remains uniform in the mesh. Divide by N , then let $N \rightarrow \infty$, using $q_N \rightarrow 0$. Equations (4.11) and (4.12) give

$$2(H_0 - H_z) \leq \frac{I}{b} - \int_0^u \frac{I - \mathbb{E}_p |\bar{w}_{1/(b+t)}(U_{1/(b+t)})|^2}{(b+t)^2} dt = \frac{I}{b+u} + \int_{1/(b+u)}^{1/b} \mathbb{E}_p |\bar{w}_r(U_r)|^2 dr.$$

Finally let $u \rightarrow \infty$. Since $1/b = z$ and the integrand is bounded by I , this proves Equation (4.9) in the claimed order of limits. $\square$

We return to the critical sequence. For $Y_0 \sim \eta$ and independent Gaussian noise, define

$$U_r = Y_0 + \sqrt{r}G, \quad \bar{w}_r(U_r) = \mathbb{E}[w(Y_0) | U_r], \quad m(r) = I^{-1} \mathbb{E} |\bar{w}_r(U_r)|^2.$$

In the strict case we also use the label L_T of Lemma 4.1, constructed with fresh Gaussian noise independent of (Y_0, G) . Thus L_T and U_r are conditionally independent given Y_0 . Set

$$p_T(r) = I^{-1} \mathbb{E} |\mathbb{E}[w(Y_0) | U_r, L_T]|^2, \quad p_0(r) = m(r).$$

Proposition 4.3 (The prediction curve). *For the strict and gap critical sequences, for every fixed $0 < r < 1/4$,*

$$m(r) \longrightarrow e^{-r}. \quad (4.20)$$

In the strict case, for every fixed finite $T \geq 0$ one also has $p_T(r) \rightarrow e^{-r}$. In the gap case the eigenfunction may be rescaled so that $I \rightarrow 0$, without changing Equation (4.20) or Equation (4.1).

Proof. For $0 \leq r < 1/4$ put

$$\theta(r) = \exp\left(-\int_0^r \frac{dt}{R_{s+t}}\right).$$

Proposition 3.8 gives $\theta(r) = e^{-r} + o(1)$, uniformly on compact intervals under consideration. The common heat flow dissipates relative entropy at one half the relative Fisher information. Thus its entropy relative to μ_{s+r} is at most $\theta(r)$ times its initial entropy, by the logarithmic Sobolev inequality at the contemporaneous base law. This assertion applies to conditional laws after averaging as well.

In the strict case write $H_r = H(\text{Law}(U_r) \mid \mu_{s+r})$ and $H_0 = I/(2\lambda)$. The entropy of U_r given L_T , relative to μ_{s+r} and averaged over the label, is at most $\theta(r)(H_0 + TI/2)$, since initially it is $H_0 + I(Y_0; L_T)$. The chain rule gives

$$\begin{aligned} \frac{I}{2} \int_0^T p_v(r) dv &= \mathbb{E}H(\text{Law}(L_T \mid U_r) \mid \gamma) \\ &= \mathbb{E}H(\text{Law}(U_r \mid L_T) \mid \mu_{s+r}) + d(T) - H_r \\ &\leq \frac{I}{2} \theta(r)(1/\lambda + T) + d(T). \end{aligned}$$

The first equality is Lemma 2.4, conditionally on U_r . Fix $v \geq 0$ and then $T > v$. Since $p_t(r)$ is nondecreasing in signal strength t , Lemma 4.1 yields

$$(T - v) \limsup p_v(r) \leq e^{-r}(1 + T).$$

The sequence limit is taken with T fixed. Allowing this fixed T to be arbitrarily large proves

$$\limsup p_v(r) \leq e^{-r}. \quad (4.21)$$

Lemma 4.2 applies to $p = \eta$ and $q = \mu_s$; the exponential-square moment was proved in the proof of Lemma 4.1. Since $H_z \leq \theta(z)H_0$, it gives

$$\int_0^z m(r) dr \geq \frac{1 - \theta(z)}{\lambda} \quad (0 < z < 1/4).$$

The reverse limiting inequality follows from Equation (4.21), $0 \leq m \leq 1$, and reverse Fatou. Hence

$$\int_0^z m(r) dr \longrightarrow 1 - e^{-z}. \quad (4.22)$$

The function m is nonincreasing in r . For $0 < h < r$ with $r + h < 1/4$, its value at r is bounded below and above by its averages on $[r, r + h]$ and $[r - h, r]$, respectively. Equation (4.22) determines the limits of both averages; letting $h \downarrow 0$ gives Equation (4.20). Finally $m(r) \leq p_T(r)$ and Equation (4.21) give the assertion for every fixed T .

We now prove the gap assertion without using additional probes. First linearize Lemma 4.2. Let q be a bounded smooth centered test under μ_s , with bounded gradient, and let

$$d\eta_\varepsilon = (1 + \varepsilon q) d\mu_s, \quad |\varepsilon| \|q\|_\infty < 1.$$

This alternative has a finite exponential-square moment. Write $\bar{q}_r = \mathbb{E}_{\mu_s}[q(Y_0) \mid U_r]$ and $\overline{Dq}_r = \mathbb{E}_{\mu_s}[Dq(Y_0) \mid U_r]$. The output density ratio is $1 + \varepsilon \bar{q}_r$, while the predicted initial relative score is exactly

$$\mathbb{E}_{\eta_\varepsilon}[D \log(1 + \varepsilon q)(Y_0) \mid U_r] = \frac{\varepsilon \overline{Dq}_r}{1 + \varepsilon \bar{q}_r}.$$

Consequently its squared norm averaged under the alternative output is $\varepsilon^2 \mathbb{E}_{\mu_{s+r}}[|\overline{Dq}_r|^2 / (1 + \varepsilon \bar{q}_r)]$. The entropy expansion $H((1 + \varepsilon q)\mu_s \mid \mu_s) = \varepsilon^2 \|q\|_2^2 / 2 + o(\varepsilon^2)$, and its output version, now give

$$\|q\|_2^2 - \|\bar{q}_z\|_2^2 \leq \int_0^z \|\overline{Dq}_r\|_2^2 dr. \quad (4.23)$$

Indeed the denominator is bounded away from zero uniformly in r , and conditional Jensen bounds the integrand by a constant times $\|Dq\|_2^2$, justifying the limit. Approximation in the form norm extends Equation (4.23) to $q = \varphi$: centered compact smooth approximants have the required boundedness, and all conditional expectation maps are L^2 contractions. To retain compact support while centering, subtract the approximant's mean times one fixed compact smooth function of mean one; this correction tends to zero in the form norm.

For clarity, we also give the heat variance identity used here. For a centered scalar or finite-dimensional vector field $g \in L^2(\mu_s)$, let $g_r = \mathbb{E}[g(Y_0) \mid U_r]$ under the base law. Then

$$\frac{d}{dr} \mathbb{E}_{\mu_{s+r}} |g_r|^2 = -\mathbb{E}_{\mu_{s+r}} |Dg_r|^2. \quad (4.24)$$

For bounded g , both the numerator density $g_r \mu_{s+r}$ and the denominator density μ_{s+r} solve the heat equation, and the quotient calculation with the square function gives $(\partial_r - \Delta/2)(\mu_{s+r} |g_r|^2) = -\mu_{s+r} |Dg_r|^2$. Integrate against expanding cutoffs, whose Laplacians tend uniformly to zero. The bound $|g_r| \leq \|g\|_\infty$ removes the cutoff terms and gives the integrated identity on positive time intervals. Its initial endpoint follows from $\|g_r(U_r) - g(Y_0)\|_2 \rightarrow 0$: for a bounded continuous test use $g(U_r)$ as a predictor and dominated convergence under the coupling $U_r = Y_0 + \sqrt{r}G$, and then approximate in $L^2(\mu_s)$ and use conditional contraction. Approximate a general L^2 field by bounded ones. Applying the same identity to their differences shows that their gradients are Cauchy in the space-time energy norm, while conditional contraction gives convergence of both endpoint norms. This proves the integrated version of Equation (4.24) in full generality.

Poincaré at μ_{s+r} and preservation of the mean therefore imply

$$\mathbb{E}_{\mu_{s+r}} |g_r|^2 \leq \theta(r) \mathbb{E}_{\mu_s} |g|^2.$$

Apply this first to φ , and then to $w - \mathbb{E}w$. The latter application and Equation (4.1) give $m(r) \leq \theta(r) + o(1)$. The former and $\|\varphi\|_2^2 = I/\lambda$, inserted into Equation (4.23), give $\int_0^z m(r) dr \geq (1 - \theta(z))/\lambda$. The same integral squeeze and monotonicity argument prove Equation (4.20). Multiplying φ by any positive scalar multiplies every squared quantity by its square, so we may finally choose these scalars to arrange $I \rightarrow 0$. $\square$

The hierarchy construction will use the following consequences of the prediction curve. The estimates distinguish conditional covariance tails, which are needed only in the strict case, from bounds on deterministic subspaces, which hold in both cases.

Lemma 4.4 (Posterior covariance and outlier subspaces). *Use the critical sequence and notation of Proposition 4.3, with $I \rightarrow 0$ in the gap case. The following assertions hold.*

- (i) For every fixed $c < \infty$, uniformly over deterministic orthogonal projections Π with $\text{rank } \Pi \leq cI$,

$$\mathbb{E}|\Pi w|^2 = o(I). \quad (4.25)$$

- (ii) For every $\epsilon > 0$, for all sufficiently far members of the sequence there is a deterministic subspace V_ϵ of dimension at most $C_\epsilon I$ such that, for every $v \perp V_\epsilon$,

$$v^*(\mathbb{E}ww^*)v \leq \epsilon|v|^2, \quad v^*(\mathbb{E}b_s b_s^*)v \leq \epsilon|v|^2, \quad v^*(\mathbb{E}M_s)v \leq \epsilon|v|^2. \quad (4.26)$$

In the gap case V_ϵ may be taken to be zero, and $\|\mathbb{E}b_s b_s^*\|_{\text{op}} + \|\mathbb{E}M_s\|_{\text{op}} = o(1)$.

- (iii) In the strict case, for every fixed $r \in (0, 1/4)$ and $\epsilon > 0$, the conditional covariance $C_r = \text{Cov}(w(Y_0) \mid U_r)$ satisfies

$$\mathbb{E} \text{tr}(C_r \mathbf{1}_{\{C_r > \epsilon I\}}) = o(I). \quad (4.27)$$

The indicator denotes the spectral projection. No assertion about additional Gaussian probes or Equation (4.27) is needed for the gap case.

Proof. We first prove the strict-case spectral estimates. If a random vector S has covariance C and mean a , the affine estimator

$$a + C(\text{Id} + C)^{-1}(S + G - a)$$

of S from $S + G$ is its least-squares affine predictor. Its mean squared norm is $|a|^2 + \text{tr}(C^2(\text{Id} + C)^{-1})$, by direct expansion of the covariances. Conditional expectation has at least this squared norm, because it is the orthogonal projection onto all functions of the observation. The same calculation holds conditionally on any side information, using fresh independent Gaussian noise.

By the Gaussian signal derivative formula and its monotonicity,

$$\mathbb{E}|\mathbb{E}[w \mid L_1]|^2 \leq 2(d(2) - d(1)) \leq 2d(2) = o(I).$$

Apply the affine prediction bound to $S = w$ and $C = \text{Cov}_\eta w$. For every fixed $\epsilon > 0$,

$$\text{tr}(C \mathbf{1}_{\{C > \epsilon I\}}) = o(I), \quad (4.28)$$

since $t^2/(1+t) \geq \epsilon t/(1+\epsilon)$ for $t > \epsilon$. For every projection of rank at most cI ,

$$\mathbb{E}|\Pi w|^2 \leq \epsilon cI + \text{tr}(C \mathbf{1}_{\{C > \epsilon I\}}) + |\mathbb{E}w|^2.$$

First take the sequence limit, then $\epsilon \downarrow 0$. Equation (4.1) proves Equation (4.25) uniformly in Π . In the gap case a projection of rank at most cI is eventually zero, which proves the same assertion.

Conditionally on U_r , the additional observation L_1 has signal w and covariance C_r . Orthogonality of successive conditional expectations and the conditional affine prediction bound yield

$$I(p_1(r) - m(r)) \geq \mathbb{E} \text{tr}(C_r^2(\text{Id} + C_r)^{-1}).$$

Proposition 4.3 makes the left side $o(I)$. The same scalar eigenvalue inequality proves Equation (4.27).

We finally bound the dimensions of the deterministic outlier spaces. Let

$$D = \mathbb{E}_\eta \mathbb{E}[XX^* \mid X + \sqrt{s}G = Y_0],$$

where the inner posterior kernel always refers to the original prior μ . Conditional Jensen and the definition of M_s imply

$$\mathbb{E}_\eta b_s b_s^* \preceq D, \quad \mathbb{E}_\eta M_s \preceq D/s. \quad (4.29)$$

For each deterministic projection Q of rank $l \geq 1$, the original subgaussian assumption gives

$$\log \mathbb{E}_\mu \exp \left(\frac{|QX|^2}{Ca^2} \right) \leq Cl. \quad (4.30)$$

Indeed choose a $1/2$ -net of the unit sphere in $\text{ran } Q$ with at most 5^l points. The norm $|QX|$ is at most twice the largest absolute linear form over this net, and bounding the exponential of a maximum by the sum proves the claim.

In the strict case, form the joint law by first drawing Y_0 from η and then drawing X from the original posterior kernel. Its entropy relative to the original joint law of $(X, X + \sqrt{s}G)$ is exactly $H_0 = I/(2\lambda)$. Data processing therefore bounds the entropy of its X -marginal relative to μ by H_0 . Applying the entropy variational inequality to Equation (4.30) gives

$$\text{tr}(QD) \leq Ca^2(H_0 + l). \quad (4.31)$$

If the spectral subspace of D above $\tau > 0$ has dimension l , then $\tau l \leq Ca^2(H_0 + l)$. For sufficiently small a this implies $l \leq 2Ca^2H_0/\tau = O_\tau(I)$; the assertion is trivial when $l = 0$. The spectral subspace of $\mathbb{E}ww^*$ above ϵ has dimension at most I/ϵ , since its trace is I . Take V_ϵ to be the span of this subspace and the spectral subspace of D above $s\epsilon$. Its dimension is at most $C_\epsilon I$, and Equation (4.29) gives all three bounds in Equation (4.26).

In the gap case, averaging the original posterior under $\eta = \mu_s$ gives $D = \mathbb{E}_\mu XX^* \preceq Ca^2 \text{Id}$. Thus Equation (4.29) gives the two claimed operator bounds directly. Also $\|\mathbb{E}ww^*\|_{\text{op}} \leq I \rightarrow 0$, so $V_\epsilon = \{0\}$ suffices eventually. This completes all assertions. $\square$

5 A symmetric hierarchy of nearly first eigenfunctions

The prediction curve from Section 4 determines much more than the norm of a conditional mean. Its Gaussian Bessel inequality is asymptotically saturated at every fixed derivative order. We use this saturation to construct symmetric tensors whose derivatives remain nearly first eigenfunctions. We then show that every tensor has negligible energy in the curvature directions of μ_s . Throughout this section, coefficient norms are unnormalized Euclidean norms, including for tensors of high order.

For a tensor field with d slots, differentiation appends a last slot. A matrix subscript, such as K_1 , specifies the slot on which it acts. Write $\|F\|_2^2 = \mathbb{E}_\eta |F|^2$ and $\langle F, G \rangle_2 = \mathbb{E}_\eta \langle F, G \rangle$. Every occurrence of A on a tensor means its componentwise action. The following formulation records precisely the inputs used in the construction; the dimension and I may vary along the sequence.

Proposition 5.1 (Symmetric tensor hierarchy). *Consider a sequence of smooth positive probability densities η on finite-dimensional Euclidean spaces, with finite second moments. Let $A = -\Delta - D \log \eta \cdot D$, and suppose that η satisfies Poincaré with constant at most λ^{-1} , where $\lambda \rightarrow 1$. Fix $s = 1/8$. Suppose smooth vector fields b, w and a symmetric matrix field K satisfy*

$$D \log \eta = -\frac{y - b(y)}{s} + \omega(y), \quad Db = \text{Id} - sK, \quad 0 \leq K \leq s^{-1} \text{Id},$$

where either $\omega = w$ throughout the sequence or $\omega = 0$ throughout the sequence. Assume $b, w \in L^2(\eta)$, w is a gradient, and, with $I = \|w\|_2^2 > 0$,

$$w \in \text{Dom } A, \quad \|(A - 1)w\|_2 = o(\sqrt{I}), \quad |\mathbb{E}_\eta w| = o(\sqrt{I}), \quad \langle w, Kw \rangle_2 = o(I). \quad (5.1)$$

Assume also the following prediction and projection properties.

(i) For $Y \sim \eta$ and an independent standard Gaussian G , set

$$U_r = Y + \sqrt{r} G, \quad h_r(u) = \mathbb{E}[w(Y) \mid U_r = u], \quad m(r) = I^{-1} \mathbb{E}|h_r(U_r)|^2.$$

For every fixed $0 < r < 1/4$, one has $m(r) \rightarrow e^{-r}$.

(ii) For every fixed $c < \infty$,

$$\sup_{\substack{\Pi^2 = \Pi = \Pi^* \text{ deterministic} \\ \text{rank } \Pi \leq cI}} \|\Pi w\|_2^2 = o(I). \quad (5.2)$$

For every $\epsilon > 0$ there is a deterministic projection Π_ϵ of rank at most $C_\epsilon I$ such that

$$\|\mathbb{E}[(1 - \Pi_\epsilon)bb^*(1 - \Pi_\epsilon)]\|_{\text{op}} + \|\mathbb{E}[(1 - \Pi_\epsilon)\omega\omega^*(1 - \Pi_\epsilon)]\|_{\text{op}} \leq \epsilon \quad (5.3)$$

eventually along the sequence.

(iii) If $\omega = w$, the conditional covariance $C_r = \text{Cov}(w(Y) \mid U_r)$ satisfies, for every fixed $r \in (0, 1/4)$ and $a > 0$,

$$\mathbb{E} \text{tr}(C_r \mathbf{1}_{\{C_r > a\}}) = o(I). \quad (5.4)$$

No conditional covariance-tail assumption is made when $\omega = 0$.

Then there are totally symmetric tensor fields T_l of order $l + 1$, for every fixed integer $l \geq 0$, with $T_0 = w$, such that

$$\|T_l\|_2^2 = (1 + o(1))I, \quad \|(A - 1)T_l\|_2 = o(\sqrt{I}), \quad (5.5)$$

$$\|DT_l - T_{l+1}\|_2 = o(\sqrt{I}).$$

$$\langle T_l, K_1 T_l \rangle_2 = o(I). \quad (5.6)$$

$$\sup_{\substack{\Pi^2 = \Pi = \Pi^* \text{ deterministic} \\ \text{rank } \Pi \leq cI}} \|\Pi_1 T_l\|_2^2 = o(I) \quad \text{for every fixed } c < \infty. \quad (5.7)$$

Every T_l belongs componentwise to $\text{Dom } A$. For $l \geq 1$ it can be chosen in $\bigcap_{p \geq 1} \text{Dom } A^p$. By symmetry, the specified slot in Equations (5.6) and (5.7) may be any slot. All assertions can be made simultaneous for any fixed finite collection of levels and parameters.

In the application, take $b = b_s$ and $K = K_s$. Proposition 3.9 supplies the drift, Poincaré, and initial estimates; Proposition 4.3 and Lemma 4.4 supply the remaining inputs. In the gap branch, $\omega = 0$ and $\|\mathbb{E}bb^*\|_{\text{op}} = o(1)$, so one may take $\Pi_\epsilon = 0$ in Equation (5.3). The normalization $I \rightarrow 0$ makes Equation (5.2) automatic: eventually the only projection of rank at most cI is zero. Thus no strict-branch probe estimate has been added to the gap-branch hypotheses.

We prove the proposition in three steps. First we construct the eigenfunction and derivative hierarchy. Second we expand its evolution along stationary diffusion and, in the strict branch, deduce operator-norm tail estimates. Finally those estimates justify a curvature integration by parts.

5.1 Gaussian Bessel saturation and the induction

We use the classical correspondence between Hermite polynomials and multiple Wiener integrals [10, Sections 3–5], with the ordered-index normalization recorded below. The saturation argument will come from the prediction curve, in addition to this orthogonality.

For a standard Gaussian vector $G \in \mathbb{R}^n$, define its symmetric Hermite tensors $\mathfrak{h}_j(G)$ by

$$e^{\xi \cdot G - |\xi|^2/2} = \sum_{j \geq 0} \frac{1}{j!} \mathfrak{h}_j(G) : \xi^{\otimes j}. \quad (5.8)$$

Here a colon contracts all indicated slots and $\mathfrak{h}_0 = 1$. Comparing coefficients in the expectation of the product of two such exponentials gives, for symmetric order- j and order- k tensors a, b ,

$$\mathbb{E}[(a : \mathfrak{h}_j(G))(b : \mathfrak{h}_k(G))] = \mathbf{1}_{\{j=k\}} j! \langle a, b \rangle. \quad (5.9)$$

This identity uses ordered coefficient indices; it explains all factorials below.

Lemma 5.2 (Bessel inequality for the prediction). *For the data of Proposition 5.1, define $g_r(y) = \mathbb{E}_G h_r(y + \sqrt{r} G)$. For each $r > 0$,*

$$\langle w, g_r \rangle_2 = m(r)I, \quad \sum_{j \geq 0} \frac{r^j}{j!} \|D^j g_r\|_2^2 \leq m(r)I. \quad (5.10)$$

In particular, every fixed derivative appearing here is in the form domain of A .

Proof. The first equality is conditional expectation:

$$\mathbb{E} \langle w(Y), h_r(U_r) \rangle = \mathbb{E} |h_r(U_r)|^2.$$

For fixed y , Gaussian integration by parts identifies the order- j Hermite coefficient of $h_r(y + \sqrt{r} G)$ as $r^{j/2} D^j g_r(y)$. Its orthogonal projection onto that chaos is

$$\frac{r^{j/2}}{j!} D^j g_r(y) : \mathfrak{h}_j(G),$$

where the vector output slot is not contracted. By Equation (5.9), the squared norm of this projection is $r^j |D^j g_r(y)|^2 / j!$. Bessel's inequality, integrated in $y \sim \eta$, proves the asserted sum.

These calculations require only $\mathbb{E} |h_r(Y + \sqrt{r} G)|^2 < \infty$. To justify the derivatives, on a compact set of starting points use positivity of η on a slightly larger compact set. Integrating translated Gaussian kernels over that larger set bounds each fixed differentiated kernel by a constant times the mixture density of U_r . The enlargement absorbs the polynomial factors from differentiation. Thus the kernel integrals are classically smooth locally; alternatively all identities first hold distributionally. The Bessel bounds for orders j and $j + 1$ put $D^j g_r$ and its weak derivative in $L^2(\eta)$, hence in the form domain by cutoff and local smoothing. The local constants need not be uniform in the dimension: only the exact integrated Bessel inequality is used quantitatively. $\square$

Two elementary Hilbert-space facts will prevent dimension factors and unjustified differentiations in the induction.

Lemma 5.3 (Cross-covariance and spectral smoothing). *If F is a square-integrable random variable with values in a finite-dimensional Hilbert space and Z is a centered Euclidean random vector with $\text{Cov } Z \leq c\text{Id}$, then*

$$|\mathbb{E}(F \otimes Z)|_{\text{HS}}^2 \leq c\mathbb{E}|F|^2. \quad (5.11)$$

Next, let A be a sequence of nonnegative Dirichlet operators with kernel the constants and spectral gap at least $\lambda \rightarrow 1$. If fields S in their form domains satisfy

$$|\mathbb{E}S|^2 = o(I), \quad \|S\|_2^2 = (1 + o(1))I, \quad \|DS\|_2^2 \leq (1 + o(1))I,$$

then $\tilde{S} = e^{1-A}S$ satisfies

$$\|\tilde{S} - S\|_2 + \|D(\tilde{S} - S)\|_2 = o(\sqrt{I}), \quad \|(A - 1)\tilde{S}\|_2 = o(\sqrt{I}). \quad (5.12)$$

Proof. The map $u \mapsto u \cdot Z$ from Euclidean space to scalar L^2 has operator norm at most $\sqrt{c}$. Apply its adjoint to each scalar coefficient of F and sum the squared bounds. This proves Equation (5.11) without a factor equal to the number of coordinates of either variable.

For the second assertion, sum the coefficient spectral measures of $S - \mathbb{E}S$ and call the resulting measure ν . Then

$$\int (x - \lambda) d\nu(x) = \|DS\|_2^2 - \lambda\|S - \mathbb{E}S\|_2^2 = o(I),$$

where the left side is nonnegative. For every fixed $\delta > 0$, the spectral mass and the spectral energy outside $[1 - \delta, 1 + \delta]$ are $o(I)$: the part below $1 - \delta$ is absent eventually, and on $x \geq 1 + \delta$ both 1 and x are bounded by a δ -dependent constant times $x - \lambda$. The constant component has squared norm $o(I)$ as well.

On the remaining interval the multipliers $e^{1-x} - 1$ and $(x - 1)e^{1-x}$ tend uniformly to zero as $\delta \downarrow 0$. Outside it, $(1 + x)|e^{1-x} - 1|^2 \leq C(1 + x)$ and $(x - 1)^2 e^{2-2x} \leq C$. Integrate these inequalities against the spectral measure, first take the sequence limit and then let $\delta \downarrow 0$. This gives Equation (5.12). The same functional calculus shows that $e^{1-A}S$ belongs to every $\text{Dom } A^p$. $\square$

Lemma 5.4 (Construction before the curvature estimate). *Under the hypotheses of Proposition 5.1, there exist totally symmetric fields T_l satisfying Equations (5.5) and (5.7), with the stated operator domains. Moreover $|\mathbb{E}T_l| = o(\sqrt{I})$ for every fixed l .*

Proof. Start with $T_0 = w$. Suppose the assertions have been constructed through level l , and put $L = l + 1$ and $V = DT_l$. Self-adjointness gives

$$\|V\|_2^2 = \langle T_l, AT_l \rangle_2 = (1 + o(1))I. \quad (5.13)$$

If Π is an admissible deterministic projection acting on an existing tensor slot, it commutes with A and differentiation. Consequently

$$\|\Pi_1 V\|_2^2 = \langle \Pi_1 T_l, A\Pi_1 T_l \rangle_2 \leq \|\Pi_1 T_l\|_2^2 + \|\Pi_1 T_l\|_2 \|(A - 1)T_l\|_2 = o(I).$$

The error is uniform over projections of rank at most cI for fixed c .

Fix $0 < r < 1/4$. Set $V_j = T_j$ for $0 \leq j \leq l$ and $V_L = V$. Successive form integration by parts gives

$$I^{-1} \langle D^j g_r, V_j \rangle_2 \longrightarrow e^{-r} \quad (0 \leq j \leq L). \quad (5.14)$$

Indeed the assertion at $j = 0$ is Equation (5.10). For $j \leq l$,

$$\langle D^{j+1} g_r, DT_j \rangle_2 = \langle D^j g_r, AT_j \rangle_2 = \langle D^j g_r, T_j \rangle_2 + o(I).$$

Replacing DT_j by T_{j+1} when $j < l$ costs $o(I)$. All test derivatives and their form energies are finite by Lemma 5.2; its bound controls every error for fixed r and j . No uniform estimate as $r \downarrow 0$ has been used.

Here is the precise Bessel budget. Equation (5.14) and $\|V_j\|_2^2/I \rightarrow 1$ imply

$$\liminf I^{-1} e^{2r} \|D^j g_r\|_2^2 \geq 1 \quad (0 \leq j \leq L).$$

Multiply Equation (5.10) by e^{2r}/I and subtract these minimum contributions. The resulting bounds are

$$\limsup \frac{\|e^r D^L g_r - V\|_2^2}{I} \leq \frac{L!}{r^L} \left(e^r - \sum_{j=0}^L \frac{r^j}{j!} \right) = O_L(r), \quad (5.15)$$

$$\limsup \frac{e^{2r} \|D^{L+1} g_r\|_2^2}{I} \leq \frac{(L+1)!}{r^{L+1}} \left(e^r - \sum_{j=0}^L \frac{r^j}{j!} \right) = 1 + O_L(r). \quad (5.16)$$

For the first line, expand the squared difference and use the correlation at order L ; its normalized limit subtracts exactly one from the normalized squared derivative norm. Thus the exponential-series tail accounts for the entire excess, rather than merely bounding the norms.

Choose $r \downarrow 0$ sufficiently slowly along the sequence in these two estimates. The field $S_0 = e^r D^L g_r$ then satisfies

$$\|S_0 - V\|_2 = o(\sqrt{I}), \quad \|DS_0\|_2^2 \leq (1 + o(1))I.$$

It is symmetric in its last L slots; V is symmetric in its first L slots. For $L \geq 2$, the permutations of those two sets of slots generate every permutation of the $L+1$ slots. Indeed their adjacent transpositions together are all the adjacent transpositions. If a permutation fixes S_0 , its action changes V by at most $2\|S_0 - V\|_2$; a permutation fixing V costs zero. Expressing each permutation as a fixed finite word therefore shows that V is $o(\sqrt{I})$ from its full symmetrization. For $L = 1$, $V = Dw$ is already symmetric because w is a gradient. Let S be the full symmetrization of S_0 . Averaging coefficient permutations is an orthogonal contraction and commutes with differentiation, so

$$\|S - V\|_2 = o(\sqrt{I}), \quad \|S\|_2^2 = (1 + o(1))I, \quad \|DS\|_2^2 \leq (1 + o(1))I. \quad (5.17)$$

The preceding projection estimate for V now holds on its last slot too, since a coefficient-slot permutation moves it to an old slot with an $o(\sqrt{I})$ error uniform over projections.

To apply Poincaré sharply we still need to center V . Write $C_y = \text{Cov}(T_l, y)$, $C_b = \text{Cov}(T_l, b)$, and $C_\omega = \text{Cov}(T_l, \omega)$, with the vector variable occupying the last slot. Testing the equation for AT_l against centered coordinates gives

$$\mathbb{E}V = C_y + E_l, \quad |E_l| = o(\sqrt{I}), \quad \mathbb{E}V = s^{-1}(C_y - C_b) - C_\omega. \quad (5.18)$$

For clarity, the first exact identity before the error is $\mathbb{E}V = \mathbb{E}[(AT_l) \otimes (y - \mathbb{E}y)]$. Poincaré gives $\text{Cov}_\eta y \leq \lambda^{-1} \text{Id}$, and Equation (5.11) bounds its error by $\lambda^{-1/2} \|(A-1)T_l\|_2$. This is the needed dimension-free estimate. The second identity follows by integration by parts using $Ay = (y - b)/s - \omega$. These coordinate tests are legitimate in the form domain; their distributional A -images are in L^2 by the stated second-moment assumptions, hence they are also in $\text{Dom } A$.

Project the last slot onto $1 - \Pi_\epsilon$. The two covariance terms involving b and ω have norm at most $C\sqrt{\epsilon I}$ by Equations (5.3) and (5.11). Subtracting the two identities in Equation (5.18), and using $s \neq 1$, therefore bounds this part of $\mathbb{E}V$ by $C_s \sqrt{\epsilon I} + o(\sqrt{I})$. On the Π_ϵ part,

$$|(\Pi_\epsilon)_{L+1} \mathbb{E}V| \leq \|(\Pi_\epsilon)_{L+1} V\|_2 = o(\sqrt{I})$$

by the projected-energy estimate and $\text{rank } \Pi_\epsilon \leq C_\epsilon I$. First take the sequence limit and then let $\epsilon \downarrow 0$. This proves $|\mathbb{E}V| = o(\sqrt{I})$, and thus $|\mathbb{E}S| = o(\sqrt{I})$.

Lemma 5.3 applies to S . Define

$$T_{l+1} = e^{1-A} S.$$

It is symmetric, lies in every operator-power domain, is $o(\sqrt{I})$ from V , and satisfies the strong near-eigenfunction estimate. Its mean is $e\mathbb{E}S = o(\sqrt{I})$. Finally, coefficient projections commute with the semigroup, and $\|e^{1-A}\|_{2 \rightarrow 2} \leq e$. Together with $S - V = o_{L^2}(\sqrt{I})$, this preserves the uniform projected-energy estimates. The induction is complete. Each step used only a finite collection of limits; diagonal choices give the assertion for all fixed levels. $\square$

5.2 Expansion along stationary diffusion

The fields just constructed retain the same norm under repeated differentiation. This makes their stochastic Taylor expansion summable with a factorial remainder. We use the ordered-integral form of the Wiener–Itô correspondence [10, Section 5]. Let Y_t be the stationary diffusion with invariant density η ,

$$dY_t = \sqrt{2} dB_t + D \log \eta(Y_t) dt, \quad Y_0 \sim \eta,$$

where Y_0 is independent of the driving Brownian motion.

Lemma 5.5 (Hermite expansion of the stationary evolution). *Let symmetric fields T_l satisfy Equation (5.5). For fixed integers $l, m \geq 0$ and fixed $t > 0$, set $N = B_t/\sqrt{t}$. Then*

$$T_l(Y_t) = e^{-t} \sum_{j=0}^m \frac{(2t)^{j/2}}{j!} T_{l+j}(Y_0) : \mathfrak{h}_j(N) + \mathcal{E}_{l,m,t}, \quad (5.19)$$

where the colon contracts the last j tensor slots and

$$\limsup \frac{\|\mathcal{E}_{l,m,t}\|_2}{\sqrt{I}} \leq \frac{(2t)^{(m+1)/2}}{\sqrt{(m+1)!}}. \quad (5.20)$$

For $0 \leq t \leq T < \infty$ the same bound holds uniformly with t replaced on the right by T and with the expansion at $t = 0$ interpreted by continuity. The limits are also simultaneous for any fixed finite set of starting levels. In particular the normalized remainder tends to zero when the sequence limit is followed by $m \rightarrow \infty$.

Proof. The operator-domain Itô formula from Lemma 2.2 gives, for each fixed d ,

$$\begin{aligned} e^v T_d(Y_v) &= T_d(Y_0) + \sqrt{2} \int_0^v e^u T_{d+1}(Y_u) : dB_u \\ &\quad + \int_0^v e^u (1 - A) T_d(Y_u) du + \sqrt{2} \int_0^v e^u (DT_d - T_{d+1})(Y_u) : dB_u. \end{aligned}$$

Stationarity, Cauchy–Schwarz for the time integral, and Itô isometry show that the last two terms are $o(\sqrt{I})$, uniformly for $v \in [0, T]$ in squared-mean norm. Substitute this formula for the integrand at the next level, and repeat $m + 1$ times. For fixed l, m, T , every replacement error remains $o(\sqrt{I})$; iterated isometry sums coefficient squares and introduces no dimension factor.

After multiplication by e^{-t} , the remaining integral has coefficient $2^{(m+1)/2}$, is over the ordered time simplex of length t , and has innermost integrand $e^{u-t} T_{l+m+1}(Y_u)$. Since $u \leq t$, its squared norm is at most

$$2^{m+1} \frac{t^{m+1}}{(m+1)!} \|T_{l+m+1}\|_2^2 = (1 + o(1)) I \frac{(2t)^{m+1}}{(m+1)!}.$$

This proves the remainder estimate and its uniform version. The other terms have coefficients evaluated at Y_0 . Symmetry in their Brownian slots turns their ordered iterated integrals into $t^{j/2} \mathfrak{h}_j(N)/j!$. To verify the factor directly, expand the Brownian exponential martingale $\exp(\xi \cdot B_t - t|\xi|^2/2)$ by iterated Itô integration and compare its coefficients with Equation (5.8). The additional $\sqrt{2}$ at each stochastic integration gives exactly $(2t)^{j/2}/j!$ in Equation (5.19). $\square$

5.3 Operator tails in the strict branch

For a tensor F , a one-slot flattening is the matrix whose row index is one specified tensor slot and whose column index is the tuple of all remaining slots. Its Hilbert–Schmidt norm is exactly $|F|$.

In the strict branch, put $Q = Dw$. The drift decomposition gives $D^2(-\log \eta) = K - Q$. Viewing DF as a matrix with its derivative slot as row index, commuting a derivative with A formally gives

$$D(AF) = A(DF) + (K - Q)DF.$$

For $F = T_l$, both Q and DF belong to L^2 , but their product QDF need not belong to L^2 . We cannot yet use this formula to deduce the curvature estimate. The conditional covariance bound will instead let us approximate DF by tests P whose uniform operator norms tend to zero. Then $\|QP\|_2 \leq \|P\|_{\text{op},\infty} \|Q\|_2 = o(\sqrt{I})$, which is the product bound needed for the justified identity below. This step is needed only when $\omega = w$; in the gap branch $Q = 0$.

Lemma 5.6 (Vanishing operator tails). *In the strict branch of Proposition 5.1, $Q = Dw$ and the fields T_l constructed in Lemma 5.4 satisfy the following property. For every fixed $\delta > 0$, each one-slot flattening F of T_l , $l \geq 1$, and also $F = Q$, obeys*

$$\mathbb{E} \text{tr}(FF^* \mathbf{1}_{\{FF^* > \delta^2\}}) = o(I). \quad (5.21)$$

Equivalently, clipping every singular value at any fixed positive cutoff changes the field by $o(\sqrt{I})$ in L^2 .

Proof. We first prove the assertion for Q . Couple the heat observations by $U_v = Y_0 + W_v$, where W is a standard Brownian motion independent of $Y_0 \sim \eta$, and retain $h_v = \mathbb{E}[w(Y_0) \mid U_v]$. Put $H_v = Dh_v(U_v)$ wherever this derivative is square integrable. The conditional covariance has the exact representation

$$C_r = \int_0^r \mathbb{E}[H_v H_v^* \mid U_r] dv, \quad \int_0^r \mathbb{E}|H_v|^2 dv = (1 - m(r))I. \quad (5.22)$$

To check the first identity and its normalization, let p_v denote the density of U_v . The entries of $p_v h_v$ solve the heat equation with generator $\Delta/2$. Direct differentiation gives the matrix identity

$$(\partial_v - \tfrac{1}{2}\Delta)(p_v h_v h_v^*) = -p_v Dh_v (Dh_v)^*.$$

Integrate it against the later heat kernel and divide by p_r . The initial term becomes $\mathbb{E}[w w^* \mid U_r]$, and the final term is $h_r h_r^*$, which proves the first formula. Taking the trace and expectation proves the second formula. For nonsmooth or unbounded initial w , perform the calculation on scalar projections with convex truncations increasing to the square, and then use polarization. Spatial cutoffs and the square-integrability of w justify the positive-time passage. At time zero, approximation of w in $L^2(\eta)$ by bounded continuous functions gives $\mathbb{E}|w - \mathbb{E}[w \mid U_v]|^2 \rightarrow 0$. Thus there is no missing boundary term in Equation (5.22).

By Equation (5.1), $\|Q\|_2^2 = \langle w, Aw \rangle_2 = (1 + o(1))I$. For almost every $v > 0$, convolution differentiation and form integration by parts give

$$\mathbb{E}\langle Q(Y_0), H_v \rangle = \langle Dw, Dg_v \rangle_2 = \langle Aw, g_v \rangle_2 = m(v)I + o(I). \quad (5.23)$$

The error is uniform in v : Jensen's inequality gives $\|g_v\|_2 \leq \sqrt{I}$, so it is bounded by $\|(A-1)w\|_2\sqrt{I}$. The convolution derivative identity follows first with cutoffs and bounded functions, then by the square-integrable derivative bounds in Equation (5.22).

Fix a small $z > 0$ with $2z < 1/4$. Combining the last two equations and using $0 \leq m(v) \leq 1$ yields the exact limiting discrepancy

$$\lim \frac{1}{I} \int_0^{2z} \mathbb{E}|Q(Y_0) - H_v|^2 dv = 2z + 1 - e^{-2z} - 2 \int_0^{2z} e^{-v} dv \quad (5.24)$$

$$= 2z - 1 + e^{-2z} = O(z^2). \quad (5.25)$$

Dominated convergence is applicable to $m(v)$, so this calculation does not require uniform convergence of the prediction curve near zero.

We select a time in $[z, 2z]$ while preserving a covariance-tail bound. For a fixed $\kappa > 0$, define

$$a(r) = I^{-1} \mathbb{E} \operatorname{tr} \left(\frac{C_r}{r} \mathbf{1}_{\{C_r/r > \kappa\}} \right).$$

For fixed $r \in [z, 2z]$, Equation (5.4) gives $a(r) \rightarrow 0$, and $a(r) \leq 1/z$ follows from $\mathbb{E} \operatorname{tr} C_r \leq I$. Consequently $\int_z^{2z} a(r) dr \rightarrow 0$. On the other hand, Equation (5.25) and Markov's inequality show that the set of times satisfying

$$\mathbb{E}|Q(Y_0) - H_r|^2 \leq CzI$$

has measure at least $z/2$, with an absolute C and for sufficiently far terms of the sequence. The set where $a(r)$ exceeds a quantity tending to zero has measure tending to zero. Select $r \in [z, 2z]$ in the intersection. Thus r may vary with the sequence index, but

$$\mathbb{E}|Q(Y_0) - H_r|^2 \leq CzI, \quad \mathbb{E} \operatorname{tr} \left(\frac{C_r}{r} \mathbf{1}_{\{C_r/r > \kappa\}} \right) = o(I). \quad (5.26)$$

All constants in this selection are for fixed z, κ .

We compare $B = H_r H_r^*$ with $C = C_r/r$ in trace norm. For matrices U, V ,

$$\|UU^* - VV^*\|_1 \leq |U - V|_{\text{HS}}(|U|_{\text{HS}} + |V|_{\text{HS}}). \quad (5.27)$$

Compare both B and C with $\mathbb{E}[QQ^* | U_r]$. For B , conditional Jensen's inequality and the first bound in Equation (5.26) give an $O(\sqrt{z})I$ bound on the expected trace-norm difference. For C , use the first identity in Equation (5.22), followed by Equation (5.27) and Cauchy-Schwarz in both probability and time. The integrated discrepancy is $O(z^2)I$, whereas

$$\int_0^r \mathbb{E}(|H_v| + |Q|)^2 dv \leq 2(1 - m(2z))I + 4z\|Q\|_2^2.$$

Here we used $r \leq 2z$ and the nonnegative integral in the second identity of Equation (5.22). The normalized limsup of the right side is $O(z)$, by prediction at the fixed time $2z$. No prediction limit at the selected varying time is needed. Division by $r \geq z$ therefore also costs at most $O(\sqrt{z})I$. We have proved, in normalized limsup,

$$\mathbb{E}\|B - C\|_1 \leq O(\sqrt{z})I. \quad (5.28)$$

Fix $\delta > 0$, take $\kappa = \delta^2/2$ above, and set $P = \mathbf{1}_{\{B > \delta^2\}}$. Positivity of C implies

$$\mathrm{tr}(BP) \leq \frac{\delta^2}{2} \mathrm{rank} P + \mathrm{tr}(C\mathbf{1}_{\{C > \delta^2/2\}}) + \|B - C\|_1.$$

Since $\delta^2 \mathrm{rank} P \leq \mathrm{tr}(BP)$, the first term is absorbed into the left side. Equations (5.26) and (5.28) bound the remaining high singular-value energy by $O(\sqrt{z})I + o(I)$. Clip H_r at δ ; its squared distance from the operator-norm ball of radius δ is at most this energy. The first estimate in Equation (5.26) then gives the same bound, up to constants, for the distance of Q from that ball. Let $z \downarrow 0$ after the sequence limit. The squared distance is $o(I)$ for every fixed $\delta > 0$. Applying this distance assertion at cutoff $\delta/2$ bounds the energy of singular values above δ by four times the squared distance, proving Equation (5.21) for Q . Since $\|T_1 - Q\|_2 = o(\sqrt{I})$, it also proves the assertion for T_1 .

It remains to propagate this estimate to higher levels. Fix $l \geq 1$ and $t > 0$. Apply Lemma 5.5 with starting level 1. Conditional on Y_0 , the Hermite chaoses are orthogonal even after applying an arbitrary measurable projection $P(Y_0)$ to the first tensor slot. The chaos containing $T_l(Y_0)$ has squared coefficient

$$c_{l,t} = e^{-2t} \frac{(2t)^{l-1}}{(l-1)!} > 0.$$

Use the expansion to order $m \geq l-1$, first take the sequence limit, and then let $m \rightarrow \infty$ in its remainder bound. The resulting inequality is

$$\liminf \frac{\mathbb{E}|P(Y_0)_1 T_1(Y_t)|^2 - c_{l,t} \mathbb{E}|P(Y_0)_1 T_l(Y_0)|^2}{I} \geq 0. \quad (5.29)$$

The remainder estimates are uniform over such projections, since their operator norms are at most one.

Choose $P(Y_0)$ to project onto the singular values greater than a fixed $\epsilon > 0$ of the first-slot flattening of $T_l(Y_0)$. Then

$$\mathbb{E} \mathrm{rank} P \leq \epsilon^{-2} \|T_l\|_2^2 = (1 + o(1)) \epsilon^{-2} I.$$

For every $\delta > 0$, let S_δ be the singular-value clipping of T_1 at δ . Pointwise,

$$|P(Y_0) S_\delta(Y_t)|_{\mathrm{HS}}^2 \leq \delta^2 \mathrm{rank} P(Y_0).$$

Stationarity controls the $o(I)$ clipping error for $T_1(Y_t)$. Hence the normalized limsup of the first expectation in Equation (5.29) is at most δ^2/ϵ^2 . Send $\delta \downarrow 0$ to obtain the tail claim at threshold ϵ for T_l . No independence between $P(Y_0)$ and $T_1(Y_t)$ was required. Tensor symmetry supplies every other one-slot flattening. $\square$

5.4 The curvature estimate and its domains

It remains to prove the curvature assertion of the hierarchy. We first justify the differentiated operator identity against a test whose product with the curvature is in L^2 . The preceding tail estimate will supply such a test in the strict branch; the gap branch will use T_{l+1} directly.

Lemma 5.7 (Weighted divergence and the differentiated operator). *Let η be a smooth positive probability density, let $A = -\Delta - D \log \eta \cdot D$, and put $H = D^2(-\log \eta)$. Let $P = (P_{i\alpha})$ have one vector slot i and a finite-dimensional coefficient index α . Suppose every coefficient belongs to $\text{Dom } A$ and $HP \in L^2(\eta)$. Then the weighted adjoint divergence*

$$(D^*P)_\alpha = -\sum_i \partial_i P_{i\alpha} - \sum_i (\partial_i \log \eta) P_{i\alpha}$$

belongs to $L^2(\eta)$ and satisfies

$$\|D^*P\|_2^2 \leq \|DP\|_2^2 + \langle P, HP \rangle_2. \quad (5.30)$$

For every coefficient field $F \in \text{Dom } A$,

$$\langle AF, D^*P \rangle_2 = \langle DF, AP \rangle_2 + \langle DF, HP \rangle_2. \quad (5.31)$$

In particular this identity requires $HP \in L^2$, not $HDF \in L^2$.

Proof. For compactly supported smooth P , commuting two first derivatives in weighted integration by parts gives

$$\|D^*P\|_2^2 = \sum_{i,j,\alpha} \langle \partial_j P_{i\alpha}, \partial_i P_{j\alpha} \rangle_2 + \langle P, HP \rangle_2. \quad (5.32)$$

Cauchy–Schwarz applied to the two arrays with i, j interchanged bounds the sum by $\|DP\|_2^2$, without a dimension factor.

For the stated P , membership in $\text{Dom } A$ gives $P \in H_{\text{loc}}^2$ by local elliptic regularity and $DP \in L^2(\eta)$. Choose smooth cutoffs $0 \leq \chi_R \leq 1$ tending to one with $\|D\chi_R\|_\infty \rightarrow 0$. Equation (5.32) applies to $\chi_R P$ by local Sobolev approximation. Since

$$\|D(\chi_R P) - DP\|_2 \rightarrow 0, \quad \langle \chi_R P, H\chi_R P \rangle_2 \rightarrow \langle P, HP \rangle_2,$$

its right side is bounded. The second convergence uses $|\langle P, HP \rangle| \in L^1$, a consequence of $P, HP \in L^2$. Weak compactness in L^2 and distributional convergence of $D^*(\chi_R P)$ identify the limit as D^*P . Lower semicontinuity proves Equation (5.30). Only first derivatives of the cutoffs have entered the estimate.

For compactly supported smooth F , differentiate the scalar operator:

$$D(AF) = A(DF) + HDF.$$

Pair this identity with P , use symmetry of H , and integrate by parts to obtain Equation (5.31). For general $F \in \text{Dom } A$, use the operator core from Lemma 2.2 to choose compact smooth F_j with $F_j \rightarrow F$ and $AF_j \rightarrow AF$ in L^2 . This convergence also holds in the form norm, since

$$\|D(F_j - F)\|_2^2 = \langle F_j - F, A(F_j - F) \rangle_2 \rightarrow 0.$$

All terms in Equation (5.31) therefore converge: the three fixed factors are D^*P , AP , and HP , each in L^2 . This proves the claimed global identity. $\square$

Completion of the proof of Proposition 5.1. Lemma 5.4 supplies the fields and all assertions except Equation (5.6). Fix $l \geq 0$ and write $F = T_l$. View DF as a matrix with its derivative slot as row index and all old tensor slots as column index. We construct a matrix field P satisfying

$$\|P - DF\|_2 + \|(A - 1)P\|_2 = o(\sqrt{I}), \quad \|DP\|_2 = O(\sqrt{I}), \quad \|QP\|_2 = o(\sqrt{I}), \quad (5.33)$$

where $Q = Dw$ in the strict branch and $Q = 0$ in the gap branch.

In the strict branch, apply Lemma 5.6 to the corresponding flattening of T_{l+1} . Choose a singular-value cutoff $\delta \downarrow 0$ sufficiently slowly that the clipped field C obeys

$$\|C - T_{l+1}\|_2 = o(\sqrt{I}), \quad \|C(y)\|_{\text{op}} \leq \delta.$$

Set $P = e^{1-A}C = eP_1C$. Markov averaging and convexity of the operator-norm ball give $\|P(y)\|_{\text{op}} \leq e\delta$. The spectral multipliers e^{1-x} , $(x-1)e^{1-x}$, and $\sqrt{x}e^{1-x}$ are bounded for $x \geq 0$. Moreover $|e^{1-x} - 1| \leq C|x-1|$ there. These bounds, the approximation of C , and the near-eigenfunction assertion for T_{l+1} give the first two estimates in Equation (5.33). Finally $\|Q\|_2^2 = (1 + o(1))I$ and

$$\|QP\|_2 \leq \|P\|_{\text{op},\infty}\|Q\|_2 = o(\sqrt{I}).$$

The clipping is only used to construct P , so it need not preserve tensor symmetry. In the gap branch take $P = T_{l+1}$ directly. The first two estimates follow from Equation (5.5), and the last is immediate from $Q = 0$. No operator-tail argument is needed in this branch.

The drift hypothesis gives $H = D^2(-\log \eta) = K - Q$ in either case. Both KP and QP are in L^2 , and the operator-domain assertion for P was retained by its construction. Lemma 5.7 therefore applies. In particular,

$$\|D^*P\|_2^2 \leq \|DP\|_2^2 + s^{-1}\|P\|_2^2 + \|P\|_2\|QP\|_2 = O(I).$$

Insert $AF = F + (A-1)F$ and $AP = P + (A-1)P$ into Equation (5.31). The terms $\langle F, D^*P \rangle_2$ and $\langle DF, P \rangle_2$ cancel. Consequently

$$\langle DF, KP \rangle_2 = \langle (A-1)F, D^*P \rangle_2 - \langle DF, (A-1)P \rangle_2 + \langle DF, QP \rangle_2 = o(I).$$

Since $\|K\|_{\text{op},\infty} \leq s^{-1}$ and $P - DF = o_{L^2}(\sqrt{I})$, this gives

$$\langle DF, KDF \rangle_2 = o(I).$$

Replacing DF by T_{l+1} costs $o(I)$ by the derivative estimate. It proves Equation (5.6) at level $l+1$ on the derivative slot, and hence on every slot by symmetry. Level zero was an initial assumption. All claims of the proposition follow. $\square$

To use many tensor degrees at once, first impose the estimates on a fixed finite list of levels, projection-rank constants, and times. Then increase that list sufficiently slowly along the original sequence. For example, one may choose $k \rightarrow \infty$ so that the normalized errors in Proposition 5.1 tend to zero uniformly through level $3k$, while $\sup_{l \leq 3k} \|T_l\|_2^2/I$ stays bounded. The same diagonal applies to any fixed number of terms in Lemma 5.5 and any fixed bounded time interval. Thus averaging over a band of degrees or shifting that band by a fixed integer does not change any of the preceding normalized estimates.

6 The reduced square root and a frozen matrix algebra

The hierarchy of Proposition 5.1 contains tensors of arbitrarily high order whose derivatives have almost the same energy as the tensors themselves. We first transfer this near equality to the positive square root of their one-slot reduction. We then describe that square root along the stationary diffusion by matrices frozen at the initial point. The construction uses weighted traces throughout; in particular, no lower bound on a nonzero eigenvalue will be required.

6.1 Sharp energy for the reduced square root

All limits in this section are along the counterexample sequence. We make the following diagonal choice explicit. For each finite integer K , impose the conclusions of Proposition 5.1 through level $3K$, the fixed-projection conclusions for rank coefficients at most K , and the estimates of Lemma 5.5 for every starting level $0 \leq l < 2K$ and every truncation order $0 \leq m \leq K$, on time intervals $0 \leq t \leq K$. Only levels through $3K$ enter these expansions, including their remainders. For each such finite list, the hierarchy defects and the excesses over the normalized Hermite remainder bounds can be made arbitrarily small. Choose an integer $k \rightarrow \infty$ slowly enough that these defects and excesses tend to zero, uniformly over the corresponding list. Further auxiliary parameters below will be chosen by the same finite-list procedure. In particular, normalized endpoint energies remain bounded when a band of k levels is shifted by a fixed number of levels.

Regard the direct sum

$$\Psi = \bigoplus_{k \leq l < 2k} \frac{T_l}{\sqrt{kI}}$$

as a matrix by separating the first tensor index from the remaining indices and the direct-sum label. Define

$$\rho = \Psi\Psi^*, \quad R = \rho^{1/2}. \quad (6.1)$$

High tensor degree provides many interchangeable slots, while averaging adjacent degrees makes each fixed shift cost only endpoint terms of order k^{-1} . These two features will force sharp energy estimates for the positive matrix field R .

Here R is a positive matrix field, distinct from the scalar log-Sobolev constant $R(\nu)$. Write $\|\cdot\|_2$ for the field norm formed with ordinary Hilbert–Schmidt coefficient norms. The hierarchy gives

$$\mathbb{E} \operatorname{tr} R^2 = \|\Psi\|_2^2 \longrightarrow 1, \quad \|D\Psi\|_2^2 \longrightarrow 1. \quad (6.2)$$

Lemma 6.1 (Centering and the square-root energy). *The field R belongs to the Dirichlet form domain and satisfies*

$$\|\mathbb{E}R\|_{\text{HS}} = o(1), \quad \liminf \|DR\|_2^2 \geq 1, \quad \|DR\|_2^2 = O(1).$$

Proof. For rectangular matrices X, Y , the Araki–Yamagami inequality [1, Theorem 1] gives the following square-root bound. We give a finite-dimensional proof by self-adjoint dilation:

$$\|\sqrt{XX^*} - \sqrt{YY^*}\|_{\text{HS}}^2 \leq 2\|X - Y\|_{\text{HS}}^2. \quad (6.3)$$

Indeed the absolute value of the dilation of X has diagonal blocks $\sqrt{XX^*}$ and $\sqrt{X^*X}$. If A, B are self-adjoint and f is scalar Lipschitz, expansion in an eigenbasis of A and an eigenbasis of B shows that $\|f(A) - f(B)\|_{\text{HS}} \leq \operatorname{Lip}(f)\|A - B\|_{\text{HS}}$. Apply this with $f(x) = |x|$. Difference quotients, followed by form approximation, prove that R is in the form domain and that $\|DR\|_2^2 \leq 2\|D\Psi\|_2^2$.

The curvature conclusion of the hierarchy yields

$$\mathbb{E} \operatorname{tr}((1 - M_s)R^2) = o(1).$$

We also have

$$\operatorname{tr}((\mathbb{E}M_s)(\mathbb{E}R^2)) = o(1). \quad (6.4)$$

To see the latter assertion in the strict branch, fix $\delta > 0$ and let P_δ be the spectral projection of $\mathbb{E}M_s$ onto (δ, ∞) . Lemma 4.4 gives $\operatorname{rank} P_\delta = O_\delta(I)$; the deterministic-projection conclusion of

Proposition 5.1 therefore gives $\text{tr}(P_\delta \mathbb{E} R^2) = o(1)$. Since $0 \leq M_s \leq 1$, the left side of Equation (6.4) is at most $o(1) + \delta \mathbb{E} \text{tr} R^2$. Send δ to zero after the sequence limit. In the gap branch use directly $\|\mathbb{E} M_s\|_{\text{op}} = o(1)$.

Let R' be evaluated at an independent point. Insert $1 - M_s + M_s$ between R and R' in $\|\mathbb{E} R\|_{\text{HS}}^2 = \mathbb{E} \text{tr}(RR')$. Hilbert–Schmidt Cauchy–Schwarz bounds the term containing $1 - M_s$ by

$$(\mathbb{E} \text{tr}((1 - M_s)R^2))^{1/2} (\mathbb{E} \text{tr} R^2)^{1/2} = o(1).$$

The other term is bounded by

$$(\mathbb{E} \text{tr}(M_s R^2))^{1/2} (\text{tr}((\mathbb{E} M_s)(\mathbb{E} R^2)))^{1/2} = o(1).$$

Poincaré for η , whose constant tends to one, now gives the lower bound on $\|DR\|_2^2$ from Equation (6.2). $\square$

The next estimate is the reason for using tensors with many symmetric slots. It compares averaging an action over those slots with the geometric, rather than arithmetic, weighting by the marginal matrix.

Lemma 6.2 (Averaging over symmetric tensor slots). *Let ψ be a finite direct sum of symmetric tensors, each with at least $k \geq 2$ slots. Let ρ be its one-slot reduction and $R = \sqrt{\rho}$. For a self-adjoint matrix H , let H_{av} act on each component by averaging the action of H over its slots. Then*

$$\|H_{\text{av}} \psi\|^2 \leq \text{tr}(RHRH) + \frac{2}{k} \text{tr}(\rho H^2). \quad (6.5)$$

Proof. For a component ψ_d with d slots, put

$$a_d = \langle \psi_d, H_{(1)}^2 \psi_d \rangle, \quad c_d = \langle \psi_d, H_{(1)} H_{(2)} \psi_d \rangle.$$

Symmetry and Cauchy–Schwarz give $|c_d| \leq a_d$, and direct expansion gives

$$\|H_{\text{av}} \psi\|^2 = \sum_d \left(\frac{a_d}{d} + \left(1 - \frac{1}{d}\right) c_d \right) \leq \sum_d c_d + \frac{2}{k} \sum_d a_d.$$

It remains to prove $|\sum_d c_d| \leq \text{tr}(RHRH)$. This correlation bound can also be deduced from the Wigner–Yanase case of Lieb’s trace-concavity theorem [15]; we give a proof by weighted interpolation.

Define the linear map C by

$$\text{tr}(XC(Y)) = \sum_d \langle \psi_d, X_{(1)} Y_{(2)} \psi_d \rangle.$$

Flattening all remaining indices exhibits $C(Y)$ as a sum of terms $A_\alpha Y^\top A_\alpha^*$. Thus C is positive and $C(1) = C^*(1) = \rho$. Restrict to the support of ρ , on which inverse powers are defined, and set

$$U(Y) = \rho^{-1/2} C(Y) \rho^{-1/2}.$$

The map U is unital and contractive in operator norm: its preceding representation is an isometric compression of copies of $Y^\top$. It also satisfies $\text{tr}(\rho U(Y)) = \text{tr}(\rho Y)$. For $0 \leq \Re z \leq 1$ introduce

$$U_z(Y) = \rho^{z/2} U(\rho^{-z/2} Y \rho^{-z/2}) \rho^{z/2}.$$

On $\Re z = 0$, multiplication on either side by imaginary powers of ρ is isometric, so U_z is operator-norm contractive. At $z = 1$,

$$U_1(Y) = C(\rho^{-1/2}Y\rho^{-1/2}), \quad U_1^*(1) = 1.$$

Its adjoint is a unital positive map with the same compression and transpose representation, hence is operator-norm contractive. Duality makes U_1 trace-norm contractive; imaginary powers give the same conclusion on $\Re z = 1$.

The interpolation used here is the finite-dimensional weighted Schatten argument; compare [3, Section III, Theorem 4]. We give the details for our positive correlation map. Given matrices of Hilbert–Schmidt norm one, continue the singular values of the input by powers $2z$, and those of the matrix paired with its output by powers $2(1 - z)$, keeping the unitary factors fixed. The bilinear trace pairing is analytic in the strip, bounded by one on both boundary lines by the preceding endpoint bounds and trace duality. The three-lines inequality bounds it by one at $z = 1/2$. Zero singular values follow by approximation. This proves $\|U_{1/2}(Y)\|_{\text{HS}} \leq \|Y\|_{\text{HS}}$. Take $Y = \rho^{1/4}H\rho^{1/4}$. Then

$$\text{tr}(YU_{1/2}(Y)) = \text{tr}(HC(H)) = \sum_d c_d, \quad \|Y\|_{\text{HS}}^2 = \text{tr}(RHRH).$$

Cauchy–Schwarz proves the claim. Compression to the support leaves the correlation unchanged: every tensor slot lies in that support. Complexification, used only for interpolation, preserves the conclusion for real tensors. $\square$

Lemma 6.3 (Saturation of the square-root energy). *There are measurable symmetric matrix fields H_i , $1 \leq i \leq n$, solving*

$$D_i R = \frac{H_i R + R H_i}{2}, \tag{6.6}$$

with zero kernel–kernel block, such that

$$\|DR\|_2^2 \longrightarrow 1, \quad \sum_i \mathbb{E} \| [H_i, R] \|_{\text{HS}}^2 = o(1), \quad \sum_i \mathbb{E} \text{tr}(R^2 H_i^2) \longrightarrow 1. \tag{6.7}$$

Moreover,

$$\sum_i \left(\|D_i R - H_i R\|_2^2 + \|D_i \Psi - (H_i)_{(1)} \Psi\|_2^2 \right) = o(1). \tag{6.8}$$

The first slot can be replaced by any existing slot in each component.

Proof. In an eigenbasis of R with eigenvalues r_a , define $(H_i)_{ab} = 2(D_i R)_{ab}/(r_a + r_b)$ when $r_a + r_b > 0$, and zero otherwise. The kernel–kernel block of $D_i R$ vanishes almost everywhere. One way to check this for a form-domain positive matrix field is to use absolutely continuous line representatives: at a differentiability point, the quadratic form on every vector in the kernel has a local minimum zero, and polarization gives the assertion. The formula is measurable and gives

$$\sum_i \|H_i R\|_{\text{HS}}^2 \leq 4\|DR\|^2.$$

Set

$$e = \|DR\|_2^2, \quad a = \sum_i \mathbb{E} \text{tr}(R^2 H_i^2), \quad c = \sum_i \mathbb{E} \| [H_i, R] \|_{\text{HS}}^2, \quad d = \|D\Psi\|_2^2.$$

The Sylvester equation implies the exact identities

$$e = a - \frac{c}{4}, \quad \sum_i \mathbb{E} \text{tr}(R H_i R H_i) = e - \frac{c}{4}. \tag{6.9}$$

Differentiating $R^2 = \Psi\Psi^*$ in the weak form sense gives

$$e = \sum_i \mathbb{E} \langle D_i \Psi, (H_i)_{(1)} \Psi \rangle = \sum_i \mathbb{E} \langle D_i \Psi, (H_i)_{\text{av}} \Psi \rangle.$$

For example, the first equality follows by pairing $D_i(R^2)$ with $H_i/2$; all products are integrable by the displayed bound. The second uses the symmetry of each $D_i \Psi$ on its existing slots. Lemma 6.2 and Cauchy–Schwarz yield

$$e^2 \leq d \left(e - \frac{c}{4} + \frac{2a}{k} \right).$$

Since $a \leq 4e$, $d \rightarrow 1$, and $\liminf e \geq 1$, this first forces $e \rightarrow 1$ and then $c \rightarrow 0$; Equation (6.9) gives $a \rightarrow 1$. Finally,

$$\sum_i \|D_i R - H_i R\|_2^2 = \frac{c}{4}, \quad \sum_i \|D_i \Psi - (H_i)_{(1)} \Psi\|_2^2 = d + a - 2e.$$

Both tend to zero. Symmetry gives the assertion for the other slots. $\square$

6.2 Bounded columns and spectral blocks

We next make the column $(H_i)_i$ bounded in operator norm. This requires a weighted tail estimate; an unweighted truncation would be inadequate when R has many small eigenvalues. Throughout the modifications below, H_i denotes the current column; the exact Sylvester identity (6.6) refers to the original column. The approximate derivative-action and balance estimates are preserved as stated in the following lemmas.

Lemma 6.4 (Balance and bounded columns). *The matrices in Lemma 6.3 satisfy*

$$\mathbb{E} \left\| \sum_i H_i \rho H_i - \rho \right\|_1 = o(1), \quad \mathbb{E} \left\| R \left(\sum_i H_i^2 \right) R - \rho \right\|_1 = o(1). \quad (6.10)$$

They can be replaced by symmetric matrix fields, still denoted H_i , so that Equation (6.8), the commutator conclusion of Equation (6.7), and Equation (6.10) hold, and

$$S := \sum_i H_i^2 \leq 4\text{Id}, \quad \mathbb{E} \|(S - 1)R\|_{\text{HS}}^2 = o(1). \quad (6.11)$$

The total change satisfies $\sum_i \mathbb{E} \|(H_i^{\text{new}} - H_i^{\text{old}})R\|_{\text{HS}}^2 = o(1)$.

Proof. We repeatedly use the dimension-free inequality

$$\|XX^* - YY^*\|_1 \leq (\|X\|_{\text{HS}} + \|Y\|_{\text{HS}})\|X - Y\|_{\text{HS}}. \quad (6.12)$$

By Equation (6.8), the first sum in Equation (6.10) is close in expected trace norm to the one-slot reduction of $D\Psi$, with the derivative index included among the column indices. The derivative coupling identifies this with

$$\frac{1}{kI} \sum_{k \leq l < 2k} T_{l+1} T_{l+1}^*.$$

The trace-norm error from the coupling is $o(1)$ by Equation (6.12) and Equation (6.2). Shifting the band changes its reduction only by its two endpoints, whose expected trace-norm cost is

$$\frac{\|T_{2k}\|_2^2 + \|T_k\|_2^2}{kI} = O(k^{-1}).$$

This proves the first balance equation. Comparing the arrays of matrices $(H_i R)_i$ and $(R H_i)_i$ proves the second by the same inequality and the summed commutator estimate.

For the original, possibly unbounded column write $\mathbf{H}v = (H_i v)_i$, $S = \mathbf{H}^* \mathbf{H}$, and $J = (1 + S)^{-1}$. Let $\mathbf{K} = \mathbf{H}R - R^{\oplus n} \mathbf{H}$, so that $\mathbb{E} \|\mathbf{K}\|_{\text{HS}}^2 = o(1)$. Algebra gives

$$[S, R] = \mathbf{H}^* \mathbf{K} - \mathbf{K}^* \mathbf{H}, \quad [R, J] = J[S, R]J.$$

The bounds

$$\|\mathbf{H}J\mathbf{H}^*\|_{\text{op}} \leq 1, \quad \|\mathbf{H}J\|_{\text{op}} \leq \frac{1}{2}, \quad \|J\|_{\text{op}} \leq 1$$

therefore imply $\|\mathbf{H}[R, J]\|_{\text{HS}} \leq 2\|\mathbf{K}\|_{\text{HS}}$. For every fixed $f(x) = a + b/(1 + x)$, it follows that

$$\mathbb{E} \|\mathbf{H}[f(S), R]\|_{\text{HS}}^2 = o(1).$$

The squared norms of $\mathbf{H}f(S)R$ and $\mathbf{H}Rf(S)$ differ by $o(1)$: the latter has bounded mean-square norm, and the preceding bound controls their difference. Testing the second balance equation against the bounded matrix $f(S)^2$ consequently gives

$$\mathbb{E} \text{tr}(R^2(S - 1)f(S)^2) = o(1).$$

Use $f = 1, 1 + J, 1 - J$ and $(S - 1)^2/(1 + S) = (S - 1) - 2(S - 1)J$ to obtain

$$\mathbb{E} \text{tr} \left(R^2 \frac{(S - 1)^2}{1 + S} \right) = o(1). \quad (6.13)$$

All these expressions are integrable: their absolute values are bounded by a constant times $\text{tr}(R^2(1 + S))$.

Let $Q = \mathbf{1}_{(4, \infty)}(S)$ and $P = 1 - Q$. On this spectral region, $(S - 1)^2/(1 + S)$ dominates both a positive constant and a positive multiple of S . Thus

$$\mathbb{E} \text{tr}(R^2 Q) + \mathbb{E} \text{tr}(R^2 S Q) = o(1).$$

The input truncation costs $\sum_i \mathbb{E} \|H_i Q R\|_{\text{HS}}^2 = \mathbb{E} \text{tr}(R^2 S Q) = o(1)$. The output truncation costs

$$\sum_i \mathbb{E} \|Q H_i R\|_{\text{HS}}^2 = \mathbb{E} \text{tr} \left(Q \sum_i H_i \rho H_i \right) \leq \mathbb{E} \text{tr}(Q \rho) + \mathbb{E} \left\| \sum_i H_i \rho H_i - \rho \right\|_1 = o(1).$$

Consequently replacing H_i by $P H_i P$ has total right- R squared cost $o(1)$. The new column obeys $\sum_i (P H_i P)^2 \leq P S P \leq 4P$.

Here is the perturbation accounting needed both now and below. If symmetric $\tilde{H}_i$ have

$$\Delta = \sum_i \mathbb{E} \|(\tilde{H}_i - H_i)R\|_{\text{HS}}^2 = o(1), \quad (6.14)$$

then symmetry gives the same bound with R on the left. Each derivative error in Equation (6.8) changes in norm by at most $\sqrt{\Delta}$, and the summed commutator norm changes by at most $2\sqrt{\Delta}$. The two balance errors change in expected trace norm by at most

$$\left(\left[\sum_i \mathbb{E} \|H_i R\|_{\text{HS}}^2 \right]^{1/2} + \left[\sum_i \mathbb{E} \|\tilde{H}_i R\|_{\text{HS}}^2 \right]^{1/2} \right) \sqrt{\Delta},$$

by Equation (6.12), and this is $o(1)$. The new summed energy is bounded by the old one plus the perturbation in norm. These statements apply to the compression just made.

For the bounded column, $[S, R] = \mathbf{H}^* \mathbf{K} - \mathbf{K}^* \mathbf{H}$ satisfies $\|[S, R]\|_2 \leq 4\|\mathbf{K}\|_2 = o(1)$. Put $T = S - 1$, so $\|T\|_{\text{op}} \leq 3$. The second balance equation says $\mathbb{E}\|RTR\|_1 = o(1)$. Finally,

$$\|TR\|_{\text{HS}}^2 = \text{tr}(TRTR) + \text{tr}(T[T, R]R).$$

The expectation of the first term is at most $3\mathbb{E}\|RTR\|_1$ in absolute value; the second is at most $3\|[T, R]\|_2\|R\|_2 = o(1)$. This proves the last assertion. $\square$

Lemma 6.5 (Commutation and exchange of indices). *For the bounded columns just constructed,*

$$\sum_{i,j} \mathbb{E}\|[H_i, H_j]R\|_{\text{HS}}^2 = o(1), \quad \mathbb{E} \sum_{i,a,b} |(H_i R)_{ab} - (H_a R)_{ib}|^2 = o(1). \quad (6.15)$$

These conclusions also hold after a symmetric perturbation satisfying Equation (6.14), provided the new column has a uniform operator bound.

Proof. Equation (6.8) implies that moving H_i between any two existing slots costs $o(1)$ in summed mean square. Three slots suffice to exchange two factors. In the following chain, each $\simeq$ means that the difference has squared norm $o(1)$ after expectation and summation over i, j :

$$\begin{aligned} (H_i)_{(1)}(H_j)_{(1)}\Psi &\simeq (H_i)_{(1)}(H_j)_{(2)}\Psi = (H_j)_{(2)}(H_i)_{(1)}\Psi \\ &\simeq (H_j)_{(2)}(H_i)_{(3)}\Psi = (H_i)_{(3)}(H_j)_{(2)}\Psi \\ &\simeq (H_i)_{(3)}(H_j)_{(1)}\Psi = (H_j)_{(1)}(H_i)_{(3)}\Psi \\ &\simeq (H_j)_{(1)}(H_i)_{(1)}\Psi. \end{aligned}$$

The equalities use commutation on different slots. Each move costs at most the one-factor error multiplied by the column bound: for any tensor v ,

$$\sum_i \|(H_i)_{(r)}v\|^2 \leq 4\|v\|^2.$$

The squared norm of the sum of the four errors is at most four times their summed squared norms. Thus the total commutator cost on Ψ is $o(1)$. Its cost on Ψ equals its cost on R , since both have reduction R^2 .

For the second conclusion, write $\Psi = RJ_0$, where $J_0 J_0^*$ is the support projection of R . The matrix-valued array $Z_{ia,b} = (H_i R)_{ab} - (H_a R)_{ib}$ has its column index in that support, so $\|ZJ_0\|_{\text{HS}} = \|Z\|_{\text{HS}}$. The entries of ZJ_0 express the exchange of the derivative index and the first tensor index in $(H_i)_{(1)}\Psi$. Equation (6.8) and the approximation of $D\Psi$ by fully symmetric T_{l+1} give the claim. For perturbed matrices, the derivative errors and the slot-moving errors have the bounds established after Equation (6.14); the same four moves use the new uniform column bound. The index-exchange norm itself changes by at most $2\sqrt{\Delta}$. $\square$

We have obtained a bounded column acting almost identically on the symmetric tensor slots. To use Gaussian matrix polynomials, we now put its weighted estimates into a tracial form. The following construction is performed only at the initial point of the diffusion, so a varying eigenbasis will never be differentiated.

Lemma 6.6 (Spectral blocks and a bounded input map). *After enlarging the initial probability space by an independent uniform variable, there are a positive block matrix $R_* = \bigoplus_{\beta} \beta \text{Id}_{\beta}$, with zero on $\ker R$, and symmetric matrices H_i supported on entries whose row, column, and outer index belong to the same positive block, with the following properties. Their total change satisfies Equation (6.14), $R_* \leq R \leq (1 + \epsilon)R_*$ on the positive support for $\epsilon \rightarrow 0$, and the column bound $S \leq 4\text{Id}$ holds. With the positive weighted trace*

$$\tau(U) = \mathbb{E} \sum_{\beta} \beta^2 \text{tr}_{\beta} U_{\beta}, \quad (6.16)$$

where the expectation also includes any fresh noises present, we have $\tau(1) \rightarrow 1$ and

$$\tau((S - 1)^2) = o(1), \quad \sum_{i,j} \tau(|[H_i, H_j]|^2) = o(1), \quad \mathbb{E} \sum_{\beta} \beta^2 \sum_{i,a,b \in \beta} |(H_i)_{ab} - (H_a)_{ib}|^2 = o(1). \quad (6.17)$$

For a vector v in a positive block define $H[v] = \sum_{i \in \beta} v_i H_i$. We can further impose, pointwise,

$$\|H[v]\|_{\text{HS}} \leq 2|v|. \quad (6.18)$$

The bound uses ordinary, unnormalized Hilbert–Schmidt norm. All the derivative-action and balance conclusions above continue to hold for these matrices in the initial coordinates.

Proof. Choose an orthogonal eigenframe O for R at the initial point. Transform all three indices: in that frame the matrices are

$$\hat{H}_i = \sum_j O_{ji} O^* H_j O.$$

This preserves the column bound, all the summed norms, and the index-exchange estimate, since these are Euclidean tensor contractions. Finite-dimensional spectral calculus and an ordered orthonormal basis completion give measurable choices, including at repeated eigenvalues.

Let $h = \log(1 + \epsilon)$ and take an independent θ uniform on $[0, h)$. Group positive eigenvalues in $[e^{\theta+mh}, e^{\theta+(m+1)h})$, $m \in \mathbb{Z}$, and round those in each nonempty block down to $\beta = e^{\theta+mh}$. The zero eigenspace is kept separate. Write

$$\kappa = \sum_i \mathbb{E} \| [H_i, R] \|_{\text{HS}}^2 = o(1), \quad a = \sum_i \mathbb{E} \| H_i R \|_{\text{HS}}^2 = O(1).$$

For $0 < \delta < h$, a pair of positive eigenvalues with $|\log(r_a/r_b)| \geq \delta$ satisfies $r_b^2 \leq C\delta^{-2}(r_a - r_b)^2$, when $\delta \leq 1$. If their logarithmic distance is smaller than δ , the probability that a block boundary separates them is at most δ/h . Consequently the right- R squared cost of deleting entries with row and column in different blocks is bounded by

$$C\delta^{-2}\kappa + \frac{\delta}{h}a + \kappa. \quad (6.19)$$

The final term covers a zero row and positive column directly by the commutator. A zero column has no right- R cost. For example, choose $\delta \rightarrow 0$ slowly enough that $\kappa/\delta^2 \rightarrow 0$, and then $h \rightarrow 0$ slowly enough that $\delta/h \rightarrow 0$.

Deleting entries whose outer index belongs to another block has vanishing cost as well. On the set where i and b have different block labels,

$$|(H_i)_{ab}r_b|^2 \leq 2|(H_i)_{ab}r_b - (H_a)_{ib}r_b|^2 + 2|(H_a)_{ib}r_b|^2.$$

Sum over the indices. The first term is the second defect of Equation (6.15); the second is a row–column block exclusion of the kind already bounded in Equation (6.19). This includes outer indices in the zero eigenspace. Within each block the resulting matrices have the form $P_\beta \hat{H}_i P_\beta$ for $i \in \beta$, and vanish otherwise. Thus

$$\sum_{i \in \beta} (P_\beta \hat{H}_i P_\beta)^2 \leq P_\beta \left(\sum_i \hat{H}_i^2 \right) P_\beta \leq 4P_\beta.$$

They are symmetric and their total change has the cost in Equation (6.14).

For every block matrix X one has the precise comparison

$$\tau(X^* X) \leq \mathbb{E} \|XR\|_{\text{HS}}^2 \leq (1 + \epsilon)^2 \tau(X^* X). \quad (6.20)$$

In particular $\tau(1) \rightarrow 1$. The perturbation bounds following Equation (6.14) prove the derivative, commutator, and balance conclusions for these matrices. The bounded-column argument at the end of Lemma 6.4 again gives $\mathbb{E} \|(S - 1)R\|_{\text{HS}}^2 = o(1)$, so Equation (6.20) proves the first tracial defect. Lemma 6.5 gives the second. For the third, all three indices now lie in one block, and the index-exchange defect is exactly weighted by r_b^2 ; replacing this by β^2 only decreases it. Matrix symmetry and exchange of i, a together give asymptotic symmetry under every permutation of the three indices.

It remains to bound the map from the outer index to matrix coefficients. On each block let

$$Fv = \sum_i v_i H_i, \quad (Gv)_{ab} = (H_b v)_a.$$

The map G has operator norm at most two because $\|Gv\|_{\text{HS}}^2 = \langle v, Sv \rangle \leq 4|v|^2$. Total symmetry just established says

$$\mathbb{E} \sum_\beta \beta^2 \|F - G\|_{\text{HS}(\mathbb{R}^{\dim \beta}, \text{HS})}^2 = o(1).$$

Clip the singular values of F to two, obtaining $F' = FB$ for a real positive contraction B on the input space. Metric projection onto the operator-norm ball in Hilbert–Schmidt norm gives $\|F - F'\|_{\text{HS}} \leq \|F - G\|_{\text{HS}}$; this also follows immediately by singular-value decomposition and the variational characterization of singular values. Thus the block-weighted squared cost is $o(1)$. Writing $H'_i = \sum_j B_{ji} H_j$ shows that every H'_i remains symmetric. For each vector v ,

$$\sum_i \|H'_i v\|^2 \leq \sum_j \|H_j v\|^2,$$

so the column bound is preserved, as is the block support. The weighted cost controls the right- R cost by Equation (6.20). Therefore the derivative and balance errors have exactly the bounds stated after Equation (6.14); index exchange changes by at most twice the perturbation norm, and the four-move argument controls the commutators. Repeating the bounded-column last step and then Equation (6.20) proves all three tracial defects for H'_i . The defining singular-value clipping proves Equation (6.18). $\square$

6.3 Gaussian moments and the frozen diffusion law

The matrices within a block need not commute exactly. The next lemma shows that their Gaussian linear combinations nevertheless have the scalar Gaussian joint moments in the weighted limit. We include the moment bounds because they also justify the unbounded exponential needed for the diffusion law.

Lemma 6.7 (Weighted Gaussian moments). *Let the block matrices and τ be those of Lemma 6.6. For independent standard Gaussian vectors $N_1, \dots, N_q$, independent of the initial data, put $D_r = H[N_r]$. Then, for every fixed word $r_1, \dots, r_m$,*

$$\tau(D_{r_1} \cdots D_{r_m}) \longrightarrow \mathbb{E}(Z_{r_1} \cdots Z_{r_m}), \quad (6.21)$$

where $Z_1, \dots, Z_q$ are independent scalar standard normals. Products of any fixed length can be reordered with error tending to zero in $L^2(\tau)$. For every integer $p \geq 1$ and real $u \geq 0$,

$$\mathbb{E}_{N_r} \operatorname{tr}_\beta D_r^{2p} \leq 4^p (2p-1)!! \operatorname{tr}_\beta 1, \quad \tau(e^{u|D_r|}) \leq 2e^{2u^2} \tau(1). \quad (6.22)$$

If $\mathfrak{h}_j$ is the tensor Hermite polynomial of Lemma 5.5 and $\mathfrak{h}_j^{(1)}$ is its one-dimensional version, then for each fixed j ,

$$\tau \left(\left| \sum_{i_1, \dots, i_j} H_{i_j} \cdots H_{i_1} \mathfrak{h}_j(N_1)_{i_1 \dots i_j} - \mathfrak{h}_j^{(1)}(D_1) \right|^2 \right) = o(1). \quad (6.23)$$

Proof. We first prove a conditional moment bound on one fixed block, allowing a general column bound $S \leq C \operatorname{Id}$. We use the Gaussian trace-moment integration-by-parts argument of [19, Lemma 6.1 and Section 7]; the weighted near-commutation limit will be proved afterward. Write $D = \sum_i N_i H_i$. Gaussian integration by parts gives

$$\mathbb{E}_N \operatorname{tr} D^{2p} = \sum_{j=0}^{2p-2} \mathbb{E}_N \sum_i \operatorname{tr}(H_i D^j H_i D^{2p-2-j}).$$

In an eigenbasis of D with eigenvalues d_a , the absolute value of a summand is bounded by

$$\sum_{i,a,b} |(H_i)_{ab}|^2 |d_b|^j |d_a|^{2p-2-j}.$$

For $p > 1$, the scalar weighted arithmetic-geometric mean inequality bounds the product of powers by

$$\frac{j}{2p-2} |d_b|^{2p-2} + \left(1 - \frac{j}{2p-2}\right) |d_a|^{2p-2}.$$

Since $\sum_{i,b} |(H_i)_{ab}|^2 = S_{aa} \leq C$ and similarly with a, b exchanged, each summand is at most $C \operatorname{tr} |D|^{2p-2}$. For $p = 1$ the assertion is $\operatorname{tr} S \leq C \operatorname{tr} 1$. Induction proves $\mathbb{E}_N \operatorname{tr} D^{2p} \leq C^p (2p-1)!! \operatorname{tr} 1$. After multiplying by the block weights, all fixed moments are uniformly bounded. Also $e^{u|x|} \leq 2 \cosh(ux)$, so the even-moment series gives

$$\tau(e^{u|D|}) \leq 2 \sum_{p \geq 0} \frac{u^{2p} C^p (2p-1)!!}{(2p)!} \tau(1) = 2e^{Cu^2/2} \tau(1).$$

This proves Equation (6.22) with $C = 4$.

For clarity, the norm $\|X\|_{p,\tau} = \tau(|X|^p)^{1/p}$ is formed on the direct sum of the blocks and then averaged over the classical random variables. Schatten Hölder followed by scalar Hölder gives

$$\|X_1 \cdots X_m\|_{p,\tau} \leq \prod_{j=1}^m \|X_j\|_{p_j,\tau}, \quad \sum_j \frac{1}{p_j} = \frac{1}{p}.$$

These bounds use the actual weights β^2 , with no factor depending on a block dimension. The finite-matrix Schatten inequality can itself be obtained by the same singular-value three-lines argument as in Lemma 6.2, interpolating the endpoint bounds with one trace norm and the other norms operator norms, and then applying duality.

For independent N_r, N_s ,

$$\|[D_r, D_s]\|_{2,\tau}^2 = \sum_{i,j} \tau(|[H_i, H_j]|^2) = o(1).$$

All its higher fixed moments are bounded by Hölder and the moment bound just proved. Interpolation between $L^2(\tau)$ and an arbitrarily high fixed $L^p(\tau)$ therefore makes this commutator tend to zero in every fixed finite power norm. Applying Hölder to the factors on either side of a commutator shows that an adjacent transposition in a word of fixed length costs $o(1)$ in $L^2(\tau)$. A finite sequence of transpositions proves the reordering assertion.

Now expand a mixed moment by Gaussian pairings, conditional on the matrices. Pairings must join indices of the same Gaussian vector. For each pair introduce its own independent standard Gaussian vector, shared only between the two factors in that pair. This represents the contracted matrix word exactly as a conditional expectation. Reorder the word until each pair is adjacent; the error is $o(1)$ even in $L^2(\tau)$. Averaging the independent vectors then gives $S^{m/2}$. For each fixed integer b , boundedness of S gives

$$\|S^b - 1\|_{2,\tau} \leq \left(\sum_{a=0}^{b-1} 4^a \right) \|S - 1\|_{2,\tau} = o(1).$$

Each pairing consequently contributes $\tau(1) + o(1) = 1 + o(1)$. Odd moments vanish. The number of admissible pairings is precisely the scalar independent-normal moment, proving Equation (6.21).

Finally expand the tensor Hermite polynomial using its defining exponential $\exp(\langle v, N \rangle - |v|^2/2)$. A term consists of free N indices and a number of signed Kronecker-delta pairs. Represent each delta pair by its own independent Gaussian vector, as above, and move its two matrix factors together. Conditional averaging is contractive in $L^2(\tau)$, so the reordering error remains $o(1)$. The resulting factors S can be replaced by one, even between fixed powers of D_1 , by Hölder and interpolation applied to the bounded matrix $S - 1$. The resulting polynomial is

$$\sum_{b=0}^{\lfloor j/2 \rfloor} \frac{(-1)^b j!}{2^b b! (j-2b)!} D_1^{j-2b} = \mathfrak{h}_j^{(1)}(D_1).$$

There are only finitely many terms for fixed j , proving Equation (6.23). $\square$

Proposition 6.8 (Frozen diffusion law). *Let Y_u be the stationary diffusion with invariant law η and driving Brownian motion B_u , and fix $t > 0$. Make the matrix construction of Lemma 6.6 at Y_0 , using an independent random block offset. In its orthonormal frame put $N = B_t/\sqrt{t}$; conditionally on the initial data and block offset, N is a standard Gaussian vector. Define, in the initial spatial coordinates,*

$$\sigma = \bigoplus_{\beta} \beta^2 \exp(\sqrt{8t} H[N] - 4t \text{Id}_{\beta}), \quad (6.24)$$

and set $\sigma = 0$ on the omitted kernel. Then

$$\mathbb{E} \|R(Y_t) - \sigma^{1/2}\|_{\text{HS}}^2 \longrightarrow 0. \quad (6.25)$$

The matrices defining σ obey the bounded-column and input-map bounds (6.11) and (6.18), the weighted defects (6.17), and the mixed moment law of Lemma 6.7.

Proof. The proof has three steps: replace the added tensor indices in the hierarchy expansion by products of the initial matrices, identify their Wick contractions, and sum the resulting scalar Hermite series. All limits involving the degree of that series are taken after the counterexample-sequence limit, with t fixed.

The derivative-action estimate and the hierarchy coupling give

$$\frac{1}{kI} \sum_{k \leq l < 2k} \sum_i \|T_{l+1,i} - (H_i)_{(1)} T_l\|_2^2 = o(1). \quad (6.26)$$

It also holds when the band is shifted by any fixed integer $a \geq 0$. Indeed the same matrices have been constructed using levels k through $2k - 1$. Removing or adding at most a levels in the error sum costs $O_C(a/k)$, since the unnormalized squared norm of each one-step error is at most

$$2\|T_{l+1}\|_2^2 + 2C\|T_l\|_2^2 = O_C(I),$$

by the bounded column and the diagonal endpoint control. This also explains why the matrices may already include the block and input clipping modifications: their cost on Ψ is the right- R cost in Equation (6.14).

Iterating Equation (6.26), for each fixed j we may replace the extra-index tensor T_{l+j} by

$$(i_1, \dots, i_j) \mapsto (H_{i_j})_{(1)} \cdots (H_{i_1})_{(1)} T_l$$

in the normalized band mean-square norm, with error $o(1)$. More explicitly, if e_a is the root mean-square error in the one-step estimate on the band shifted by a , the root error after j steps is at most

$$\sum_{a=0}^{j-1} C^{a/2} e_{j-1-a}.$$

Here multiplication by a column of H_i costs at most $\sqrt{C}$ in the summed coefficient norm. Each $e_a = o(1)$ at fixed a , so the displayed bound vanishes. Contraction with $\mathfrak{h}_j(N)$ multiplies squared coefficient norm by at most $j!$: it first symmetrizes the contracted indices, an orthogonal projection, and Gaussian Hermite orthogonality then gives the factor $j!$.

Apply Lemma 5.5 to the band of levels in Ψ . For each fixed m , the preceding replacements and Equation (6.23) give

$$\Psi(Y_t) = P_{m,t}(D)\Psi(Y_0) + \mathcal{E}_{m,t}, \quad D = H[N],$$

where

$$P_{m,t}(x) = e^{-t} \sum_{j=0}^m \frac{(2t)^{j/2}}{j!} \mathfrak{h}_j^{(1)}(x)$$

and

$$\limsup \|\mathcal{E}_{m,t}\|_2 \leq \frac{(2t)^{(m+1)/2}}{\sqrt{(m+1)!}}. \quad (6.27)$$

To justify the conversion of a tracial polynomial error to the norm of its action on $\Psi(Y_0)$, use $\|X\Psi(Y_0)\|_{\text{HS}} = \|XR\|_{\text{HS}}$ and Equation (6.20). The band version of the remainder bound follows from the uniform finite-list diagonal fixed at the start of the section. Its right side tends to zero as $m \rightarrow \infty$.

The spectral measures of D averaged with τ have total mass tending to one, converge in moments to the standard Gaussian by Equation (6.21), and have uniformly bounded exponential

moments at every fixed linear parameter by Equation (6.22). These facts imply weak convergence to the standard Gaussian and convergence of the integrals of every continuous function bounded by a polynomial times a fixed linear exponential. Here is a direct justification: the second moment gives tightness; the higher moment bounds give uniform integrability of each polynomial, so every subsequential limit has the Gaussian moments. Its exponential moments determine its characteristic function by its power series, hence determine the law. Using a larger exponential parameter makes the tails of any of the specified test functions uniformly small.

The scalar Hermite generating identity gives, with Z standard normal,

$$e^{-t} \sum_{j \geq 0} \frac{(2t)^{j/2}}{j!} \mathfrak{h}_j^{(1)}(Z) = \exp(\sqrt{2t} Z - 2t)$$

in squared mean. In fact its squared tail norm is $e^{-2t} \sum_{j > m} (2t)^j / j!$. Define the matrix multiplier

$$M = \exp(\sqrt{2t} D - 2t \text{Id}).$$

The just-proved convergence of spectral integrals, applied with each fixed m , yields

$$\lim \tau(|M - P_{m,t}(D)|^2) = e^{-2t} \sum_{j > m} \frac{(2t)^j}{j!}, \quad \tau(M^2) \rightarrow 1. \quad (6.28)$$

Together with Equations (6.27) and (6.20), first sending the sequence index and then m to infinity, this proves

$$\mathbb{E} \|\Psi(Y_t) - M\Psi(Y_0)\|_{\text{HS}}^2 = o(1).$$

There is no operator-norm bound on M in this argument; only its weighted trace moments are used.

Write $\Psi(Y_0) = R J_0$, with $J_0 J_0^*$ equal to the support projection. On each block $(R - R_*)^2 \leq \epsilon^2 R_*^2$, and R_* is scalar there. By cyclicity and positivity of trace,

$$\mathbb{E} \|M(R - R_*) J_0\|_{\text{HS}}^2 \leq \epsilon^2 \tau(M^2) = o(1).$$

Thus $\Psi(Y_t)$ is close in squared mean to $M R_* J_0$. Its reduction is exactly

$$(M R_* J_0)(M R_* J_0)^* = M R_*^2 M = \bigoplus_{\beta} \beta^2 \exp(\sqrt{8t} H[N] - 4t \text{Id}_{\beta}) = \sigma.$$

Finally Equation (6.3) transfers the approximation of these rectangular matrices to the approximation of their positive reduced square roots, proving Equation (6.25). $\square$

The resulting description retains the weights β^2 and the ordinary trace on every block. It supplies both a bounded matrix column and the input bound of Equation (6.18), while allowing any number of positive blocks and eigenvalues arbitrarily close to zero. These are the properties needed for the covariance observations in Section 7.

7 Covariance observations and the contradiction

The matrices constructed in Section 6 describe fluctuations whose information content is incompatible with the logarithmic Sobolev inequality of the stationary law. We compare two bounds for the same Gaussian covariance observation. The first follows from logarithmic Sobolev and is

universal after integration over its scale. The second uses the frozen matrices to recover a scalar lognormal covariance experiment. Integration over scale is essential: it assigns to every spectral block exactly the weight used by the tracial limit, regardless of the sizes of its eigenvalues.

Throughout this section, limits refer to the counterexample sequence. All observation times, caps, compact parameter intervals, and polynomial degrees are fixed before taking that limit. We suppress its index. Recall that Y_t is stationary with law η , that $R(\eta) \leq \lambda^{-1}$ with $\lambda \rightarrow 1$, and that R without a measure argument denotes the positive matrix field. In particular, Equations (6.2) and (6.7) give

$$\mathbb{E} \operatorname{tr} R^2 \rightarrow 1, \quad \mathbb{E} |DR|^2 \rightarrow 1. \quad (7.1)$$

To distinguish the original Sylvester matrices from the subsequently modified block matrices, denote the former by $\hat{H}_i$. Equations (6.6) and (6.7) assert

$$D_i R = \frac{1}{2}(\hat{H}_i R + R \hat{H}_i), \quad \sum_i \mathbb{E} \|\hat{H}_i, R\|_{\text{HS}}^2 = o(1). \quad (7.2)$$

Only the upper bound below uses these original matrices.

Fix $L \geq 1$ and define

$$f_L(q) = \frac{q}{1 + q/L}, \quad C_T(y) = \operatorname{Id} + f_L(TR(y)^2), \quad Z_T = C_T(Y_t)^{1/2}G, \quad T > 0, \quad (7.3)$$

where G is an independent standard Gaussian vector. Thus $\operatorname{Id} \leq C_T \leq (1 + L)\operatorname{Id}$. We write $H(P | Q)$ for relative entropy and $\mathfrak{l}(U; V)$ for mutual information, including the usual averaged conditional version when a conditioning variable is displayed.

7.1 The integrated information upper bound

Lemma 7.1 (Integrated spectral multipliers). *For $v, w \geq 0$, not both zero, put*

$$M_L(v, w) = (v + w)^2 \int_0^\infty \frac{dT}{(1 + Tv^2/L)(1 + Tw^2/L)(1 + (1 + 1/L)Tv^2)(1 + (1 + 1/L)Tw^2)}. \quad (7.4)$$

Set $M_L(0, 0) = 0$. Then

$$M_L(v, w) \leq 4 \log(1 + L) \quad \text{for all } v, w \geq 0, \quad M_L(v, w) \leq 9 \quad \text{if } \frac{1}{2} \leq v/w \leq 2. \quad (7.5)$$

The second bound concerns positive v, w and is independent of L .

Proof. Write $a = 1/L$ and $b = 1 + 1/L$, so $b - a = 1$. If $r = \max(v, w) > 0$, discard the two denominator factors associated with the smaller eigenvalue and substitute $q = Tr^2$. This gives

$$M_L(v, w) \leq 4 \int_0^\infty \frac{dq}{(1 + aq)(1 + bq)} = 4 \log(b/a) = 4 \log(1 + L).$$

For the identity in the middle, integration up to Q gives $\log(1 + bQ) - \log(1 + aQ)$, which tends to $\log(b/a)$. If v, w are positive and comparable as in the statement, let $r_0 = \min(v, w)$. Discard the two factors with coefficient a , and use $b \geq 1$. The remaining integral is at most $\int_0^\infty (1 + Tr_0^2)^{-2} dT = r_0^{-2}$. Since $v + w \leq 3r_0$, this proves the second bound. $\square$

Proposition 7.2 (Universal integrated upper bound). *There is an absolute constant C_0 such that, for every fixed $t > 0$ and $L \geq 1$, the observations in Equation (7.3) satisfy*

$$\limsup \int_0^\infty \mathfrak{l}(Y_t; Z_T) \frac{dT}{T^2} \leq C_0. \quad (7.6)$$

In particular, one may take $C_0 = 9/4$.

Proof. Let $p_T(z \mid y)$ denote the density of the centered Gaussian with covariance $C_T(y)$, and let $p_T(z)$ be its marginal density. For each output z , the posterior density with respect to η is $q_z(y) = p_T(z \mid y)/p_T(z)$. Applying $2\lambda H \leq \mathcal{I}$ to this density and then averaging in z gives

$$2\lambda \mathbb{I}(Y_t; Z_T) \leq \mathbb{E} |D_y \log p_T(Z_T \mid Y_t)|^2. \quad (7.7)$$

Here and below the expectation on the right includes the observation noise. The posterior normalization does not depend on y and therefore does not contribute to the derivative.

We compute that derivative explicitly. For any positive covariance C and its derivative $D_i C$, differentiation of its determinant and inverse yields

$$D_i \log p(z \mid y) = \frac{1}{2} \left(z^* C^{-1} (D_i C) C^{-1} z - \text{tr}(C^{-1} D_i C) \right).$$

Conditional on y , set $B_i = C^{-1/2} (D_i C) C^{-1/2}$ and $g = C^{-1/2} z$. The vector g is standard Gaussian. Expanding its fourth moments gives $\mathbb{E}(g^* B_i g - \text{tr} B_i)^2 = 2 \text{tr} B_i^2$; indeed the three Gaussian pairings give $(\text{tr} B_i)^2 + 2 \text{tr} B_i^2$ before subtracting the mean. Consequently Equation (7.7) becomes

$$\mathbb{I}(Y_t; Z_T) \leq \frac{1}{4\lambda} \mathbb{E} \sum_i \text{tr} (C_T^{-1} D_i C_T C_T^{-1} D_i C_T). \quad (7.8)$$

These calculations also apply to the Sobolev matrix field R . To see this without assuming additional regularity, fix T, L and the finite dimension of one member of the sequence. The map $R \mapsto C_T$ has bounded derivative on positive semidefinite matrices, and its eigenvalues lie in $[1, 1 + L]$. For each fixed z , the map $C \mapsto p(z \mid C)^{1/2}$ has bounded derivative on this covariance interval. The Sobolev chain rule therefore puts $q_z^{1/2}$ in the form domain; $p_T(z) > 0$ and is a harmless fixed normalization. The Sobolev approximation at the opening of Section 2 extends the logarithmic Sobolev inequality to this form-domain function. The squared derivatives integrate in z to the nonnegative finite expression in Equation (7.8). Approximation in the form domain and Tonelli thus justify both that equation and the subsequent scale integration. The finite-dimensional bounds used for form-domain membership do not enter its numerical coefficient.

In an eigenbasis of R , write v, w for the row and column eigenvalues. The covariance function is

$$c_T(v) = 1 + f_L(Tv^2) = \frac{1 + bTv^2}{1 + aTv^2}, \quad a = 1/L, \quad b = 1 + 1/L.$$

Its divided difference, including its derivative at $v = w$, is

$$\frac{c_T(v) - c_T(w)}{v - w} = \frac{T(v + w)}{(1 + aTv^2)(1 + aTw^2)}.$$

Multiplying its square by $c_T(v)^{-1} c_T(w)^{-1}$ and integrating against dT/T^2 gives exactly the multiplier in Equation (7.4) for the squared entry $|(D_i R)_{vw}|^2$. When $v = w = 0$, the covariance derivative is zero, which explains the value assigned to $M_L(0, 0)$.

It remains to obtain a bound independent of the fixed cap. For noncomparable pairs with at least one positive eigenvalue, Equation (7.2) gives entrywise

$$|(D_i R)_{vw}|^2 = \frac{(v + w)^2}{4(v - w)^2} |[\hat{H}_i, R]_{vw}|^2 \leq \frac{9}{4} |[\hat{H}_i, R]_{vw}|^2.$$

This includes a pair with exactly one zero eigenvalue. Their total expected derivative energy is therefore $o(1)$. The kernel–kernel derivative is zero almost everywhere, as established in the Sylvester construction, and in any case has zero covariance multiplier. Lemma 7.1 now yields

$$\int_0^\infty \mathfrak{l}(Y_t; Z_T) \frac{dT}{T^2} \leq \frac{1}{4\lambda} \left(9\mathbb{E}|DR|^2 + 4\log(1+L) o(1) \right).$$

Take the sequence limit at this fixed L , and use Equation (7.1). Stationarity makes the bound independent of t . This proves the proposition. In particular, the cap-dependent bound on noncomparable pairs is used only before its vanishing error has been removed. $\square$

7.2 Stability on the entire scale interval

We next transfer observations to the frozen covariance of Proposition 6.8. A bound on a compact interval of T would not suffice, because the useful scale of a block depends on its eigenvalue. The following comparison holds on all positive scales.

Lemma 7.3 (Integrated covariance stability). *For $T > 0$ let $h_T(v) = \sqrt{1 + f_L(Tv^2)}$, $v \geq 0$, and put*

$$K_L = \int_0^\infty \frac{dq}{(1 + q/L)^3 (1 + (1 + 1/L)q)}.$$

Then $0 < K_L \leq \log(1 + L)$, and for arbitrary positive semidefinite matrices A, B of the same size,

$$\int_0^\infty \|h_T(A) - h_T(B)\|_{\text{HS}}^2 \frac{dT}{T^2} \leq K_L \|A - B\|_{\text{HS}}^2. \quad (7.9)$$

Let σ be the frozen covariance in Proposition 6.8, and couple

$$\tilde{Z}_T = (\text{Id} + f_L(T\sigma))^{1/2} G$$

to Z_T using the same independent Gaussian. For every fixed t, L ,

$$\int_0^\infty \mathbb{E}|Z_T - \tilde{Z}_T|^2 \frac{dT}{T^2} = o(1). \quad (7.10)$$

Proof. Direct differentiation gives

$$h'_T(v) = \frac{Tv}{(1 + Tv^2/L)^{3/2} (1 + (1 + 1/L)Tv^2)^{1/2}}.$$

For $v > 0$, substitution $q = Tv^2$ therefore shows

$$\int_0^\infty |h'_T(v)|^2 \frac{dT}{T^2} = K_L.$$

Discarding two of the three factors $1 + q/L$ bounds K_L by the integral evaluated in Lemma 7.1, namely $\log(1 + L)$. If $v \geq w \geq 0$, the fundamental theorem of calculus and Cauchy–Schwarz give

$$|h_T(v) - h_T(w)|^2 \leq (v - w) \int_w^v |h'_T(r)|^2 dr.$$

Integrating in T and using Tonelli proves the scalar bound $K_L(v - w)^2$. The value of the derivative integral at the single endpoint $r = 0$ does not affect this argument, so it covers $w = 0$.

Choose an orthonormal eigenbasis (e_a) of A with eigenvalues v_a and an orthonormal eigenbasis (f_b) of B with eigenvalues w_b . The entry of their functional-calculus difference between these two bases is

$$\langle e_a, (h_T(A) - h_T(B))f_b \rangle = (h_T(v_a) - h_T(w_b))\langle e_a, f_b \rangle.$$

Multiply the scalar bound by $|\langle e_a, f_b \rangle|^2$ and sum over a, b . The sum on the right is $K_L \|A - B\|_{\text{HS}}^2$, proving Equation (7.9). Finally, conditional on all data except G , Gaussian isometry gives

$$\mathbb{E}_G |Z_T - \tilde{Z}_T|^2 = \|h_T(R(Y_t)) - h_T(\sigma^{1/2})\|_{\text{HS}}^2.$$

Apply the matrix bound and the squared-mean convergence in Proposition 6.8. The resulting factor K_L is fixed while the sequence limit is taken. $\square$

7.3 Conditional information and blockwise tests

Let $\mathcal{D}$ be the initial position, random spectral partition, chosen orthonormal frame, and frozen block matrices from Proposition 6.8. Block formulas below use components in this $\mathcal{D}$ -measurable frame. Every information term $\mathfrak{l}(Y_t; Z_T)$ continues to refer to the original observation in ambient coordinates; conditional on $\mathcal{D}$, conversion to the block coordinates is invertible. Orthogonal invariance makes the components of the fresh Gaussian noise conditionally standard and independent of the diffusion data. Conditional on $\mathcal{D}$, $N = B_t/\sqrt{t}$ is standard Gaussian. On a positive block with rounded eigenvalue β , write $H[v] = \sum_{i \in \beta} v_i H_i$. The frozen covariance is

$$\sigma_\beta = \beta^2 \exp(\sqrt{8t} H[N] - 4t \text{Id}). \quad (7.11)$$

All H_i here are the modified matrices supported on their initial blocks. Their bounds, from Equations (6.18) and (6.17), will be used in the form

$$S = \sum_i H_i^2 \leq C \text{Id}, \quad \|H[v]\|_{\text{HS}}^2 \leq C_1 |v|^2, \quad (7.12)$$

$$\tau(\text{Id}) \longrightarrow 1, \quad \tau((S - \text{Id})^2) = o(1), \quad \sum_{i,j} \tau(|[H_i, H_j]|^2) = o(1). \quad (7.13)$$

The constants $C, C_1 \geq 1$ are absolute. The entries of the three-index array $(H_i)_{ab}$ are also asymptotically symmetric in all three indices in the weighted squared norm. As before,

$$\tau(U) = \mathbb{E} \sum_\beta \beta^2 \text{tr}_\beta U_\beta,$$

where expectation includes any Gaussian variables appearing in U . No trace in this definition or in Equation (7.12) is normalized by block dimension.

Lemma 7.4 (Conditional Ornstein–Uhlenbeck score). *Let N be a standard Gaussian vector conditional on $\mathcal{D}$, and let Z be any observation for which $\mathfrak{l}(N; Z \mid \mathcal{D}) < \infty$. With independent standard Gaussian G' set*

$$N_u = \alpha N + \sqrt{1 - \alpha^2} G', \quad \alpha = e^{-u}, \quad u > 0.$$

The relative score of its posterior, with respect to the standard Gaussian marginal, is

$$D \log \frac{p_{N_u|Z, \mathcal{D}}}{\gamma}(N_u) = \frac{\alpha}{1 - \alpha^2} \mathbb{E}[N - \alpha N_u \mid N_u, Z, \mathcal{D}]. \quad (7.14)$$

For every compact interval $\mathcal{J} \subset (0, \infty)$,

$$\mathsf{I}(N; Z \mid \mathcal{D}) \geq \int_{\mathcal{J}} \frac{\alpha^2}{(1 - \alpha^2)^2} \mathbb{E} |\mathbb{E}[N - \alpha N_u \mid N_u, Z, \mathcal{D}]|^2 du. \quad (7.15)$$

For the covariance observation in Equation (7.3),

$$\mathsf{I}(Y_t; Z_T) \geq \mathsf{I}(N; Z_T \mid \mathcal{D}). \quad (7.16)$$

Proof. Put $q = 1 - \alpha^2$. Given a posterior law $\pi(dn)$ of N at fixed $(Z, \mathcal{D})$, the density at y after adding the OU noise is the integral of $(2\pi q)^{-d/2} \exp(-|y - \alpha n|^2/(2q))$ against π . Differentiating this kernel and dividing by its integral gives

$$D \log p(y \mid Z, \mathcal{D}) = - \frac{y - \alpha \mathbb{E}[N \mid N_u = y, Z, \mathcal{D}]}{q}.$$

The score of the standard Gaussian is $-y$. Subtracting it proves Equation (7.14). Kernel differentiation is justified first with N truncated; on compact positive time intervals its derivatives are bounded by Gaussian kernels times polynomials, and truncation can then be removed by conditional second moments.

The OU semigroup has generator $\Delta - y \cdot D$. For a positive posterior density r_u relative to γ , integration by parts gives

$$\frac{d}{du} \int r_u \log r_u d\gamma = - \int \frac{|Dr_u|^2}{r_u} d\gamma.$$

This is the entropy dissipation calculation of Lemma 2.3; positive-time smoothing, truncation of the density and spatial cutoffs justify it for finite-entropy posteriors. Averaging over $(Z, \mathcal{D})$ and integrating over $\mathcal{J}$ bounds its nonnegative right side by the initial conditional information. Substitution of the score formula proves Equation (7.15).

For the last assertion, observation noise is independent of the initial data and diffusion path, and its law depends on them only through Y_t . Hence $(\mathcal{D}, N) \rightarrow Y_t \rightarrow Z_T$ is a Markov chain. The chain rule and data processing give

$$\mathsf{I}(N; Z_T \mid \mathcal{D}) \leq \mathsf{I}(\mathcal{D}, N; Z_T) \leq \mathsf{I}(Y_t; Z_T).$$

These informations are finite for each member of the sequence: the covariance interval $[\text{Id}, (1+L)\text{Id}]$ bounds the output entropy from above by that of a Gaussian of covariance $(1+L)\text{Id}$ and bounds its conditional entropy from below by that of a standard Gaussian. Thus $\mathsf{I}(Y_t; Z_T) \leq (d/2) \log(1+L)$ in dimension d . This finiteness assertion is used only to justify the identities. $\square$

Lemma 7.5 (Block tests and the scale Jacobian). *Fix compact intervals $\mathcal{J}, \mathcal{X} \subset (0, \infty)$, and write $\kappa(u) = \alpha^2/(1 - \alpha^2)^2$, where $\alpha = e^{-u}$. Let $P(u, x, a, b)$ be a real polynomial, evaluated on matrices with any one fixed ordering of each monomial. On each positive block define*

$$A_\beta = H_\beta[N_\beta], \quad A_{u,\beta} = H_\beta[(N_u)_\beta], \quad B_{\beta,x} = H_\beta[(Z_{x/\beta^2})_\beta],$$

and $P_{\beta,u,x} = P(u, x, A_{u,\beta}, B_{\beta,x})$. Then

$$\begin{aligned} \int_0^\infty \mathsf{I}(Y_t; Z_T) \frac{dT}{T^2} &\geq \int_{\mathcal{J}} \int_{\mathcal{X}} \kappa(u) \mathbb{E} \sum_{\beta} \beta^2 \text{tr}_{\beta} (2(A_{\beta} - \alpha A_{u,\beta}) P_{\beta,u,x} \\ &\quad - C_1 P_{\beta,u,x}^* P_{\beta,u,x}) \frac{dx}{x^2} du. \end{aligned} \quad (7.17)$$

Here the constant C_1 is the absolute constant in Equation (7.12).

Proof. For a block, consider the linear map $\mathcal{H} : v \mapsto H[v]$ between its Euclidean coordinate space and its matrix space with the ordinary Hilbert–Schmidt norm. Its adjoint is

$$(\mathcal{H}^*Q)_i = \text{tr}(H_i Q),$$

because H_i is real symmetric. Taking the adjoint in Equation (7.12) gives

$$\sum_i |\text{tr}(H_i Q)|^2 \leq C_1 \text{tr}(Q^* Q). \quad (7.18)$$

This step has no factor involving the dimension of the block.

At fixed u, T , select precisely those positive blocks for which $x = T\beta^2 \in \mathcal{X}$. On a selected block set

$$V_i = \text{tr}_\beta(H_i P(u, T\beta^2, H[(N_u)_\beta], H[(Z_T)_\beta])), \quad i \in \beta,$$

and set all other components, including those in the kernel, equal to zero. This vector is measurable with respect to $(N_u, Z_T, \mathcal{D})$. If $m = \mathbb{E}[N - \alpha N_u \mid N_u, Z_T, \mathcal{D}]$, completing the square gives

$$\mathbb{E}|m|^2 \geq 2\mathbb{E}(N - \alpha N_u) \cdot V - \mathbb{E}|V|^2.$$

The coordinates of distinct selected blocks are orthogonal. Their costs therefore add, and Equation (7.18) bounds the cost by C_1 times the sum of their unnormalized squared matrix norms. Their linear terms contract exactly to

$$(N - \alpha N_u) \cdot V = \sum_{\beta \text{ selected}} \text{tr}_\beta((H[N_\beta] - \alpha H[(N_u)_\beta])P).$$

Substitute these two statements in Lemma 7.4 and integrate in T .

For each realized positive block separately, the change of variable $x = T\beta^2$ has the exact Jacobian

$$T = \frac{x}{\beta^2}, \quad dT = \frac{dx}{\beta^2}, \quad \boxed{\frac{dT}{T^2} = \beta^2 \frac{dx}{x^2}}. \quad (7.19)$$

Thus the block selection becomes $x \in \mathcal{X}$, and the coefficient of each block trace becomes precisely β^2 . This proves Equation (7.17).

For completeness, all its polynomial terms are integrable on the compact parameter rectangle. The moment bounds of Lemma 6.7 control A_β and $A_{u,\beta}$ under the weighted trace. Conditional on the initial data, diffusion path, and OU noise, $B_{\beta,x}$ is a Gaussian matrix series whose vector covariance is at most $(1 + L)\text{Id}$. If its coefficients are $K_j = \sum_i Q_{ij} H_i$ for a square root Q of that covariance, then for every vector v

$$\sum_j |K_j v|^2 \leq (1 + L) \sum_i |H_i v|^2 \leq C(1 + L)|v|^2.$$

The same Gaussian even-moment proof therefore gives, for every fixed integer $p \geq 1$,

$$\mathbb{E}_G \text{tr}_\beta |B_{\beta,x}|^{2p} \leq [C(1 + L)]^p (2p - 1)!! \text{tr}_\beta \text{Id}. \quad (7.20)$$

Trace Hölder bounds each mixed word by its factors' moments. After multiplying by β^2 and summing, the right side is a constant times $\tau(\text{Id})$, which is finite and bounded along the sequence. The measure $\kappa(u) du dx/x^2$ is finite on the chosen rectangle. Tonelli applies to the square terms, while these moment bounds justify Fubini for the linear terms. One can equivalently first truncate V in the conditional-mean inequality and then remove that truncation in this integrated squared norm. $\square$

7.4 Transfer of the polynomial tests to scalar observations

The scale substitution has now produced the tracial weights of the matrix limit. We next identify that limit for the matrices containing the observation. Two replacements are needed: first replace Z_T by its frozen version on the whole scale interval, and then use the approximate symmetry of the three-index array to move a bounded matrix function through $H[\cdot]$.

Lemma 7.6 (Replacement in integrated polynomial tests). *In the notation of Lemma 7.5, put $\tilde{B}_{\beta,x} = H_\beta[(\tilde{Z}_{x/\beta^2})_\beta]$. For every fixed polynomial in $u, x, A_\beta, A_{u,\beta}, B_{\beta,x}$ and their adjoints, its weighted trace integral over $\mathcal{J} \times \mathcal{X}$ changes by $o(1)$ if $B_{\beta,x}$ is replaced by $\tilde{B}_{\beta,x}$.*

Proof. The flattening bound and the inverse substitution in Equation (7.19) give

$$\begin{aligned} & \mathbb{E} \sum_{\beta} \beta^2 \int_{\mathcal{X}} \|B_{\beta,x} - \tilde{B}_{\beta,x}\|_{\text{HS}}^2 \frac{dx}{x^2} \\ &= \int_0^\infty \mathbb{E} \sum_{\beta: T\beta^2 \in \mathcal{X}} \|H_\beta[(Z_T - \tilde{Z}_T)_\beta]\|_{\text{HS}}^2 \frac{dT}{T^2} \\ &\leq C_1 \int_0^\infty \mathbb{E} |Z_T - \tilde{Z}_T|^2 \frac{dT}{T^2} = o(1), \end{aligned}$$

by Lemma 7.3. All projected errors are summed before this comparison; no small block weight is divided out. Multiplication by the bounded weight $\kappa(u)$ and integration in the compact interval $\mathcal{J}$ preserve $o(1)$.

To pass from this squared norm to arbitrary fixed products, include the parameter integrals in the finite tracial functional

$$\mathcal{T}(U) = \int_{\mathcal{J}} \int_{\mathcal{X}} \kappa(u) \mathbb{E} \sum_{\beta} \beta^2 \text{tr}_{\beta} U_{\beta,u,x} \frac{dx}{x^2} du.$$

Every matrix in the statement has bounded $L^p(\mathcal{T})$ norm for every fixed finite p . For B this was proved in Equation (7.20); for $\tilde{B}$ the same proof applies because its covariance is also at most $(1+L)\text{Id}$. The remaining matrices are covered by the Gaussian moment lemma. For any $q > p \geq 2$, interpolation with $1/p = \theta/2 + (1-\theta)/q$ gives

$$\|B - \tilde{B}\|_{L^p(\mathcal{T})} \leq \|B - \tilde{B}\|_{L^2(\mathcal{T})}^\theta \|B - \tilde{B}\|_{L^q(\mathcal{T})}^{1-\theta} = o(1).$$

For lower powers, Hölder and boundedness of $\mathcal{T}(\text{Id})$ give the same conclusion. Expand the difference of two words of length r by replacing one factor at a time. Trace Hölder with all exponents r bounds each resulting trace by the product of its factors' $L^r(\mathcal{T})$ norms. The changed factor tends to zero and the others remain bounded. There are only finitely many such terms. This proves the claim, including polynomial squared norms and linear terms with $A - \alpha A_u$. $\square$

For fixed t, L , define the scalar function

$$g_x(v) = \sqrt{1 + f_L(xe^{\sqrt{8t}v - 4t})}, \quad x > 0, v \in \mathbb{R}. \quad (7.21)$$

Then $1 \leq g_x \leq \sqrt{1+L}$. By Equation (7.11), the frozen observation on a block at $T = x/\beta^2$ is exactly $g_x(A_\beta)G_\beta$. Thus $\tilde{B}_{\beta,x} = H_\beta[g_x(A_\beta)G_\beta]$.

Lemma 7.7 (Polynomial density under exponential moments). *Let ν be a finite positive measure on $\mathbb{R}^k$ with $\int e^{a|w|} d\nu(w) < \infty$ for every $a > 0$. Polynomials are dense in $L^p(\nu)$ for every $1 \leq p < \infty$.*

Proof. Suppose a continuous linear functional on $L^p(\nu)$ annihilates all polynomials. Its representing density F belongs to $L^{p'}(\nu)$, where $p' = p/(p-1)$ and $p' = \infty$ if $p = 1$. Hölder and the hypothesis show that the finite signed measure $F d\nu$ has exponential moments of every linear magnitude. For each $\xi \in \mathbb{R}^k$, the function

$$z \mapsto \int e^{z\xi \cdot w} F(w) d\nu(w)$$

is entire: on every compact set of complex z , its derivatives are dominated by an integrable exponential. All its derivatives at zero vanish because $(\xi \cdot w)^j$ is a polynomial. Its Taylor series and hence the entire function are zero. Evaluating at $z = i$ proves that the Fourier transform of $F d\nu$ vanishes. Fourier uniqueness for finite measures can here be seen by convolving with a centered Gaussian: the convolution has zero Fourier transform and an integrable Fourier inversion formula, so is zero for every positive Gaussian variance; letting that variance tend to zero shows the original measure is zero. Thus $F = 0$ almost everywhere. Duality, or equivalently separation of a proper closed subspace, proves the density assertion. $\square$

Lemma 7.8 (Nonlinear mixed trace law). *Let n, g', g be independent scalar standard Gaussians and let $n_u = \alpha n + \sqrt{1 - \alpha^2} g'$. For every fixed mixed polynomial, the weighted trace moments of*

$$(A_\beta, A_{u,\beta}, \tilde{B}_{\beta,x})$$

converge to the corresponding scalar moments of

$$(n, n_u, g_x(n)g).$$

The assertion holds after integration over compact positive u, x intervals against $\kappa(u) du dx/x^2$, and permits fixed polynomial dependence on u, x . Adjoints in a matrix word become the identical real scalar factors in its limit.

Proof. First we prove the matrix comparison

$$H[g_x(A)G] - g_x(A)H[G] \longrightarrow 0 \quad \text{in weighted squared Hilbert–Schmidt norm.} \quad (7.22)$$

All notation in this calculation is on one block. Define

$$J(v)_{ab} = (H_b v)_a, \quad D(v) = H[v] - J(v), \quad \Delta_{abi} = (H_i)_{ab} - (H_b)_{ai}.$$

Put $q = g_x(A)$ and $B = H[G]$. The exact columnwise identity is

$$H[qG] - qB = D(qG) + K - qD(G), \quad K_{\cdot b} = [H_b, q]G. \quad (7.23)$$

Indeed the b th column of $J(qG) - qJ(G)$ is $H_b qG - qH_b G$. Conditional Gaussian averaging therefore bounds the square of the left side by

$$6(1+L) \sum_{a,b,i} |\Delta_{abi}|^2 + 3 \sum_b \|[H_b, g_x(A)]\|_{\text{HS}}^2. \quad (7.24)$$

For the first and last terms in Equation (7.23), we used $\|q\|_{\text{op}} \leq \sqrt{1+L}$ and the Gaussian isometry for the linear map D .

The function g_x is Lipschitz with a bound depending only on t, L , uniformly for all $x > 0$. In fact, with $a = \sqrt{8t}$ and $r = x e^{av-4t}/L$, differentiation gives

$$|g'_x(v)| = \frac{aLr}{2(1+r)^2 g_x(v)} \leq \frac{aL}{8}.$$

In an eigenbasis of A with eigenvalues r_a , the squared commutator norm is

$$\|[H_b, g_x(A)]\|_{\text{HS}}^2 = \sum_{a,c} |(H_b)_{ac}|^2 |g_x(r_c) - g_x(r_a)|^2 \leq \text{Lip}(g_x)^2 \|[H_b, A]\|_{\text{HS}}^2.$$

This is a scalar Lipschitz estimate for matrix entries in the Hilbert–Schmidt norm. Averaging $A = \sum_i N_i H_i$ over N gives

$$\mathbb{E}_N \sum_b \|[H_b, A]\|_{\text{HS}}^2 = \sum_{b,i} \|[H_b, H_i]\|_{\text{HS}}^2.$$

Multiply Equation (7.24) by the block weight, sum, and average. The total symmetry defect and Equation (7.13) prove Equation (7.22), uniformly in x at fixed t, L .

Every fixed higher moment of both sides of this comparison is bounded. Conditional on A and the initial data, $H[g_x(A)G]$ is a Gaussian matrix series with coefficient-square sum at most $(1+L)S$, by the quadratic-form argument used in Equation (7.20). Also the Schatten ideal inequality gives

$$\|g_x(A)H[G]\|_p \leq \sqrt{1+L} \|H[G]\|_p$$

on each block. Interpolation and telescoping products, exactly as in Lemma 7.6, show that these two matrices may be interchanged in every fixed polynomial trace.

It remains to identify traces containing $g_x(A)H[G]$. Lemma 6.7 gives the joint scalar Gaussian moment law for $A = H[N]$, $H[G']$, and $H[G]$. In particular, $A_u = \alpha A + \sqrt{1-\alpha^2}H[G']$ has the required joint law with them. To include the bounded continuous function $g_x(A)$, fix x and a mixed word of length r . Choose a polynomial p approximating g_x in a sufficiently high finite $L^q(\gamma_1)$ norm, where γ_1 is the scalar standard Gaussian law. Choose q larger than all the Hölder exponents needed for this fixed word, and take the approximation error at most 1; then $\|p\|_{L^q(\gamma_1)} \leq \sqrt{1+L} + 1$. Such approximations exist by the polynomial-density argument in Lemma 7.7, applied to one Gaussian coordinate. The scalar spectral measures of A under τ converge to γ_1 by the moment and exponential-moment bounds in Lemma 6.7. Their higher moments are uniformly bounded. Consequently, for even q ,

$$\tau(|g_x(A) - p(A)|^q) \longrightarrow \mathbb{E}|g_x(n) - p(n)|^q. \quad (7.25)$$

To justify this last passage directly, cut the scalar spectral variable off at $|v| \leq M$, use weak convergence for the bounded continuous integrand there, and bound the complement by a higher uniform moment of A . Since g_x is bounded and p has fixed degree, that tail tends to zero as $M \rightarrow \infty$.

Replace each occurrence of $g_x(A)$ by $p(A)$. Telescoping the word and applying trace Hölder bounds the replacement error by a constant times the L^q error in Equation (7.25), with all other factors controlled by fixed higher moments. For factors $p(A)$, the preceding L^q bound and moment convergence give a bound independent of how accurately p approximates g_x . Thus the error constant stays bounded as that approximation is improved. The remaining word is polynomial in the three independent Gaussian matrix series, so its trace has the scalar Gaussian limit. Let the scalar approximation error tend to zero. This proves the claimed limit at each fixed u, x ; the scalar third coordinate is $g_x(n)g$.

Finally, the moment bounds for the original, bounded function g_x are uniform on the compact parameter rectangle, as are the bounds on polynomial coefficients in u, x and on α . They dominate each mixed trace by an integrable constant. Dominated convergence thus permits parameter integration. This last argument uses the uniform bounds for g_x itself, and does not require choosing one polynomial approximant uniformly in x . $\square$

The preceding lemmas identify the polynomial tests of the actual observation with those of the scalar variance mixture. We can now optimize the tests and integrate the scalar OU dissipation. The constant in the resulting comparison is exactly the constant of the unnormalized flattening estimate.

Proposition 7.9 (Scalar information lower bound). *For every fixed $t > 0$ and $L \geq 1$, let g_x be defined by Equation (7.21), and let n, g be independent scalar standard Gaussians. Then*

$$\liminf \int_0^\infty I(Y_t; Z_T) \frac{dT}{T^2} \geq C_1^{-1} \int_0^\infty I(n; g_x(n)g) \frac{dx}{x^2}. \quad (7.26)$$

The constant C_1 is universal and is independent of t, L .

Proof. Fix compact positive intervals $\mathcal{J}, \mathcal{X}$, and let n, g', g be independent scalar standard Gaussians. Write $n_u = \alpha n + \sqrt{1 - \alpha^2} g'$, and fix a scalar polynomial $P(u, x, a, b)$. Give its matrix version any fixed ordering of the factors of each monomial. Lemmas 7.5, 7.6, and 7.8, in that order, imply

$$\begin{aligned} \liminf \int_0^\infty I(Y_t; Z_T) \frac{dT}{T^2} \\ \geq \int_{\mathcal{J}} \int_{\mathcal{X}} \kappa(u) \mathbb{E}[2(n - \alpha n_u)P(u, x, n_u, g_x(n)g) \\ - C_1 P(u, x, n_u, g_x(n)g)^2] \frac{dx}{x^2} du. \end{aligned} \quad (7.27)$$

If the matrix polynomial is not symmetric, its prelimit cost is $\text{tr}(P^*P)$ as in Equation (7.17). The mixed trace law takes this cost to the real scalar P^2 , so symmetry of the chosen ordering is unnecessary.

The sequence limit in Equation (7.27) has been taken for each fixed polynomial. We now optimize the tests in one squared-mean space containing the compact parameters. Give (u, x) the finite measure $\kappa(u) du dx/x^2$ on $\mathcal{J} \times \mathcal{X}$, independently of the scalar Gaussians n, g', g , and put

$$W = (u, x, n_u, g_x(n)g), \quad r = n - \alpha n_u. \quad (7.28)$$

The first two coordinates of W are bounded. At each fixed u, x , the variable n_u is standard Gaussian, and $|g_x(n)g| \leq \sqrt{1 + L}|g|$. Cauchy–Schwarz therefore gives all linear exponential moments for its finite law, uniformly on the compact parameter rectangle. Lemma 7.7 shows that polynomials in the four coordinates of W are dense in this L^2 space.

Since $\mathbb{E}[r^2 | u, x] = 1 - \alpha^2 \leq 1$, the conditional mean $m(W) = \mathbb{E}[r | W]$ belongs to the same space. For the finite joint measure just defined, completion of the square gives

$$\int (2rP(W) - C_1 P(W)^2) = C_1^{-1} \int m^2 - C_1 \int (P - m/C_1)^2.$$

Density makes the infimum of the final squared error zero, using single four-variable polynomials in the integrated space. Taking their supremum and then disintegrating with respect to (u, x) turns the right side of Equation (7.27) into

$$C_1^{-1} \int_{\mathcal{J}} \int_{\mathcal{X}} \frac{\alpha^2}{(1 - \alpha^2)^2} \mathbb{E} |\mathbb{E}[n - \alpha n_u | n_u, g_x(n)g]|^2 \frac{dx}{x^2} du. \quad (7.29)$$

The conditional expectation here is taken at each fixed u, x . All approximation constants used before this optimization may depend on the fixed polynomial, t, L , and the compact intervals. The coefficient C_1 depends only on Equation (7.18).

We verify the scalar information identity including its endpoints. Fix $x > 0$ and write $z_x = g_x(n)g$. Its information is finite: conditional variance lies in $[1, 1 + L]$, so Gaussian entropy comparison gives

$$0 \leq I(n; z_x) \leq \frac{1}{2} \log(1 + L). \quad (7.30)$$

Indeed $\mathbb{E}z_x^2 \leq 1 + L$ bounds its entropy above by $\frac{1}{2} \log(2\pi e(1 + L))$, while its conditional entropy is $\frac{1}{2} \log(2\pi e) + \mathbb{E} \log g_x(n)$ and is at least $\frac{1}{2} \log(2\pi e)$. Apply the scalar version of Lemma 7.4 and its entropy dissipation identity between times a and b , where $0 < a < b < \infty$. This gives

$$I(n_a; z_x) - I(n_b; z_x) = \int_a^b \frac{\alpha^2}{(1 - \alpha^2)^2} \mathbb{E} |\mathbb{E}[n - \alpha n_u \mid n_u, z_x]|^2 du.$$

As $a \downarrow 0$, the joint pair (n_a, z_x) converges in law to (n, z_x) . Lower semicontinuity of relative entropy, applied to the joint law and the product of its marginals, gives $I(n; z_x) \leq \liminf_{a \downarrow 0} I(n_a; z_x)$. The reverse inequality follows from the Markov chain $z_x \rightarrow n \rightarrow n_a$ and data processing. Thus the initial endpoint is exactly $I(n; z_x)$. At the other endpoint, the same chain gives

$$0 \leq I(n_u; z_x) \leq I(n_u; n) = -\frac{1}{2} \log(1 - \alpha^2) \rightarrow 0.$$

The equality follows by subtracting the conditional Gaussian entropy of variance $1 - \alpha^2$ from that of the standard Gaussian marginal. We have proved

$$I(n; z_x) = \int_0^\infty \frac{\alpha^2}{(1 - \alpha^2)^2} \mathbb{E} |\mathbb{E}[n - \alpha n_u \mid n_u, z_x]|^2 du. \quad (7.31)$$

Finally exhaust both positive parameter axes by compact intervals in Equation (7.29). Its integrand is nonnegative. Monotone convergence and Equation (7.31) prove Equation (7.26). In particular, the polynomial optimization and the exhaustion take place after the counterexample-sequence limit, so neither requires a uniform approximation over unbounded parameters. $\square$

7.5 Divergence of the scalar scale integral

The upper and lower comparisons are now in the same information units, with universal constants C_0, C_1 . To contradict them, it suffices to let the fixed diffusion time and the fixed covariance cap grow together. The following elementary test detects the large values of the scalar lognormal variance.

Lemma 7.10 (Scalar information divergence). *Let n, g be independent scalar standard Gaussians and, for $t \geq 1$, define*

$$\ell = \exp(\sqrt{8t}n - 4t), \quad z_x = \sqrt{1 + f_t(x\ell)}g.$$

There are absolute constants $c > 0$ and $h_0 \geq 1$ such that, for all $t \geq h_0$,

$$\int_0^\infty I(n; z_x) \frac{dx}{x^2} \geq c \mathbb{E}[\ell \mathbf{1}_{\{\ell \geq e^{2t}\}}] \log(t/h_0) \geq \frac{c}{2} \log(t/h_0). \quad (7.32)$$

Proof. The Gaussian moment generating formula gives $\mathbb{E}\ell = 1$. Let $P_{x,n}$ be the conditional law of z_x given n , and let Q_x be its marginal law. Thus

$$I(n; z_x) = \mathbb{E}_n H(P_{x,n} \mid Q_x).$$

Set $p_0 = \mathbb{P}\{|g| \geq \sqrt{2}\} > 0$. Suppose first that $\ell \geq e^{2t}$, and choose a scale with $h = x\ell \in [h_0, t]$. Since $h \leq t$,

$$\text{Var}(z_x \mid n) = 1 + \frac{h}{1 + h/t} \geq h/2.$$

It follows that the event $A_h = \{|z| \geq \sqrt{h}\}$ has conditional probability $p = P_{x,n}(A_h) \geq p_0$.

To bound its marginal probability, use an independent variable ℓ' with the law of ℓ . On $x\ell' \leq 1$, the conditional variance of the mixture component is at most 2, so its probability of A_h is at most $2e^{-h/4}$ by the Gaussian exponential bound. On the complementary event, Markov's inequality and $\mathbb{E}\ell' = 1$ give probability at most x . Hence

$$q = Q_x(A_h) \leq 2e^{-h/4} + x \leq 2e^{-h/4} + te^{-2t} \leq 3e^{-h/4}. \quad (7.33)$$

The last inequality holds for $t \geq 1$ and $h \leq t$, because $te^{-2t} \leq e^{-t/4} \leq e^{-h/4}$.

Apply data processing to the event indicator. The binary relative entropy is bounded below by

$$\begin{aligned} H(P_{x,n} \mid Q_x) &\geq p \log(p/q) + (1-p) \log((1-p)/(1-q)) \\ &\geq p \log(1/q) - \log 2 \\ &\geq p_0(h/4 - \log 3) - \log 2. \end{aligned}$$

Choose once and for all

$$h_0 \geq \max\{1, 8 \log 3 + 8 \log 2/p_0\}, \quad c = p_0/8.$$

The preceding expression is then at least ch for $h \in [h_0, t]$. This entropy estimate is made separately for each conditional law $P_{x,n}$; the chosen event may therefore depend on n through $h = x\ell$.

Nonnegativity of relative entropy permits restricting the scale integral to these choices of n and x . Tonelli and the change of variable $x = h/\ell$ yield

$$\begin{aligned} \int_0^\infty \mathbb{I}(n; z_x) \frac{dx}{x^2} &\geq \mathbb{E} \left[\mathbf{1}_{\{\ell \geq e^{2t}\}} \int_{h_0/\ell}^{t/\ell} H(P_{x,n} \mid Q_x) \frac{dx}{x^2} \right] \\ &\geq c \mathbb{E} \left[\ell \mathbf{1}_{\{\ell \geq e^{2t}\}} \int_{h_0}^t \frac{dh}{h} \right] \\ &= c \mathbb{E} [\ell \mathbf{1}_{\{\ell \geq e^{2t}\}}] \log(t/h_0). \end{aligned}$$

Under the probability change of measure with density ℓ , completion of the square makes n a Gaussian of mean $\sqrt{8t}$ and variance 1. Therefore $\log \ell$ under that measure is Gaussian of mean $4t$ and variance $8t$. If Φ is the standard Gaussian distribution function, then

$$\mathbb{E}[\ell \mathbf{1}_{\{\ell \geq e^{2t}\}}] = \Phi(\sqrt{t}/2) \geq \frac{1}{2}.$$

This proves both inequalities in Equation (7.32). $\square$

Proof of Theorem 1.1. Suppose that the asserted dimension-free bound fails. Lemma 3.1 gives compact centered counterexamples with normalized logarithmic Sobolev constant 1 and linear subgaussian parameter tending to zero. Proposition 3.8 shows that their positive-time regularized constants converge to 1. Proposition 3.9 supplies the stationary laws and near-eigenfunction fields at the fixed regularization time $s = 1/8$, in either the strict entropy branch or the spectral gap branch. Proposition 4.3 gives the exact prediction curve in both cases. Proposition 5.1 then gives

the symmetric tensor hierarchy, and Proposition 6.8 constructs its matrix square root and frozen block description. These are precisely the inputs used in the present section.

For each fixed $t > 0$ and $L \geq 1$, combine Propositions 7.2 and 7.9. They give

$$\int_0^\infty \mathfrak{I}\left(n; \sqrt{1 + f_L(xe^{\sqrt{8t}n-4t})} g\right) \frac{dx}{x^2} \leq C_0 C_1, \quad (7.34)$$

where $C_0 C_1$ is independent of both fixed parameters. Choose an arbitrarily large but finite $t \geq h_0$ and the fixed cap $L = t$. Lemma 7.10 makes the left side at least $(c/2) \log(t/h_0)$, which exceeds $C_0 C_1$ for a sufficiently large choice of t . This is a contradiction.

All counterexample limits used to obtain Equation (7.34) were taken with this t, L fixed. Only afterward was their numerical size chosen. Thus no uniform estimate on the intermediate errors as $t, L \rightarrow \infty$ is needed. The contradiction excludes an unbounded ratio $R(\mu)/a^2$ over the measures in the theorem. Finally the convention $\text{Ent}_\mu(f^2) \leq 2R(\mu) \int |Df|^2 d\mu$ absorbs the factor 2 into a universal constant and gives the asserted inequality for every smooth compactly supported real test function. $\square$

References

- [1] Huzihiro Araki and Shigeru Yamagami. An inequality for Hilbert–Schmidt norm. *Communications in Mathematical Physics*, 81(1):89–96, 1981.
- [2] Dominique Bakry and Michel Émery. Diffusions hypercontractives. In *Séminaire de probabilités, XIX, 1983/84*, volume 1123 of *Lecture Notes in Mathematics*, pages 177–206. Springer, Berlin, 1985.
- [3] Salman Beigi. Sandwiched Rényi divergence satisfies data processing inequality. *Journal of Mathematical Physics*, 54(12):122202, 2013. Cited theorem numbers refer to arXiv:1306.5920v6.
- [4] Pierre Bizeul. On the log-Sobolev constant of log-concave vectors. *Journal of Functional Analysis*, 290(9):111368, 2026. Cited theorem numbers refer to arXiv:2306.12997v3, 16 February 2026; first posted in 2023.
- [5] Sergey G. Bobkov. Isoperimetric and analytic inequalities for log-concave probability measures. *The Annals of Probability*, 27(4):1903–1921, 1999.
- [6] Yuansi Chen and Ronen Eldan. Localization schemes: A framework for proving mixing bounds for Markov chains. *Duke Mathematical Journal*, 174(8), 2025. Cited section and equation numbers refer to arXiv:2203.04163v2, 6 June 2022.
- [7] Ronen Eldan, Frederic Koehler, and Ofer Zeitouni. A spectral condition for spectral gap: fast mixing in high-temperature Ising models. *Probability Theory and Related Fields*, 182(3-4):1035–1051, 2022. Cited theorem and equation numbers refer to arXiv:2007.08200v2, 6 August 2021.
- [8] Leonard Gross. Logarithmic Sobolev inequalities. *American Journal of Mathematics*, 97(4):1061–1083, 1975.

- [9] Dongning Guo, Shlomo Shamai (Shitz), and Sergio Verdú. Mutual information and minimum mean-square error in Gaussian channels. *IEEE Transactions on Information Theory*, 51(4):1261–1282, April 2005. Cited theorem and equation numbers refer to arXiv:cs/0412108v1.
- [10] Kiyosi Itô. Multiple Wiener integral. *Journal of the Mathematical Society of Japan*, 3(1):157–169, 1951.
- [11] Young-Heon Kim and Emanuel Milman. A generalization of Caffarelli’s contraction theorem via (reverse) heat flow. *Mathematische Annalen*, 354(3):827–862, 2012.
- [12] Bo’az Klartag and Joseph Lehec. Isoperimetric inequalities in high-dimensional convex sets. *Bulletin of the American Mathematical Society*, 62(4):575–642, 2025. Cited theorem numbers refer to arXiv:2406.01324v2, 29 November 2024.
- [13] Bo’az Klartag and Eli Putterman. Spectral monotonicity under Gaussian convolution. *Annales de la Faculté des Sciences de Toulouse. Mathématiques*, 32(5):939–967, 2023.
- [14] Michel Ledoux. Concentration of measure and logarithmic Sobolev inequalities. In *Séminaire de probabilités, XXXIII*, volume 1709 of *Lecture Notes in Mathematics*, pages 120–216. Springer, Berlin, 1999.
- [15] Elliott H. Lieb. Convex trace functions and the Wigner–Yanase–Dyson conjecture. *Advances in Mathematics*, 11(3):267–288, 1973.
- [16] Emanuel Milman. Isoperimetric and concentration inequalities: Equivalence under curvature lower bound. *Duke Mathematical Journal*, 154(2):207–239, 2010. Cited theorem numbers refer to arXiv:0902.1560v2, 1 December 2009.
- [17] Felix Otto and Cédric Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis*, 173(2):361–400, 2000.
- [18] O. S. Rothaus. Hypercontractivity and the Bakry–Emery criterion for compact Lie groups. *Journal of Functional Analysis*, 65(3):358–367, 1986.
- [19] Joel A. Tropp. Second-order matrix concentration inequalities. *Applied and Computational Harmonic Analysis*, 44(3):700–736, 2018. Cited section and lemma numbers refer to the author preprint revised 3 August 2016.
- [20] Jeff Viaclovsky. Differential analysis, MIT 18.156: Lecture notes. MIT OpenCourseWare, Spring 2004, 2004. Lectures 3, 10, 16, 18, 19, and 24; notes prepared from the instructor’s handwritten notes by Zuoqin Wang (Lectures 0–11) and Yanir Rubinstein (Lectures 12–24).